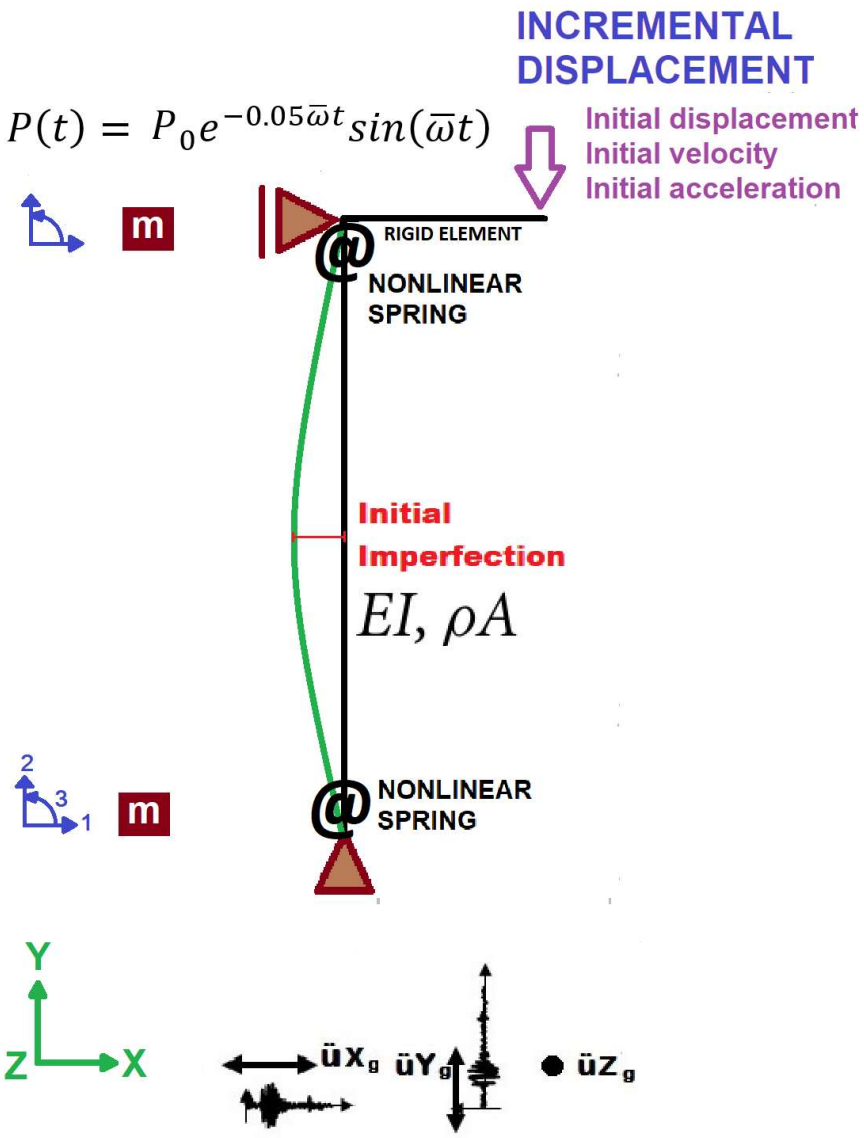
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST-BUCKLING PHENOMENA A COLUMN WITH SEMI-RIGID CONNECTIONS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


INCREMENTAL DISPLACEMENT
$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$
Initial displacement
Initial velocity
Initial acceleration
m
RIGID ELEMENT
NONLINEAR SPRING
Initial Imperfection
$EI, \rho A$
2
3
1
m
NONLINEAR SPRING
Y
Z
X
$\ddot{u}X_g$
$\ddot{u}Y_g$
$\ddot{u}Z_g$

$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

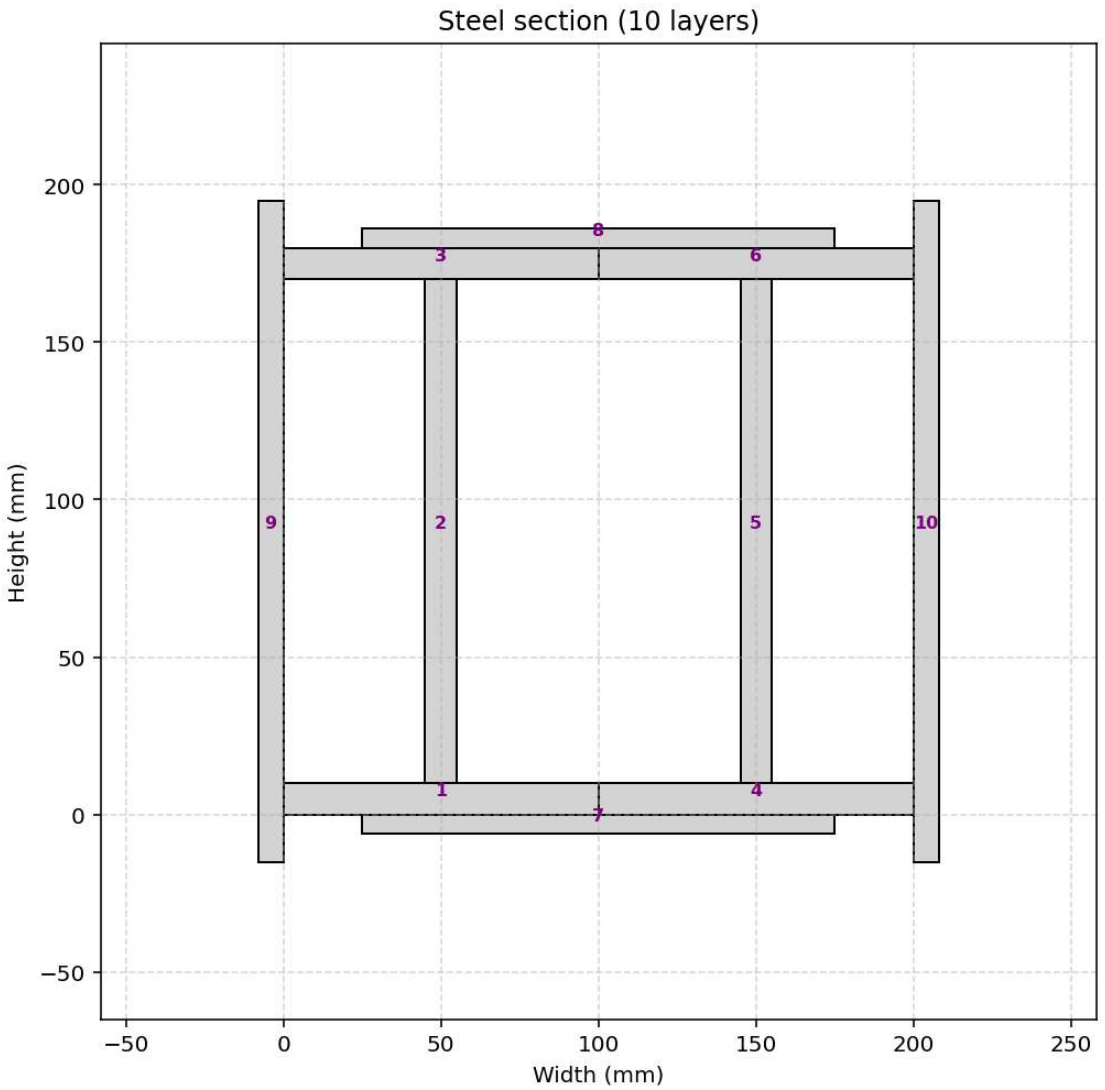
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

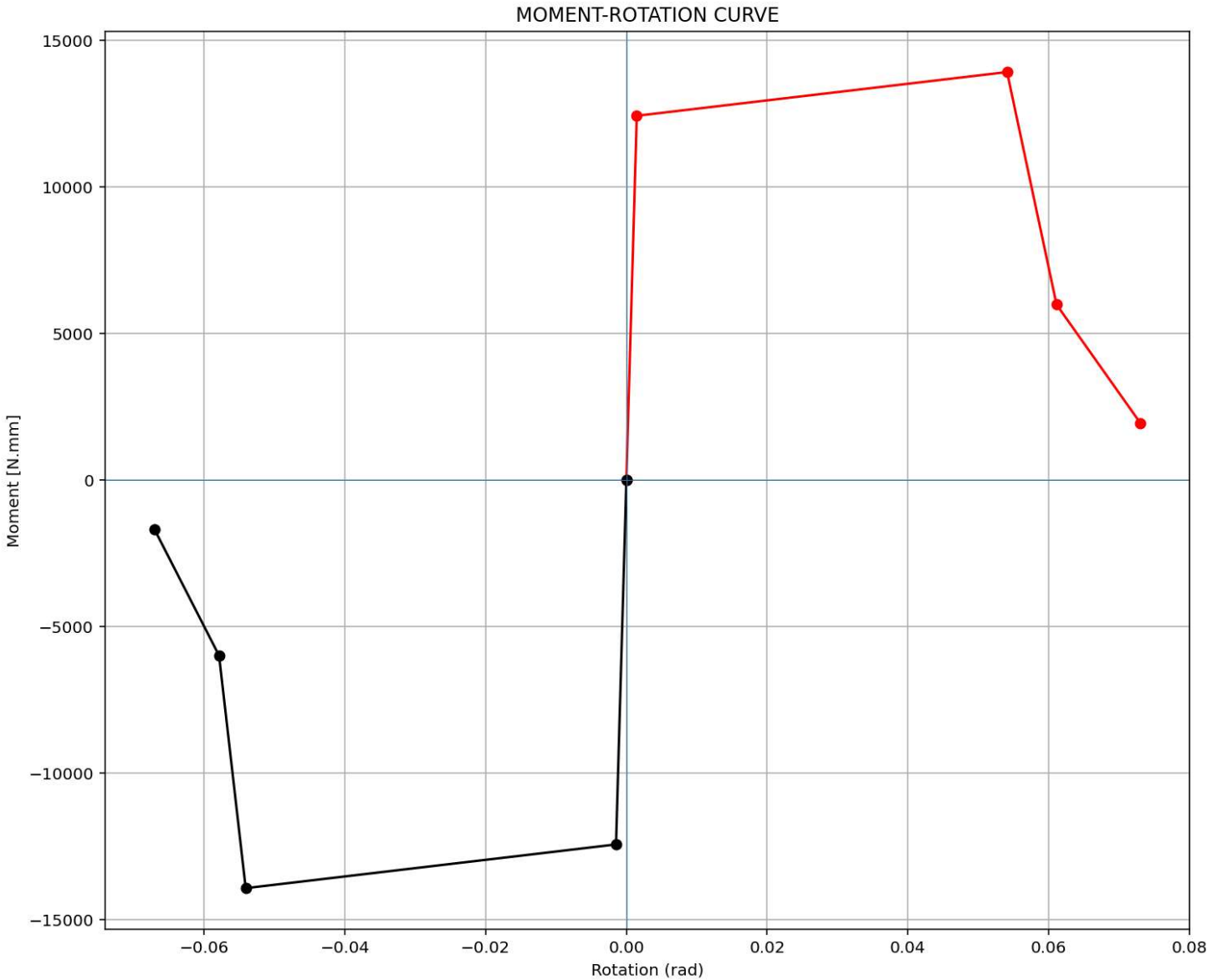
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

boundary conditions

$$v(0,t) = v(L,t) = 0, \quad \left.\frac{\partial^2 v(x,t)}{\partial x^2}\right|_{x=0} = \left.\frac{\partial^2 v(x,t)}{\partial x^2}\right|_{x=L} = 0.$$


Steel section (10 layers)
Height (mm)
Width (mm)
200
150
100
50
0
−50
−50
0
50
100
150
200
250
1
2
3
4
5
6
7
8
9
10


MOMENT-ROTATION CURVE
15000
10000
5000
0
−5000
−10000
−15000
Moment [N.mm]
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Rotation (rad)

Spyder (Python 3.12)

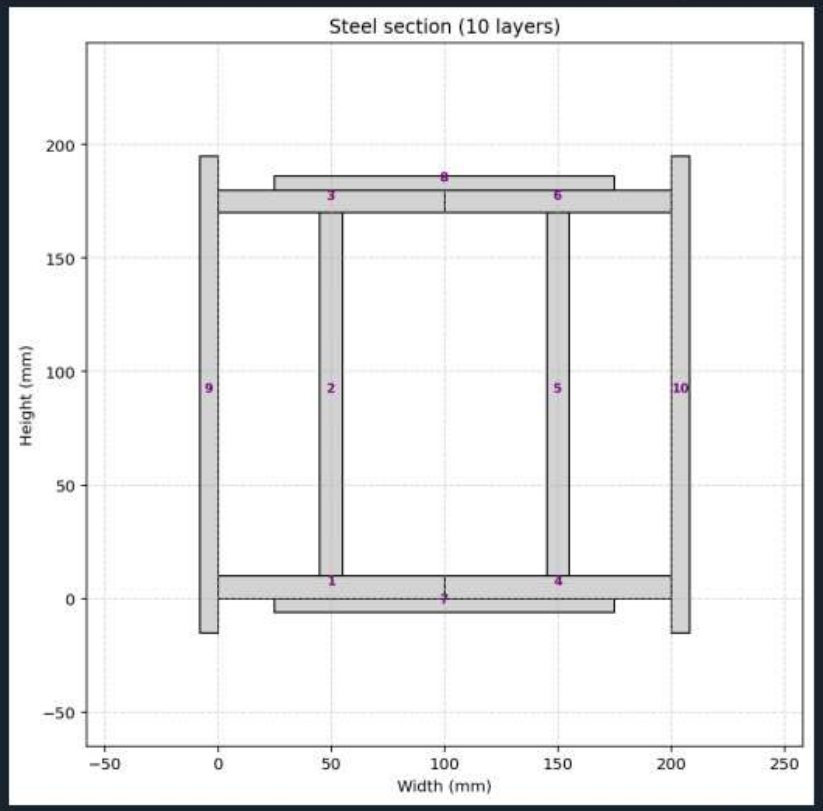
File Edit Search Source Run Debug Consoles Projects Tools View Help

...ON_CAPACITY_INDEX\55_STEEL_CONCRETE_COLUMN_POST-BUCKLING+

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...P(t)_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_STEEL_CONCRET...-BUCKLING_8_ANA.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1  ##########################################################################
2  #                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
3  # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST-BUCKLING PH
4  #         AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STA
5  #                                DYNAMIC TIME-HISTORY ANALYSIS
6  #--------------------------------------------------------------------------
7  #                                                  P(t) = P0 sin(wt)
8  #                                          P(t) = P0 exp(-0.05wt) sin(wt)
9  #--------------------------------------------------------------------------
10 #          ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
11 #                                         OF ENERGY DISSIPATION CAPACITY IND
12 #--------------------------------------------------------------------------
13 #                     THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
14 #                                   EMAIL: salar.d.ghashghaei@gmail.com
15 ##########################################################################
16 """
17 Nonlinear Seismic Performance Assessment of a column:
18 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
19
20 This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
21  for performance-based earthquake engineering. The 2D model incorporates bo
22  (elastic-perfectly plastic or hysteretic steel with strain hardening)
23  and geometric nonlinearity to capture P-delta effects
24  and large displacements—essential for collapse assessment.
25
26 Eight analysis protocols are implemented:
27 (1) [PERIOD] : Structural Period
28 (2) [STATIC] : Gravity load analysis establishing dead load state
29 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
30  and plastic mechanism identification
31 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
32  pinching behavior, and energy dissipation degradation
33 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External
34 (6) [DYNAMIC EXTERNAL TIME-DEPENDENT LOADING] : Dynamic Analysis of Externa
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 811, Col 44 UTF-8 CRLF RW Mem 48%

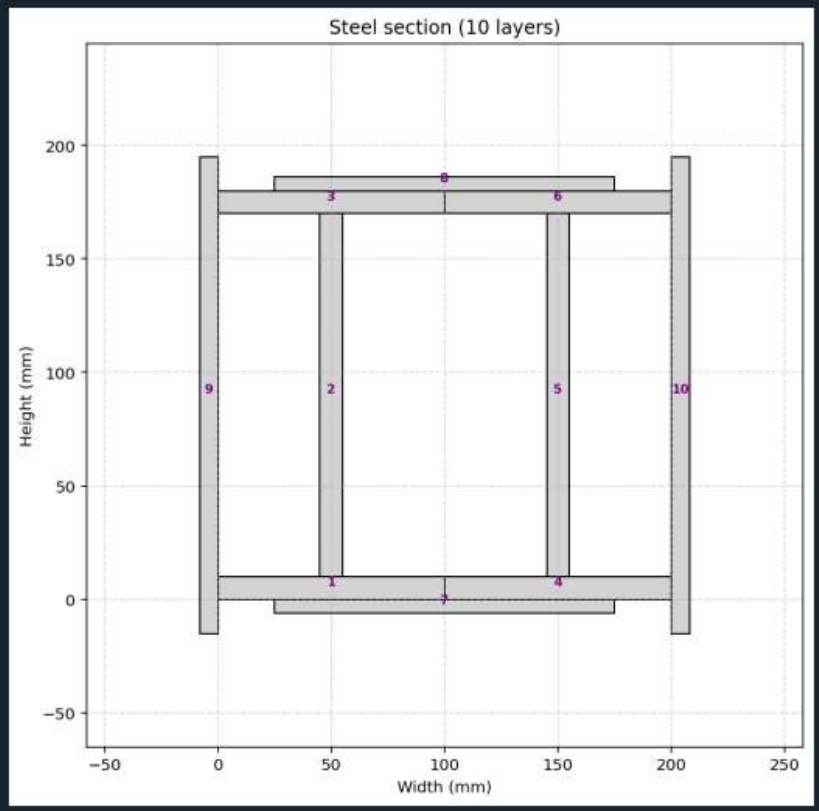
Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...LE_I_STEEL_SECTION_DIFF_WIDTH_HEIGHT_FUN_EXTRA.py

P(t)_STEEL_CONCRET...-BUCKLING_8_ANA.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
21      # Define parameters (units: mm, N)
22      # --------------------------------------
23      # STEEL
24      fy = 240          # [N/mm²] Steel Yield Strength
25      Es = 2e5          # [N/mm²] Modulus of Elasticity
26      ey = fy/Es        # [mm/mm] Steel Yield Strain
27      fu = 1.1818*fy    # [N/mm²] Steel Ultimate Strength
28      esu = 0.15        # [mm/mm] Steel Ultimate Strain
29      Esh = (fu - fy)/(esu - ey)
30      Bs = Esh / Es
31
32      steelTag = secTag + 100
33      if STEEL_TYPE == 'ELASTIC':
34          ops.uniaxialMaterial('Elastic', steelTag, Es)
35      if STEEL_TYPE == 'INELASTIC':
36          pinchX = 0.8   # Pinching factor in X direction
37          pinchY = 0.5   # Pinching factor in Y direction
38          damage1 = 0.0  # Damage due to ductility
39          damage2 = 0.0  # Damage due to energy
40          beta = 0.1     # Stiffness degradation parameter
41          ops.uniaxialMaterial('Hysteretic', steelTag,
42                               fy, ey,
43                               fu, esu,
44                               0.2*fu, 1.1*esu,
45                               -fy, -ey,
46                               -fu, -esu,
47                               -0.2*fu, -1.1*esu,
48                               pinchX, pinchY,
49                               damage1, damage2, beta)
50          # INFO LINK: https://opensees.berkeley.edu/wiki/index.php/Hysteretic
51
52
53      # -------------------- Fiber section -------------------------
54      ops.section('Fiber', secTag)
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 5 UTF-8 CRLF RW Mem 48%

STATIC ANALYSIS

Spyder (Python 3.12)

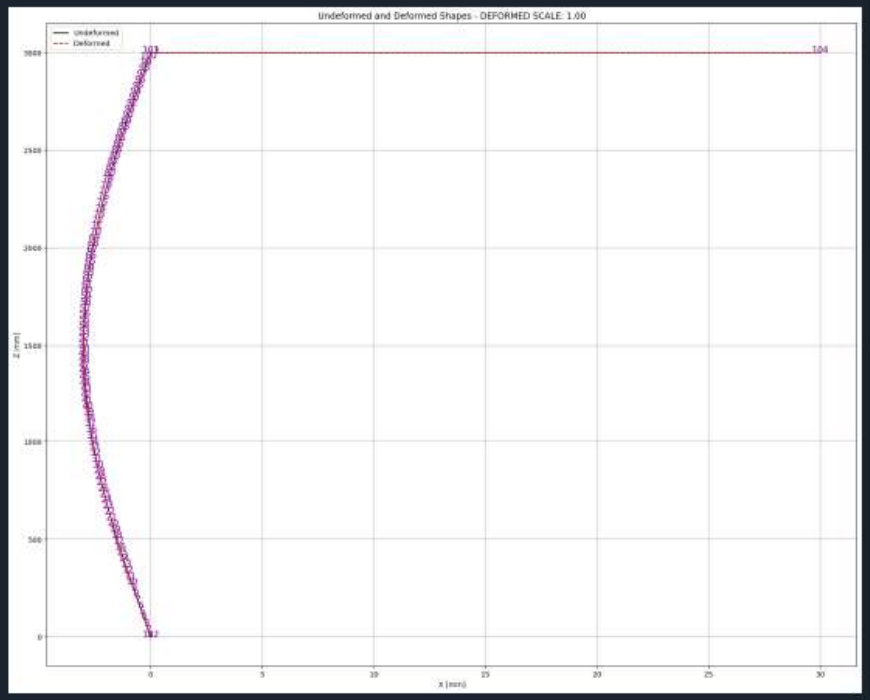
File Edit Search Source Run Debug Consoles Projects Tools View Help

...NCRETE_COLUMN_POST-BUCKLING+\EXAMPLE_02_AXIAL_AND_MOMENT

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

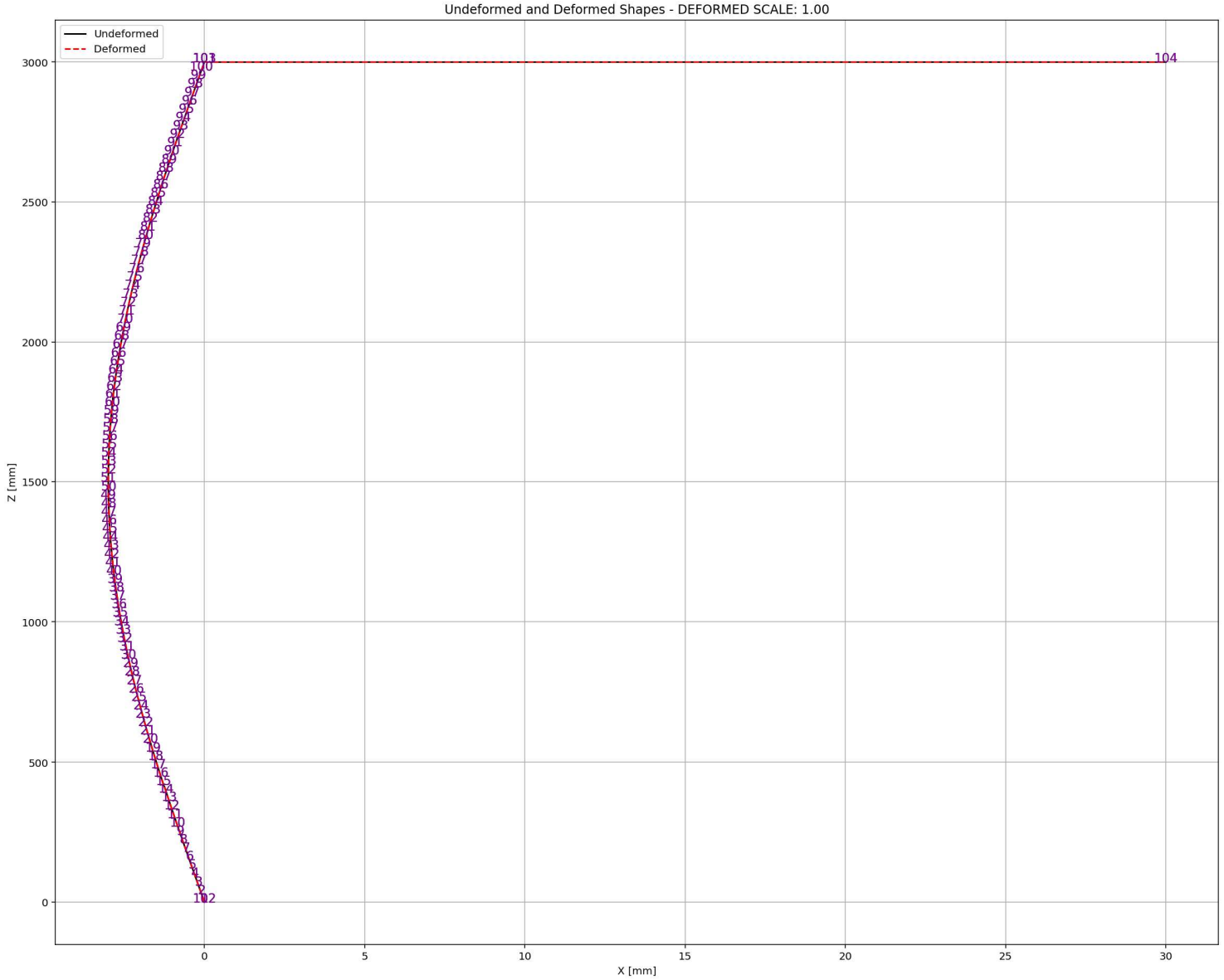
```
1189  # STATIC ANALYSIS
1190  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1191  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1192  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1193  ANAL_TYPE = 'STATIC'
1194
1195  DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1196  (reactionX, reactionY, reactionZ, disp_mid,
1197   ele_axialforce, ele_shearforce, ele_momentforce,
1198   node_displacementsX, node_displacementsY,
1199   dispX, dispY) = DATA
1200
1201  S01.PLOT_2D_FRAME(deformed_scale=1.0)
1202  #%%-------------------------------------------------
1203  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1204  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1205  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1206  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1207  ANAL_TYPE = 'PUSHOVER'
1208
1209  DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1210  (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1211   ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1212   node_displacementsX_PUSH, node_displacementsY_PUSH,
1213    dispX_PUSH, dispY_PUSH,
1214   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1215   STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH, NA
1216
1217
1218  XDATA = disp_mid_PUSH
1219  YDATA = reactionY_PUSH
1220  XLABEL = 'Displacement [mm]'
1221  YLABEL = 'Base Reaction [N]'
1222  TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
```



23 %

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 780, Col 47 UTF-8 CRLF RW Mem 49%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
5
10
15
20
25
30
103
104
102

PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

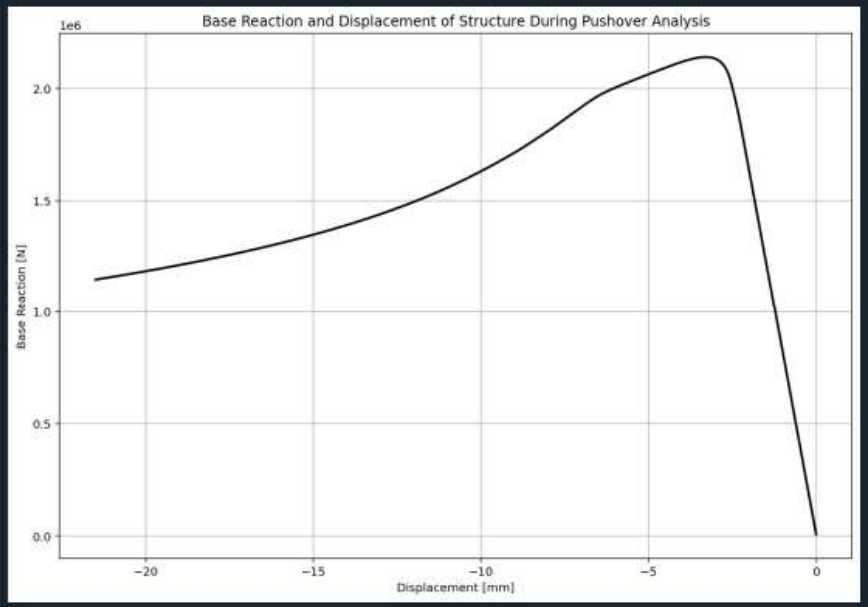
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1203    # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1204    #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1205    ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1206    MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1207    ANAL_TYPE = 'PUSHOVER'
1208
1209    DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1210    (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1211     ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1212     node_displacementsX_PUSH, node_displacementsY_PUSH,
1213      dispX_PUSH, dispY_PUSH,
1214     PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1215     STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH, NA
1216
1217
1218    XDATA = disp_mid_PUSH
1219    YDATA = reactionY_PUSH
1220    XLABEL = 'Displacement [mm]'
1221    YLABEL = 'Base Reaction [N]'
1222    TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
1223    COLOR = 'black'
1224    SEMILOGY = False
1225    PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1226
1227    DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionX_PUSH), SLO
1228    (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_S
1229
1230    # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1231    plt.figure(0, figsize=(12, 8))
1232    plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1233    plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1234    plt.title('Period of Structure During Pushover Analysis')
1235    plt.ylabel('Structural Period [s]')
1236    plt.xlabel('Displacement [mm]')
```

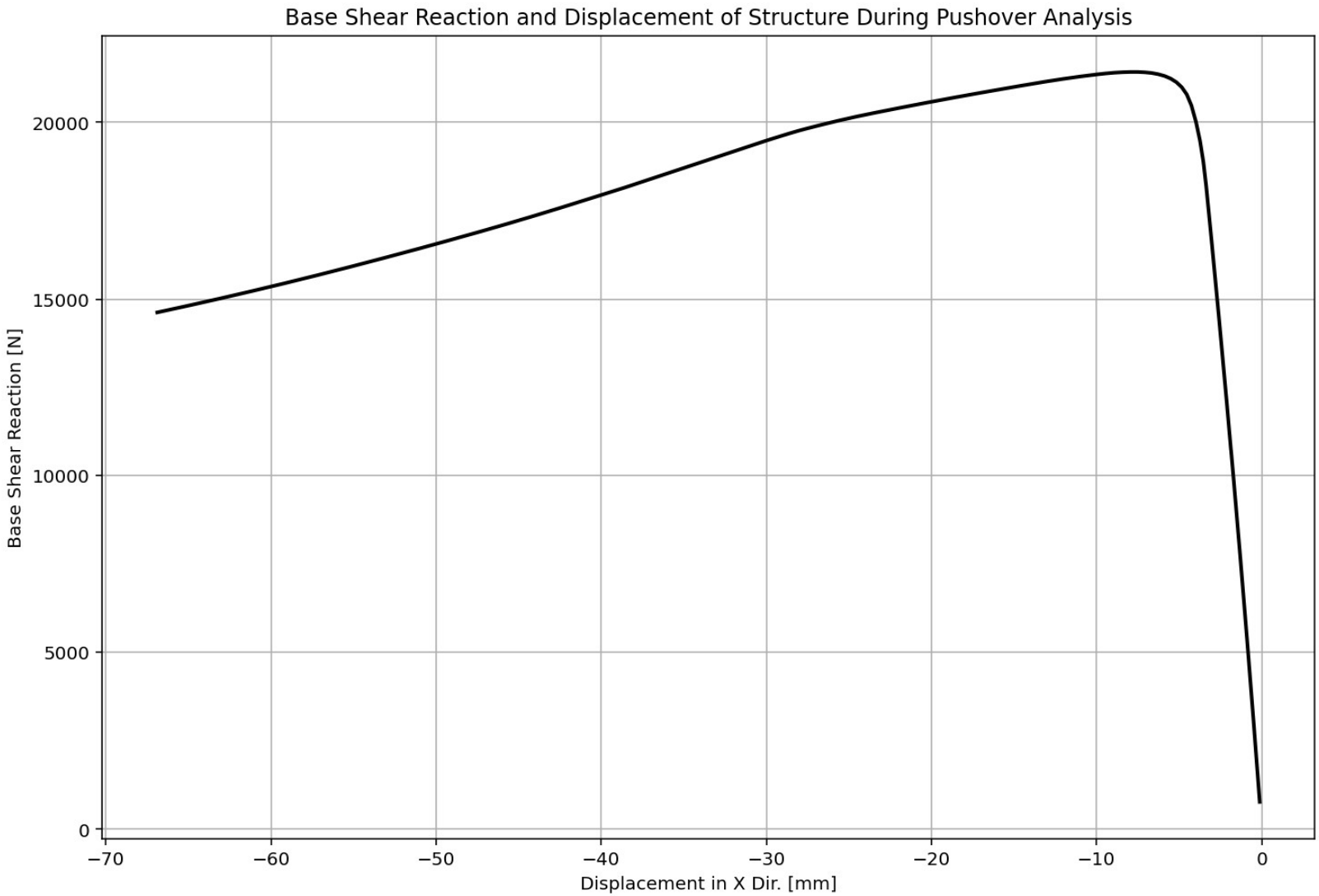
...NCRETE_COLUMN_POST-BUCKLING+\EXAMPLE_02_AXIAL_AND_MOMENT

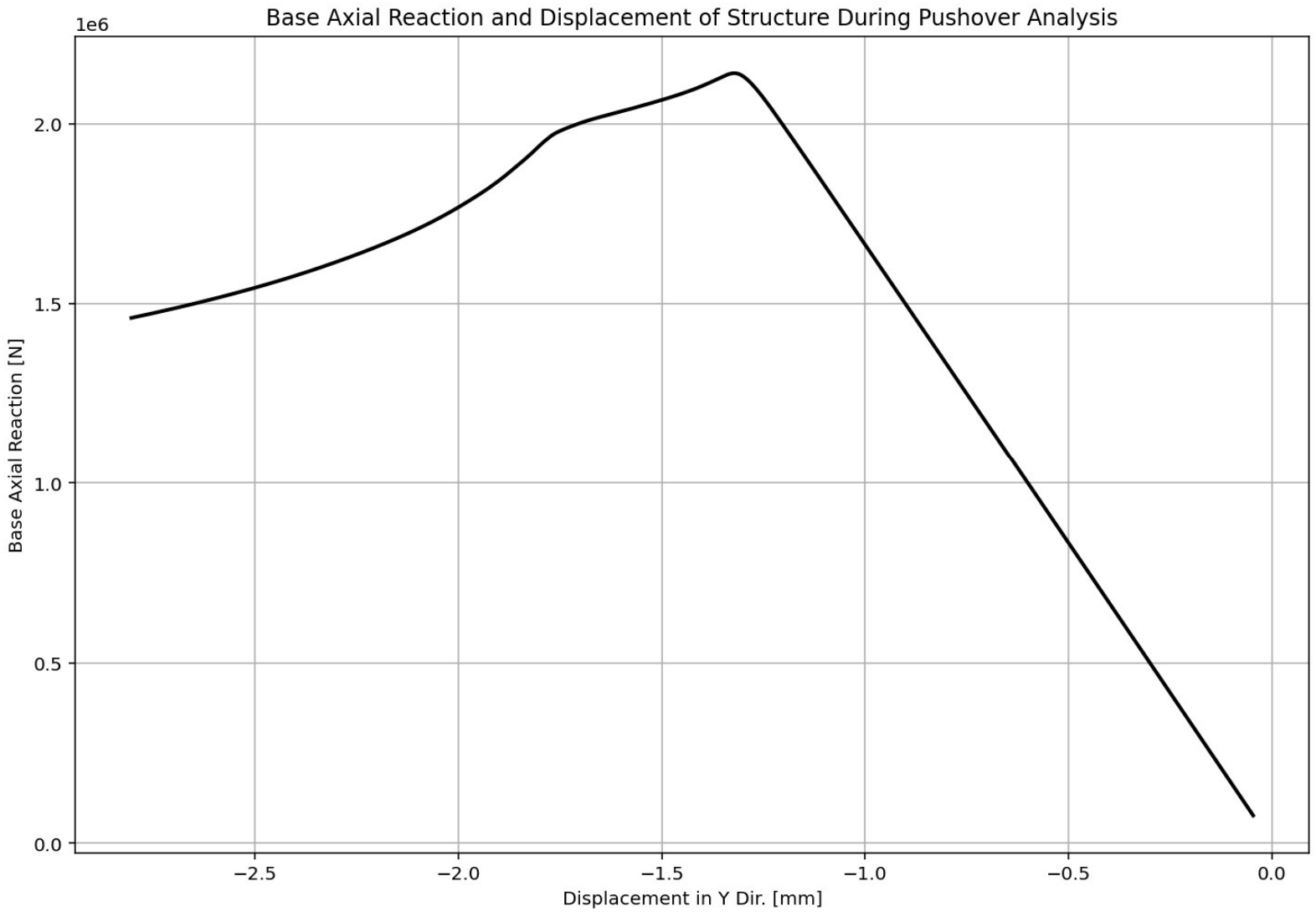
38 %

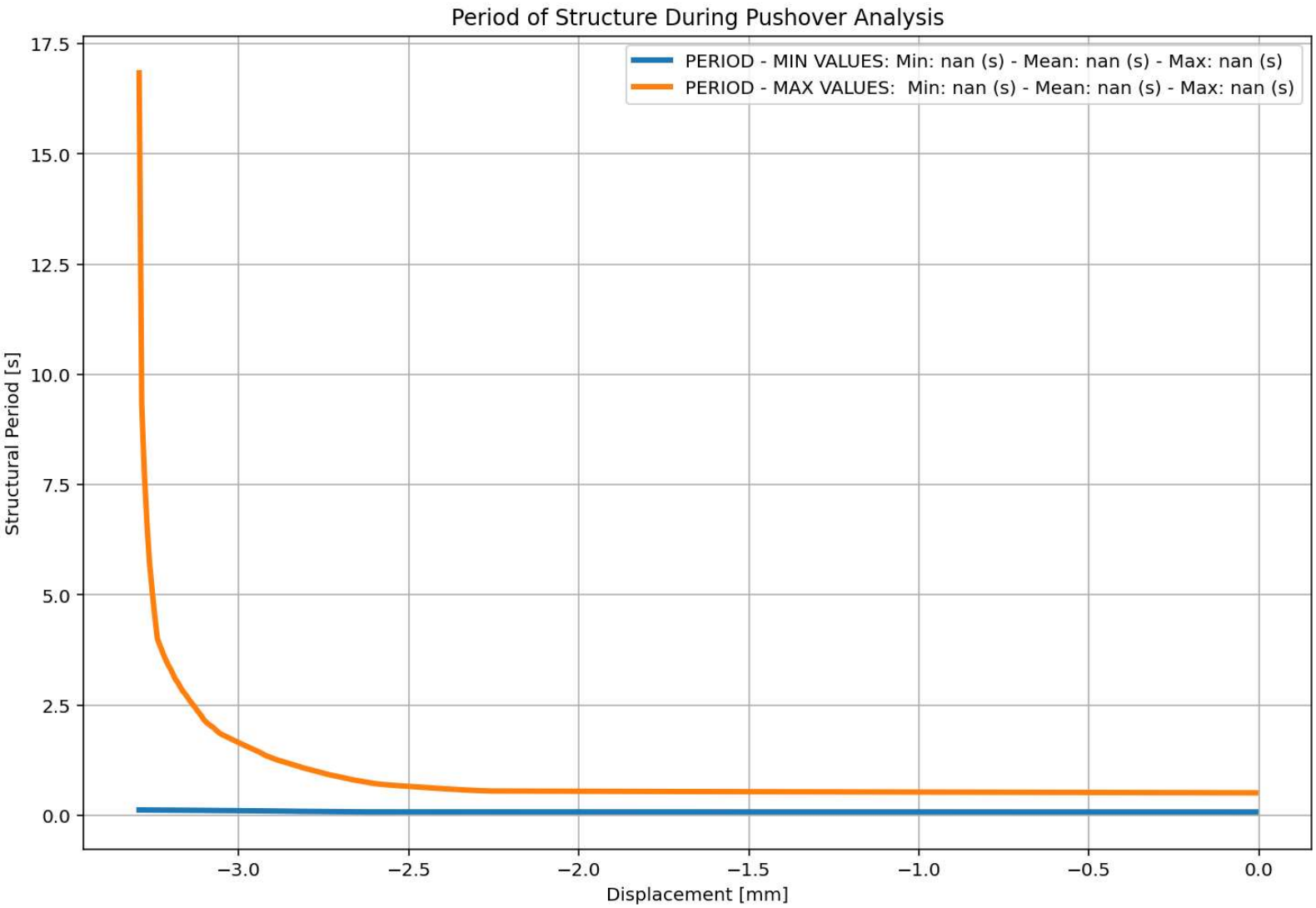


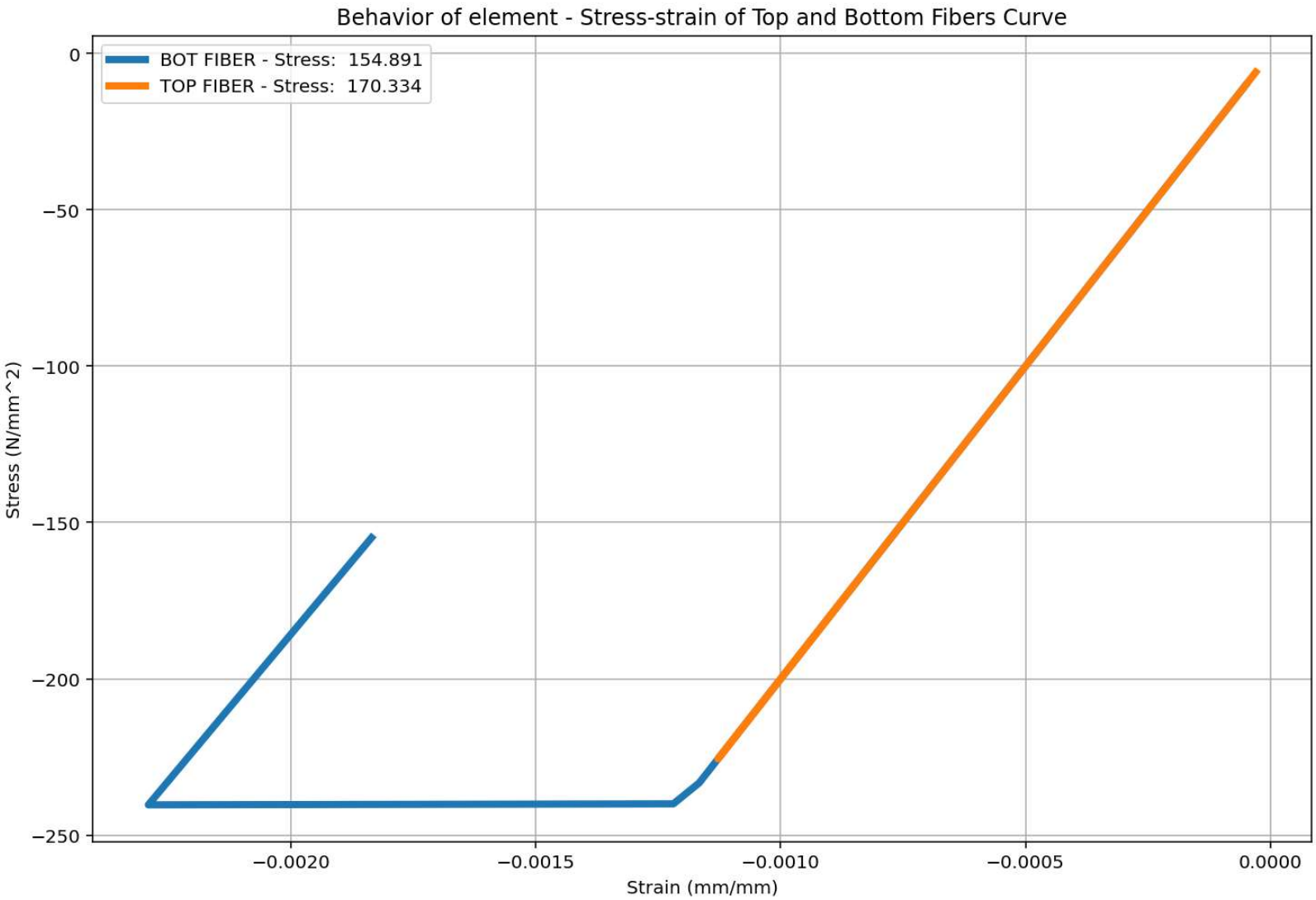
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

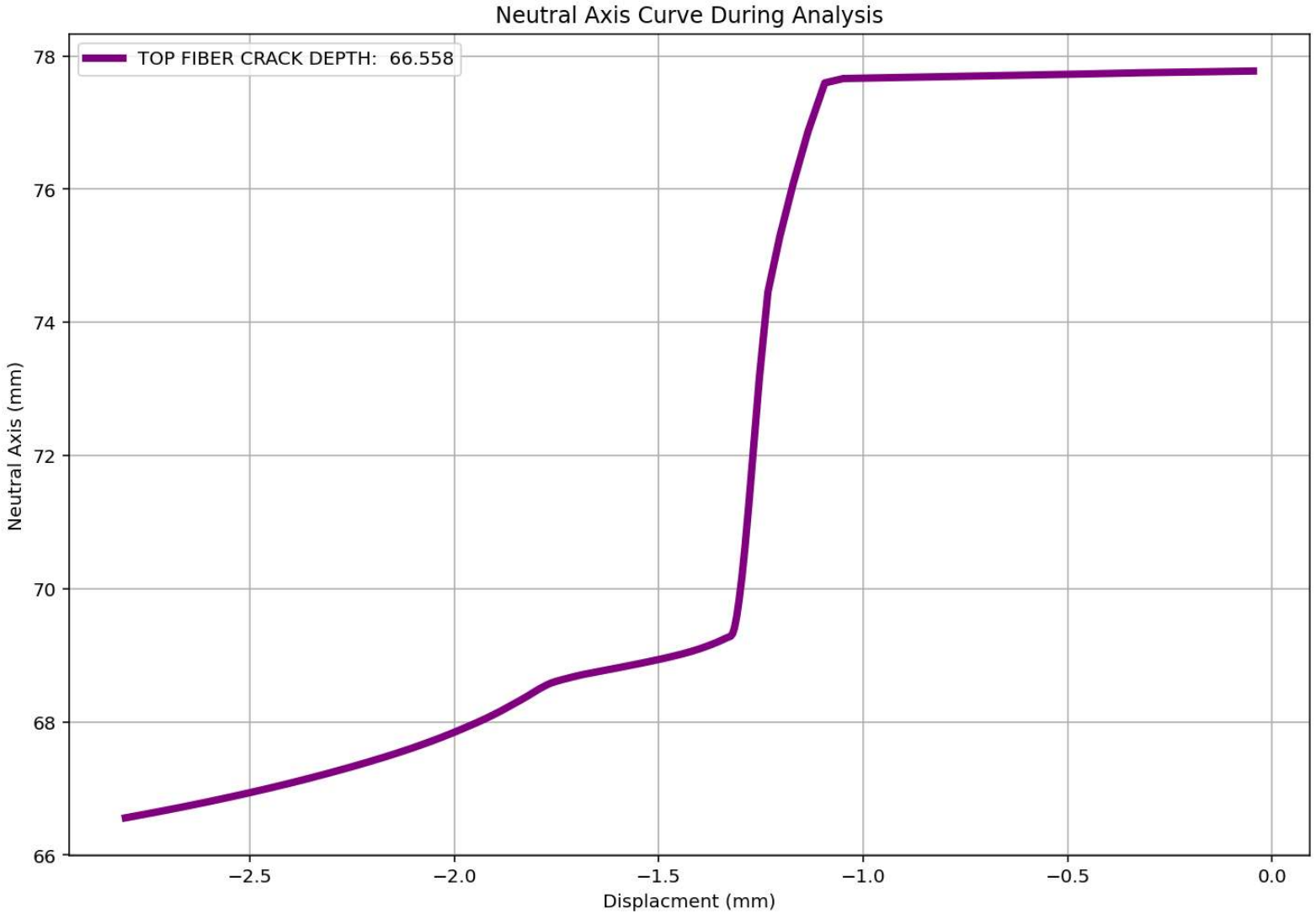
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 780, Col 47 | UTF-8 | CRLF | RW | Mem 49%

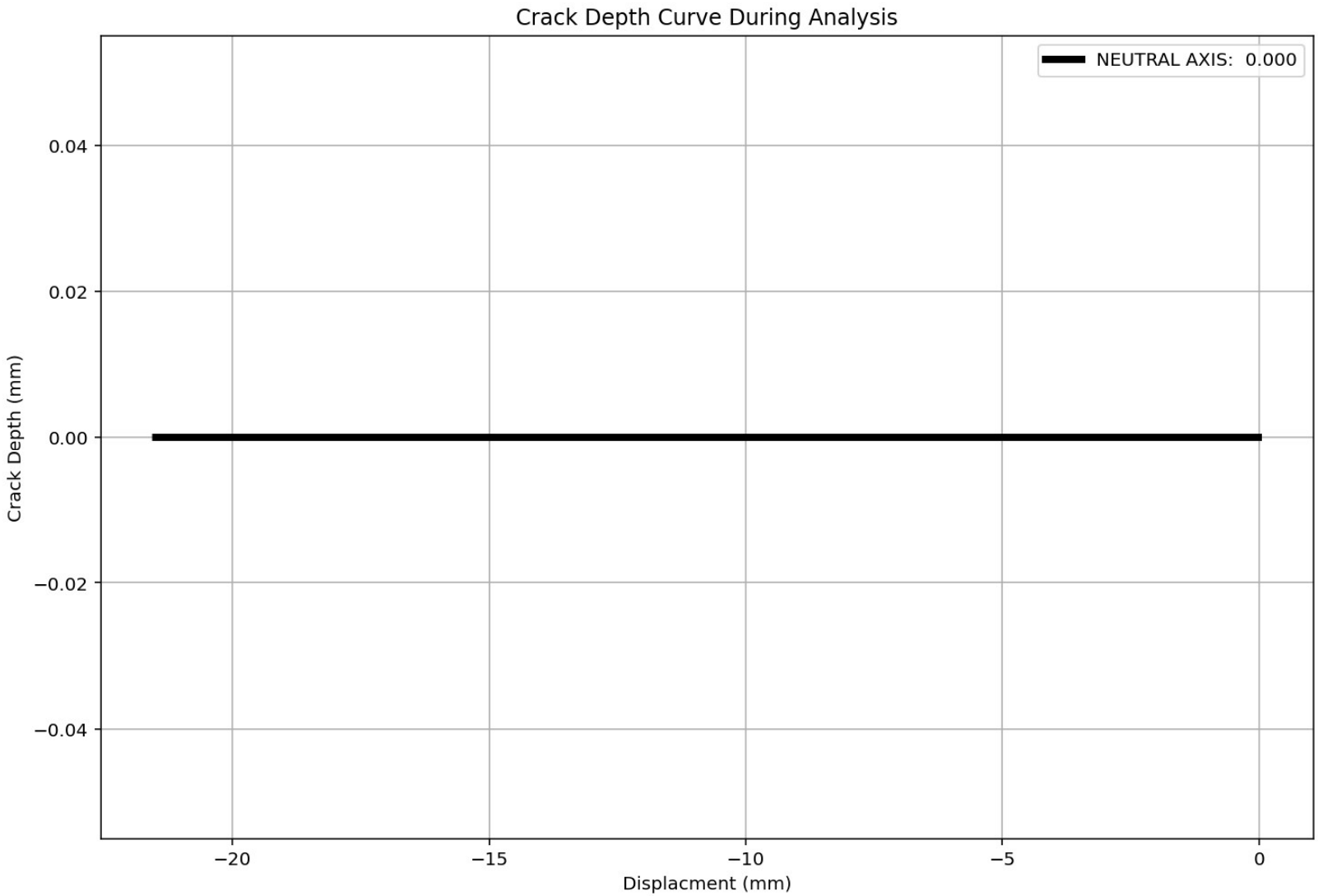

Base Shear Reaction and Displacement of Structure During Pushover Analysis
Base Shear Reaction [N]
20000
15000
10000
5000
0
−70
−60
−50
−40
−30
−20
−10
0
Displacement in X Dir. [mm]

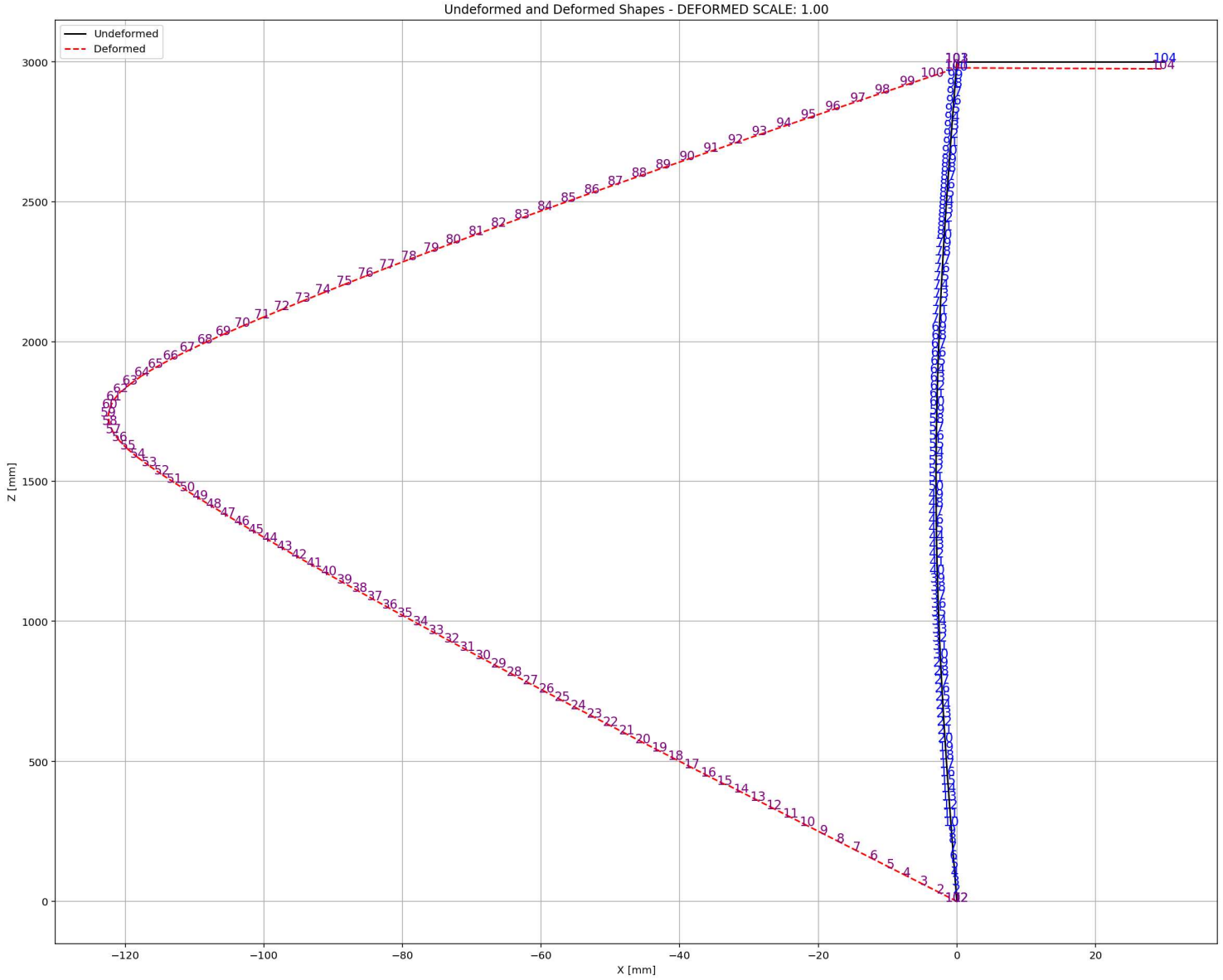

Base Axial Reaction and Displacement of Structure During Pushover Analysis
1e6
2.0
1.5
1.0
0.5
0.0
Base Axial Reaction [N]
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Displacement in Y Dir. [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Displacement [mm]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 154.891
TOP FIBER - Stress: 170.334
0
−50
−100
−150
−200
−250
Stress (N/mm^2)
−0.0020
−0.0015
−0.0010
−0.0005
0.0000
Strain (mm/mm)


Neutral Axis Curve During Analysis
TOP FIBER CRACK DEPTH: 66.558
Neutral Axis (mm)
78
76
74
72
70
68
66
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Displacment (mm)


Crack Depth Curve During Analysis
NEUTRAL AXIS: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−20
−15
−10
−5
0
Displacment (mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
3000
2500
2000
1500
1000
500
0
−120
−100
−80
−60
−40
−20
0
20

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

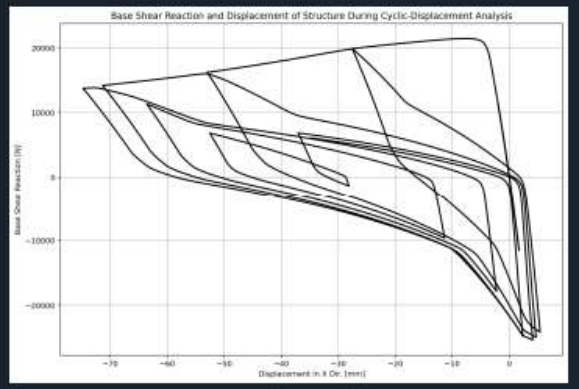
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

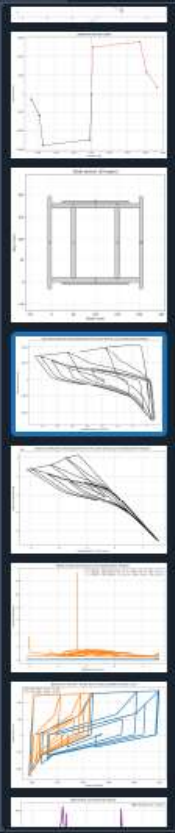
P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py

```
1305  # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1306  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1307  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
1308  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1309  ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1310
1311  DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS, MODE_NUM, SEMI_RIGID)
1312  (reactionX_CP, reactionY_CP, reactionZ_CP, disp_mid_CP,
1313   ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1314   node_displacementsX_CP, node_displacementsY_CP,
1315    dispX_CP, dispY_CP,
1316   PERIOD_MIN_CP, PERIOD_MAX_CP,
1317   STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINt_CP, CURVATURE_CP, NA_CP, CD_CP) = DATA
1318
1319
1320  XDATA = dispX_CP
1321  YDATA = reactionX_CP
1322  XLABEL = 'Displacement in X Dir. [mm]'
1323  YLABEL = 'Base Shear Reaction [N]'
1324  TITLE = 'Base Shear Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1325  COLOR = 'black'
1326  SEMILOGY = False
1327  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1328
1329
1330  XDATA = dispY_CP
1331  YDATA = reactionY_CP
1332  XLABEL = 'Displacement in Y Dir. [mm]'
1333  YLABEL = 'Base Axial Reaction [N]'
1334  TITLE = 'Base Axial Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1335  COLOR = 'black'
1336  SEMILOGY = False
1337  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1338
```

C:\Users\Dell

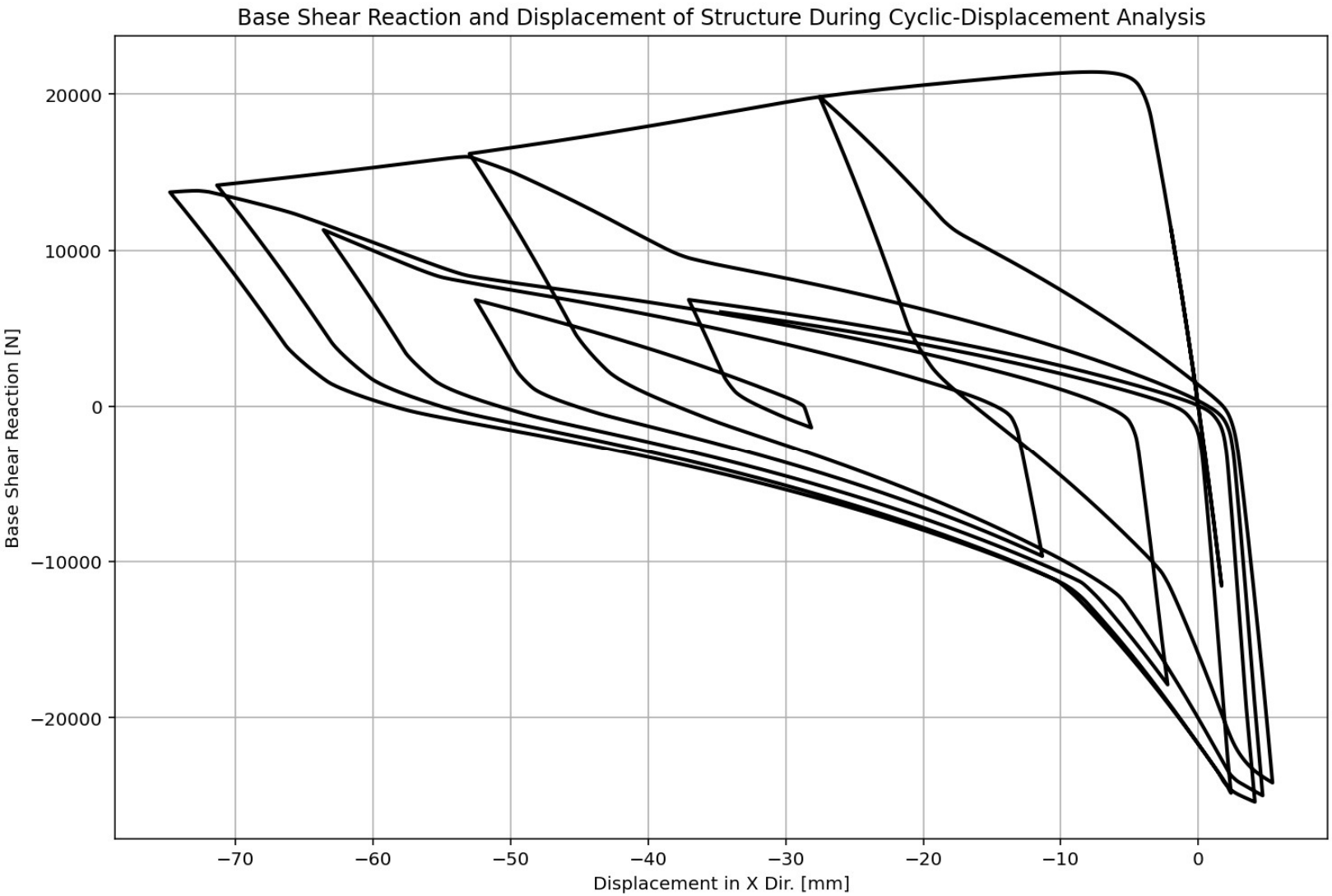
24 %

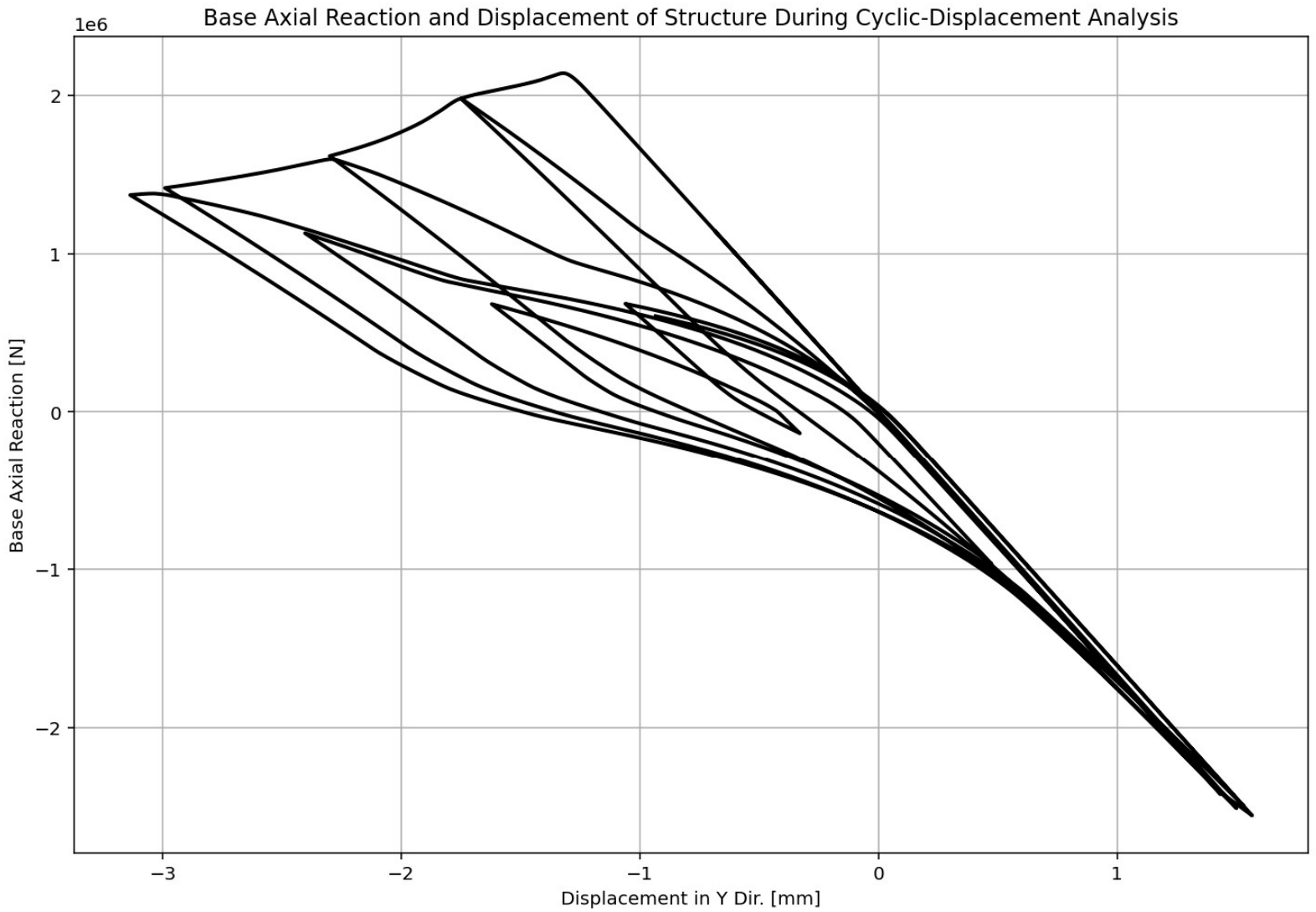


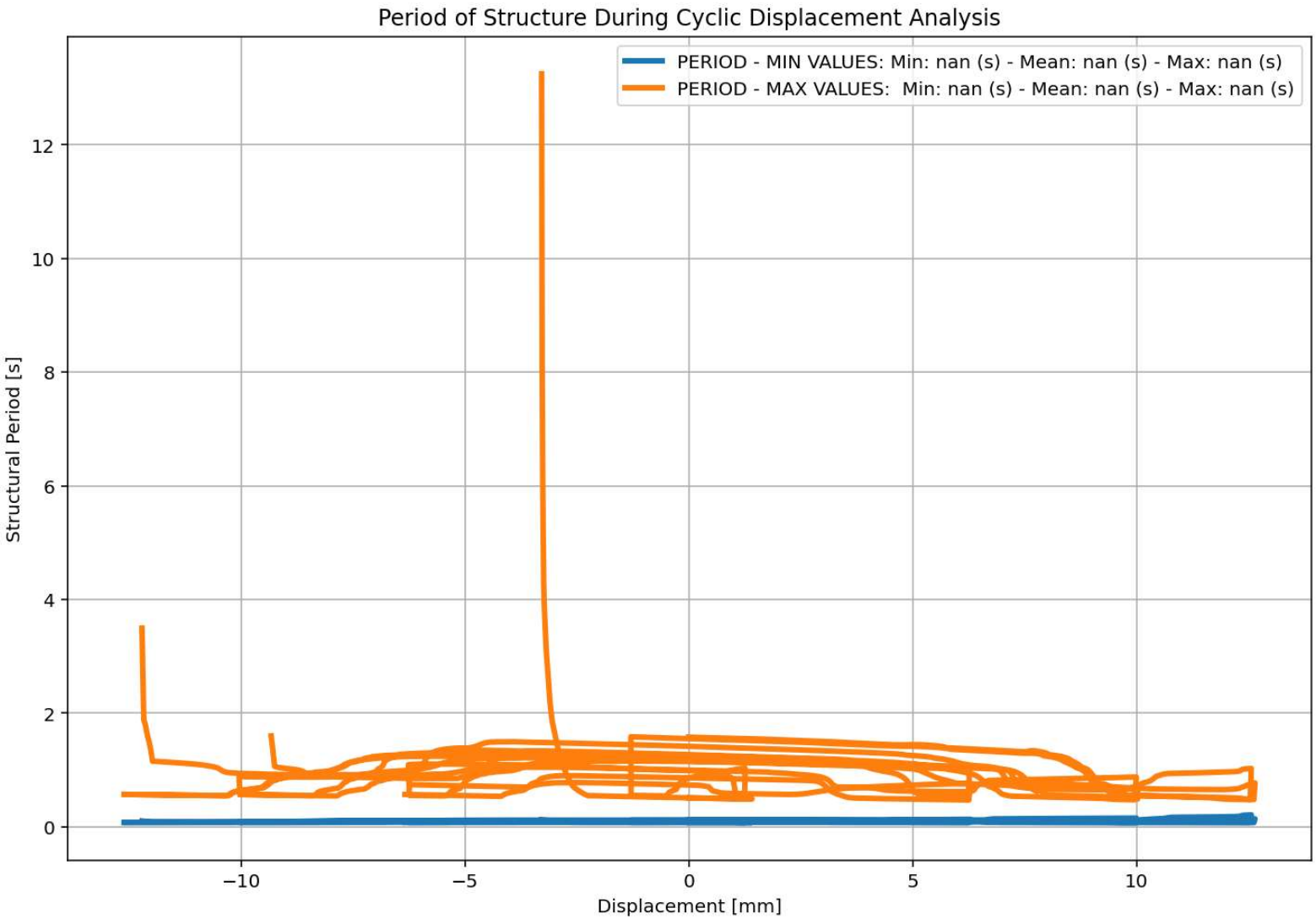


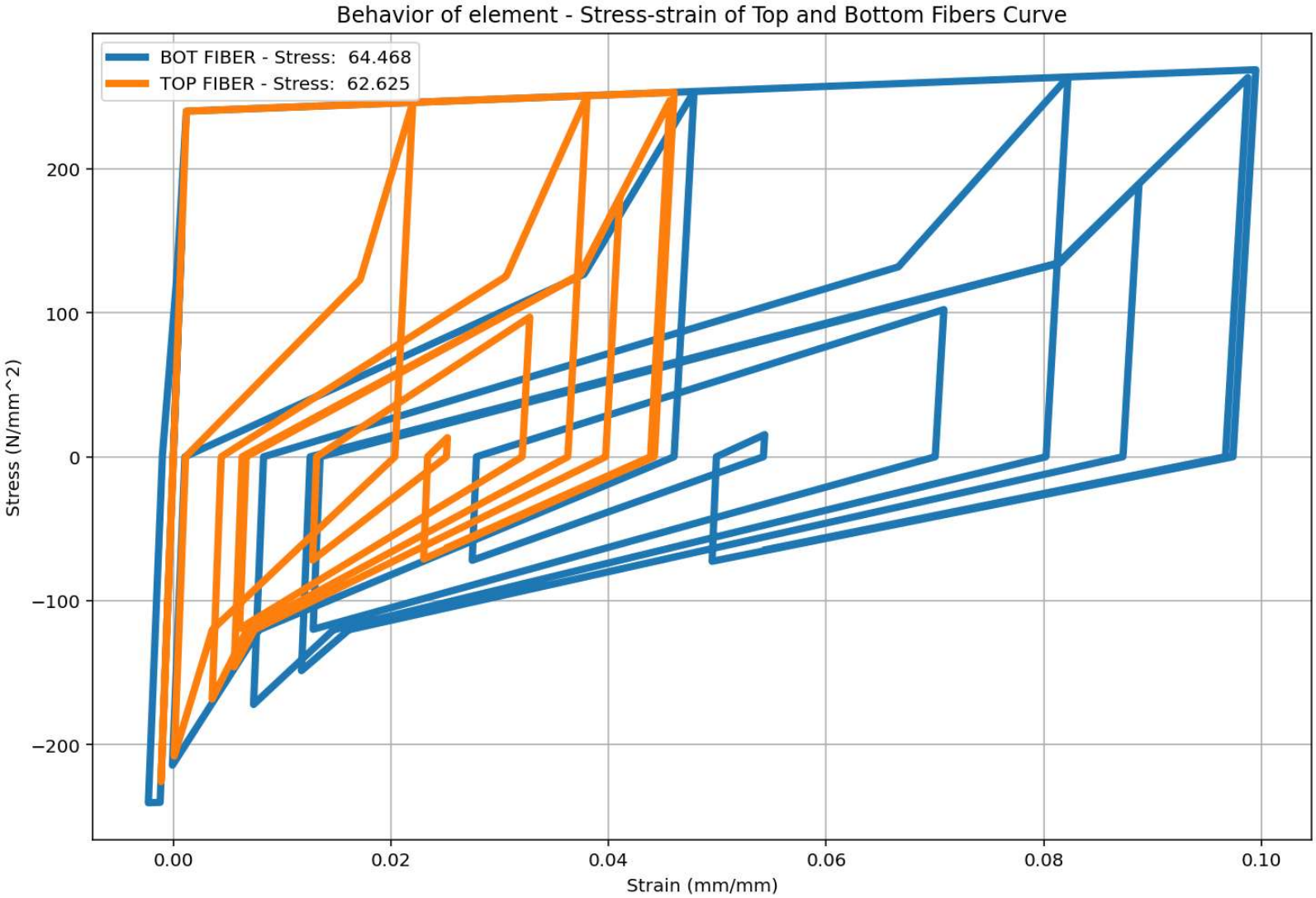
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

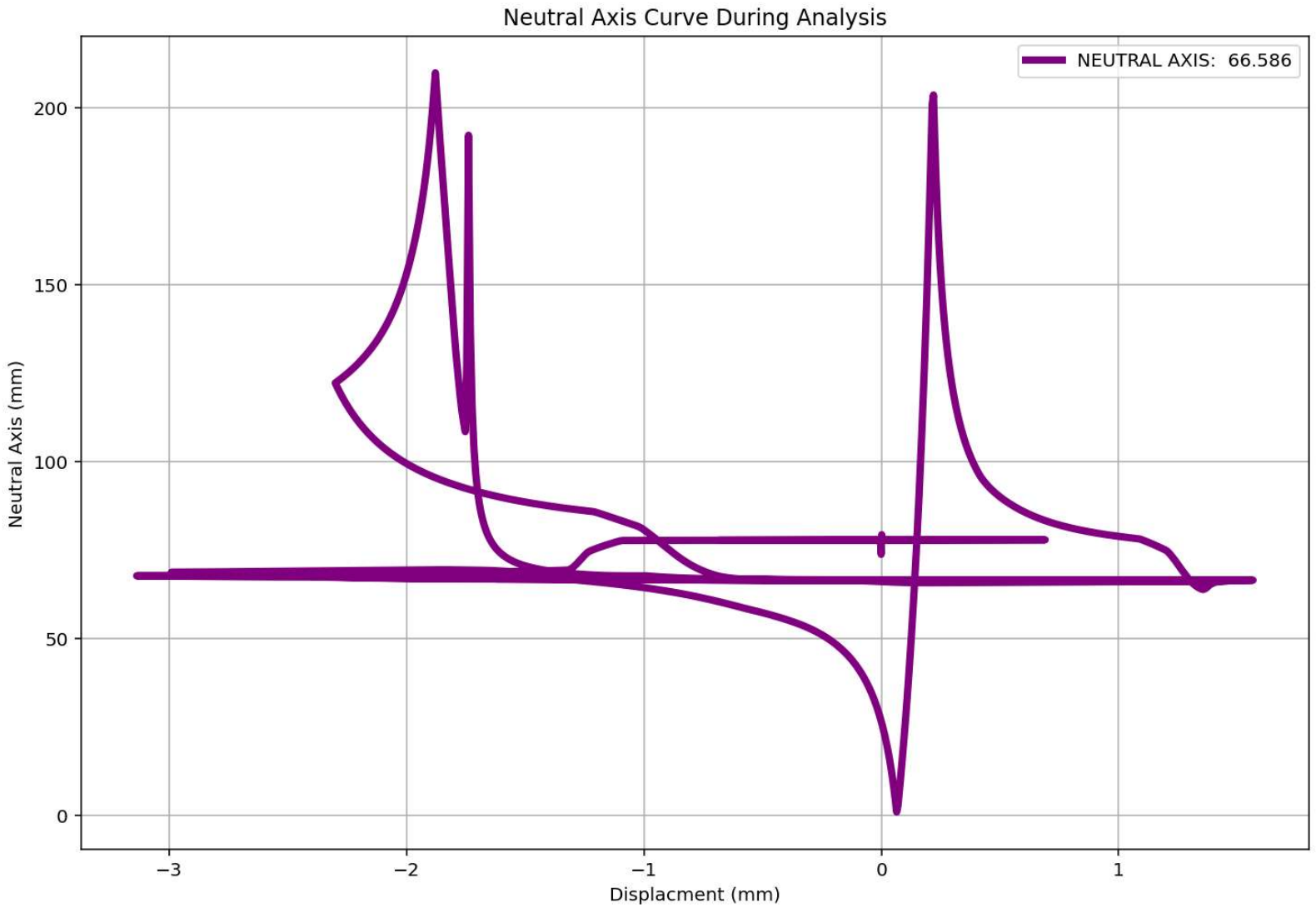
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1, Col 1 | UTF-8 | CRLF | RW | Mem 51%

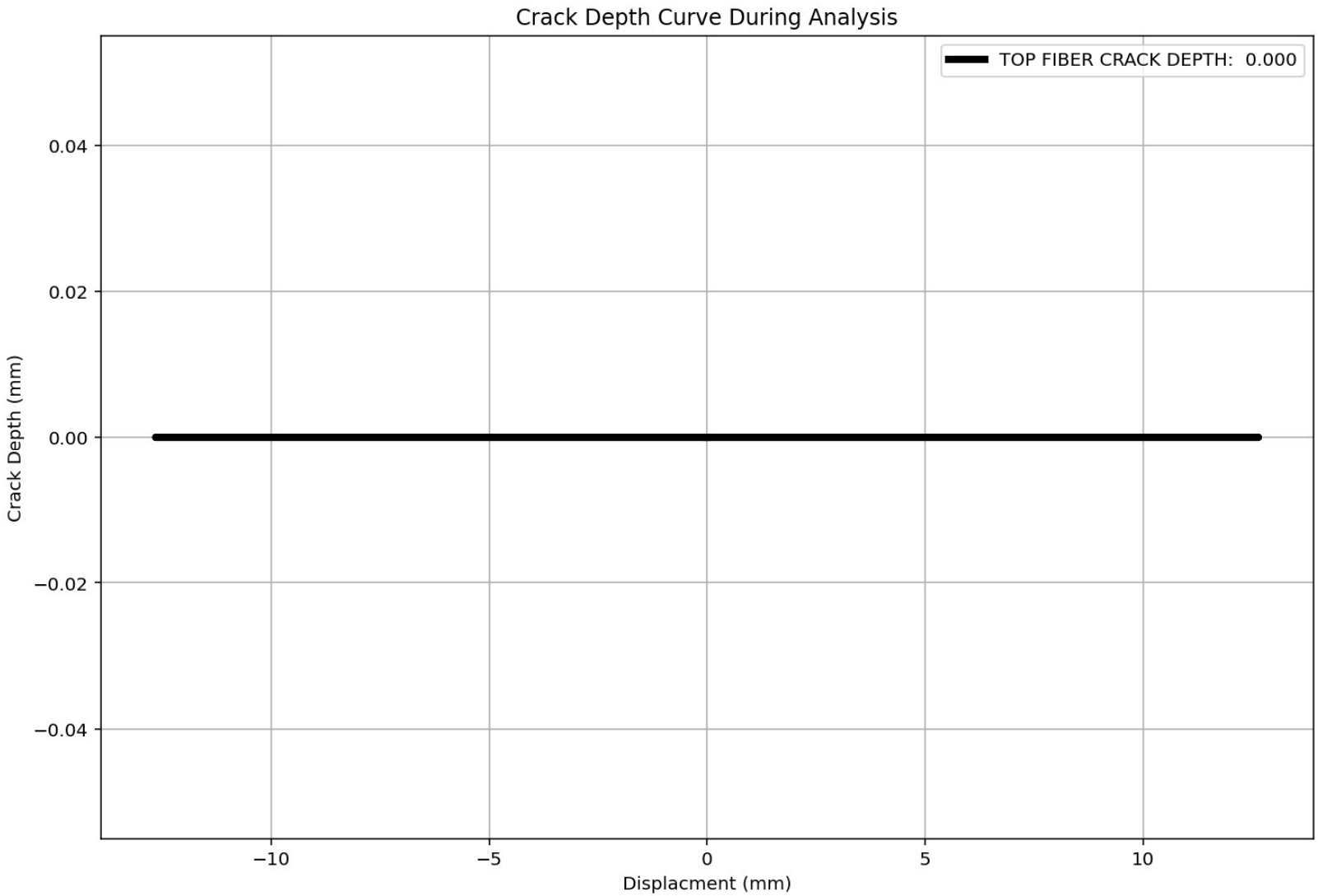

Base Shear Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Shear Reaction [N]
20000
10000
0
−10000
−20000
−70
−60
−50
−40
−30
−20
−10
0
Displacement in X Dir. [mm]

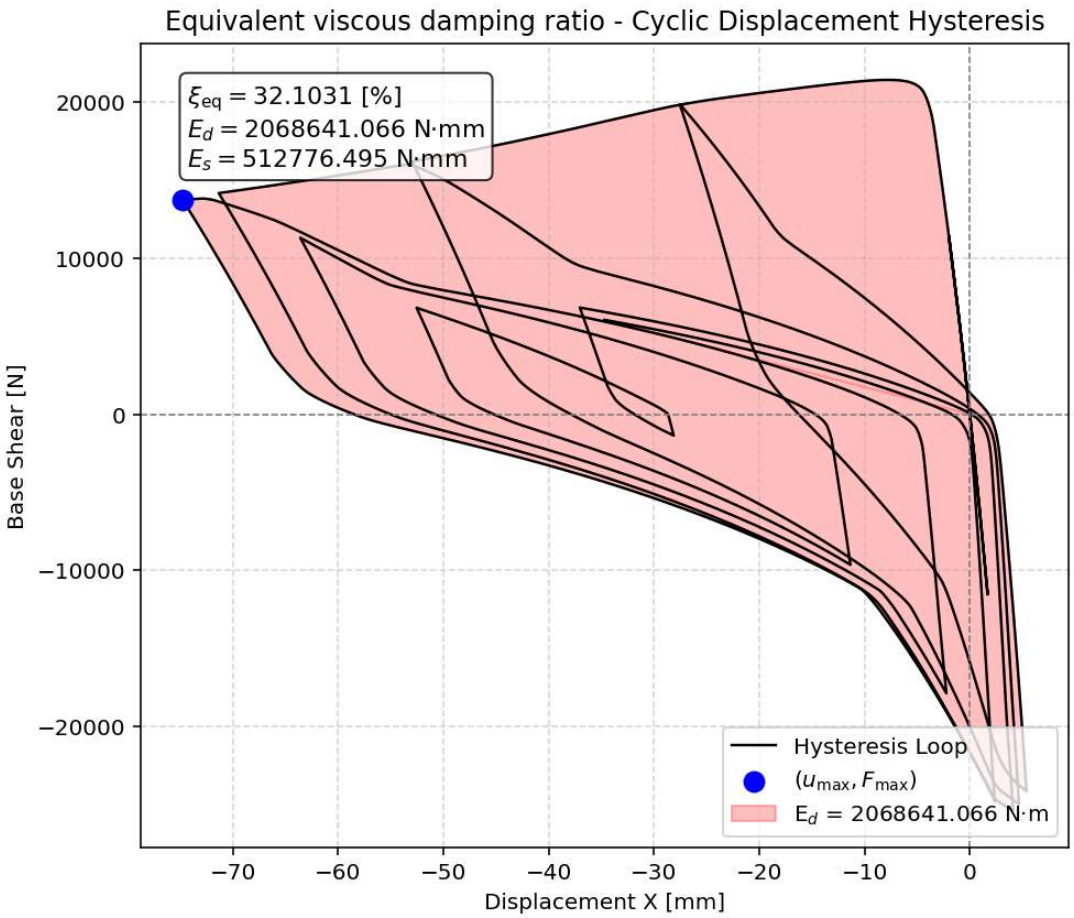

Base Axial Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
2
1
0
−1
−2
Base Axial Reaction [N]
−3
−2
−1
0
1
Displacement in Y Dir. [mm]

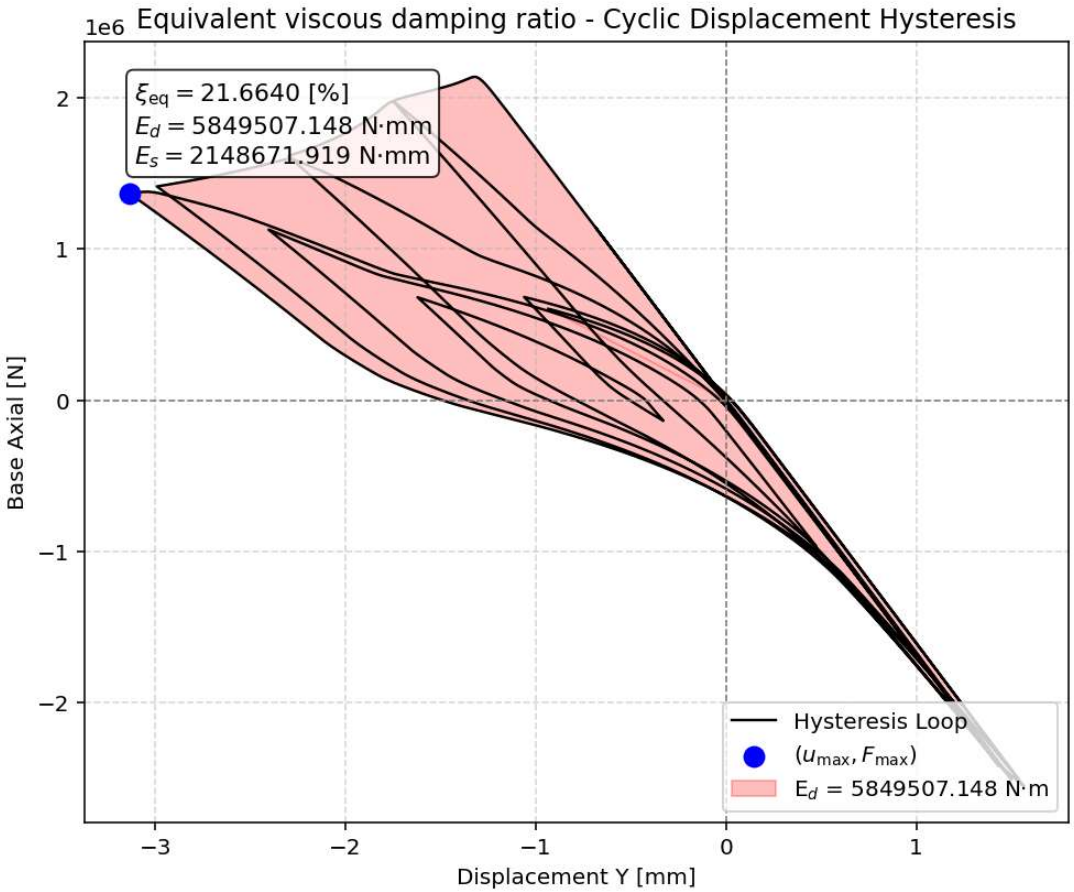

Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
0
2
4
6
8
10
12
−10
−5
0
5
10
Displacement [mm]

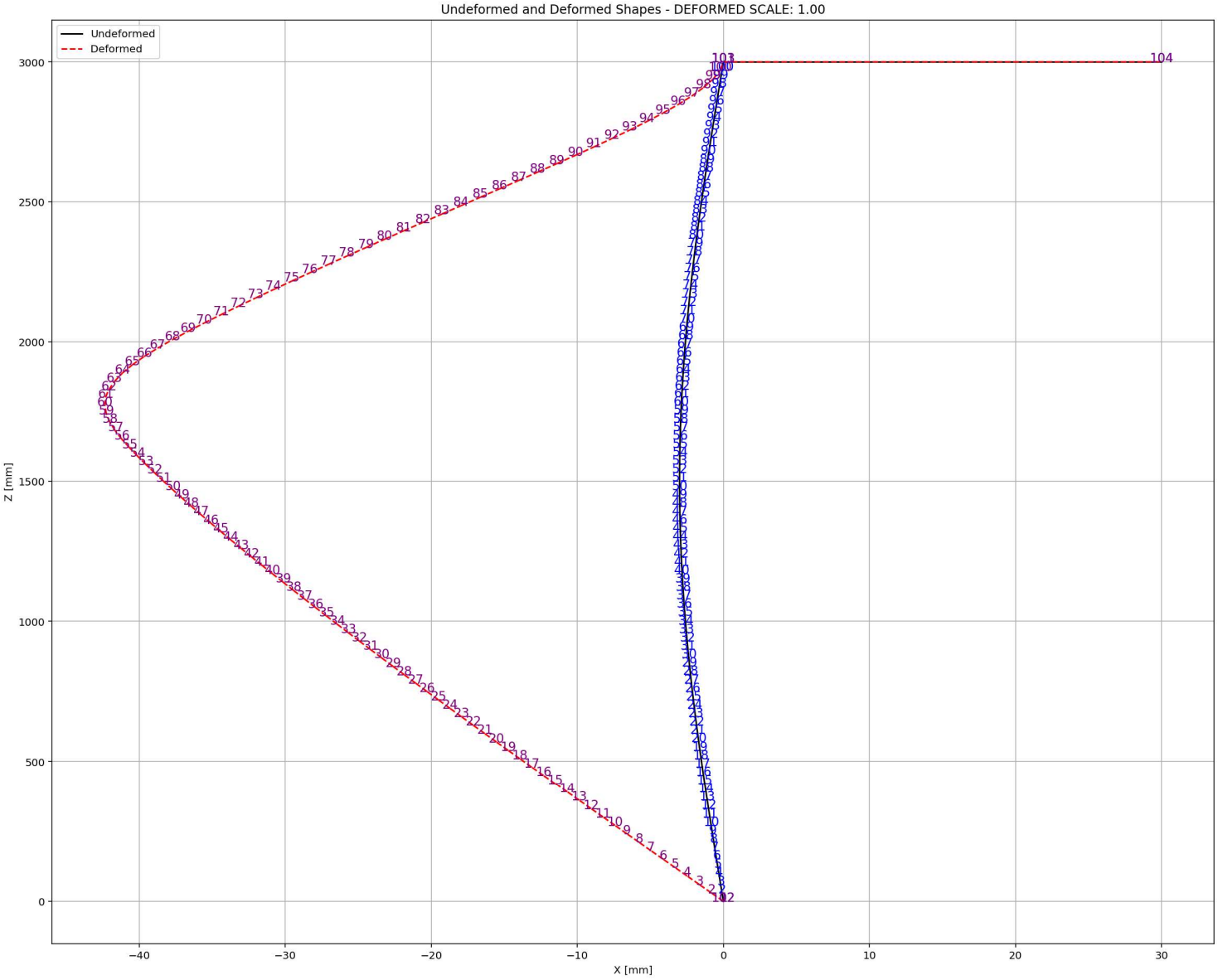

Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 64.468
TOP FIBER - Stress: 62.625
Stress (N/mm^2)
200
100
0
−100
−200
0.00
0.02
0.04
0.06
0.08
0.10
Strain (mm/mm)


Neutral Axis Curve During Analysis
NEUTRAL AXIS: 66.586
Neutral Axis (mm)
200
150
100
50
0
−3
−2
−1
0
1
Displacment (mm)


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−10
−5
0
5
10
Displacment (mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 32.1031$ [%]
$E_d = 2068641.066$ N·mm
$E_s = 512776.495$ N·mm
20000
10000
0
−10000
−20000
Base Shear [N]
−70
−60
−50
−40
−30
−20
−10
0
Displacement X [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2068641.066 N·m


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
1e6
$\xi_{eq} = 21.6640$ [%]
$E_d = 5849507.148$ N·mm
$E_s = 2148671.919$ N·mm
Base Axial [N]
Displacement Y [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 5849507.148 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
3000
2500
2000
1500
1000
500
0
−40
−30
−20
−10
0
10
20
30
104

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

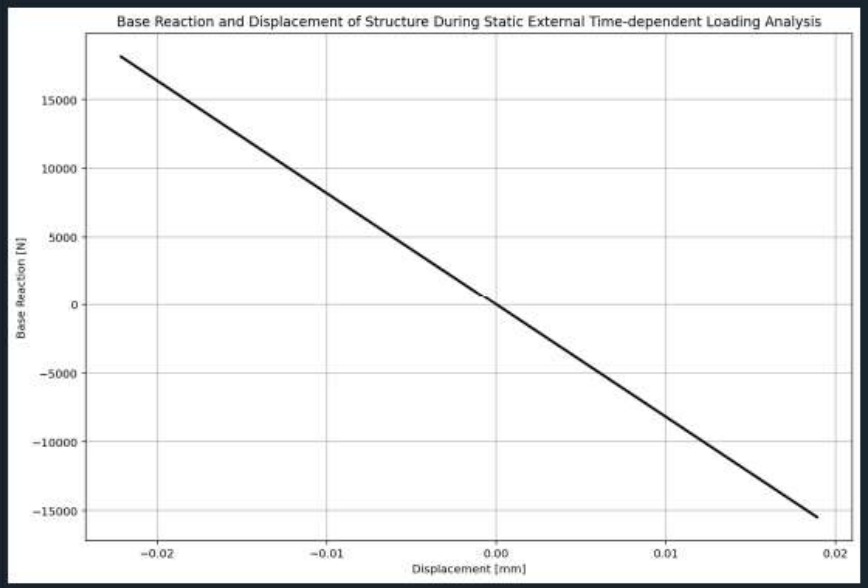
...NCRETE_COLUMN_POST-BUCKLING+\EXAMPLE_02_AXIAL_AND_MOMENT

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

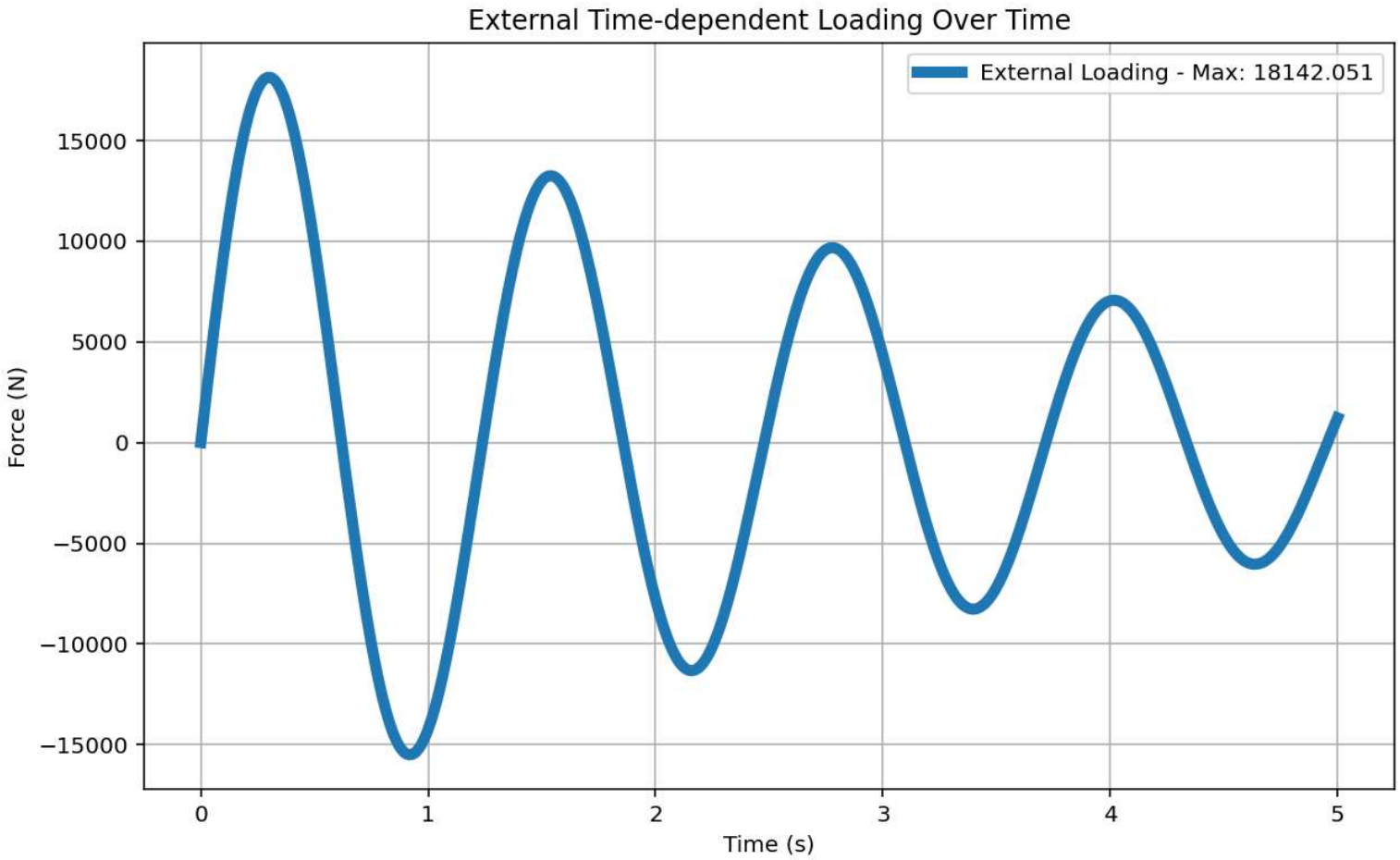
```
1348  # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1349  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1350  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1351  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1352  ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1353
1354  DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1355
1356  (reactionX_ETDLS, reactionY_ETDLS, reactionZ_ETDLS, disp_mid_ETDLS,
1357   ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1358   node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1359    dispX_ETDLS, dispY_ETDLS,
1360   PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1361   STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDL
1362
1363
1364  XDATA = disp_mid_ETDLS
1365  YDATA = reactionY_ETDLS
1366  XLABEL = 'Displacement [mm]'
1367  YLABEL = 'Base Reaction [N]'
1368  TITLE = 'Base Reaction and Displacement of Structure During Static External
1369  COLOR = 'black'
1370  SEMILOGY = False
1371  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1372
1373  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1374  plt.figure(0, figsize=(12, 8))
1375  plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1376  plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1377  plt.title('Period of Structure During Static External Time-dependent Loadin
1378  plt.ylabel('Structural Period [s]')
1379  plt.xlabel('Displacement [mm]')
1380  #plt.semilogy()
1381  plt.grid()
```

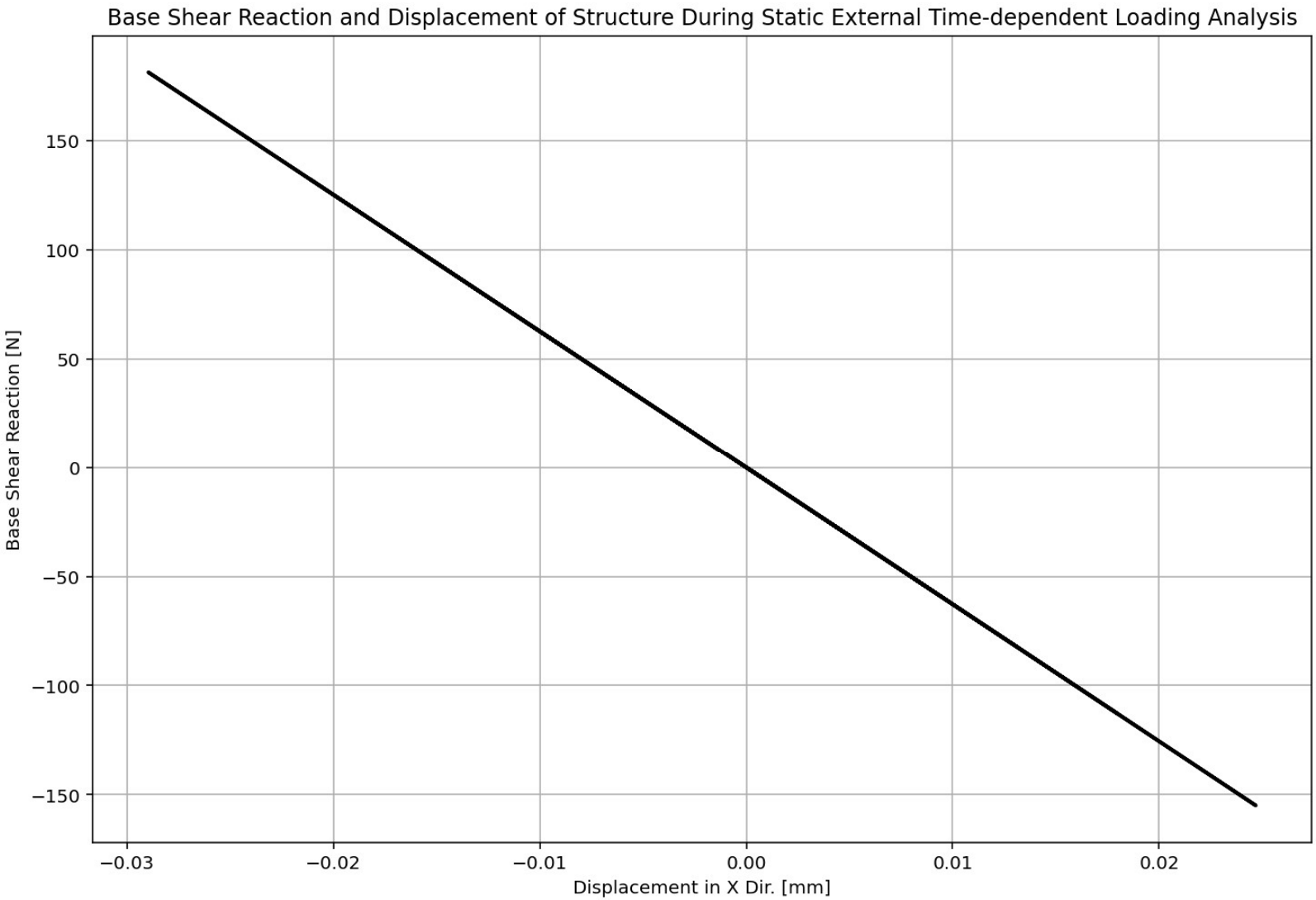
37 %

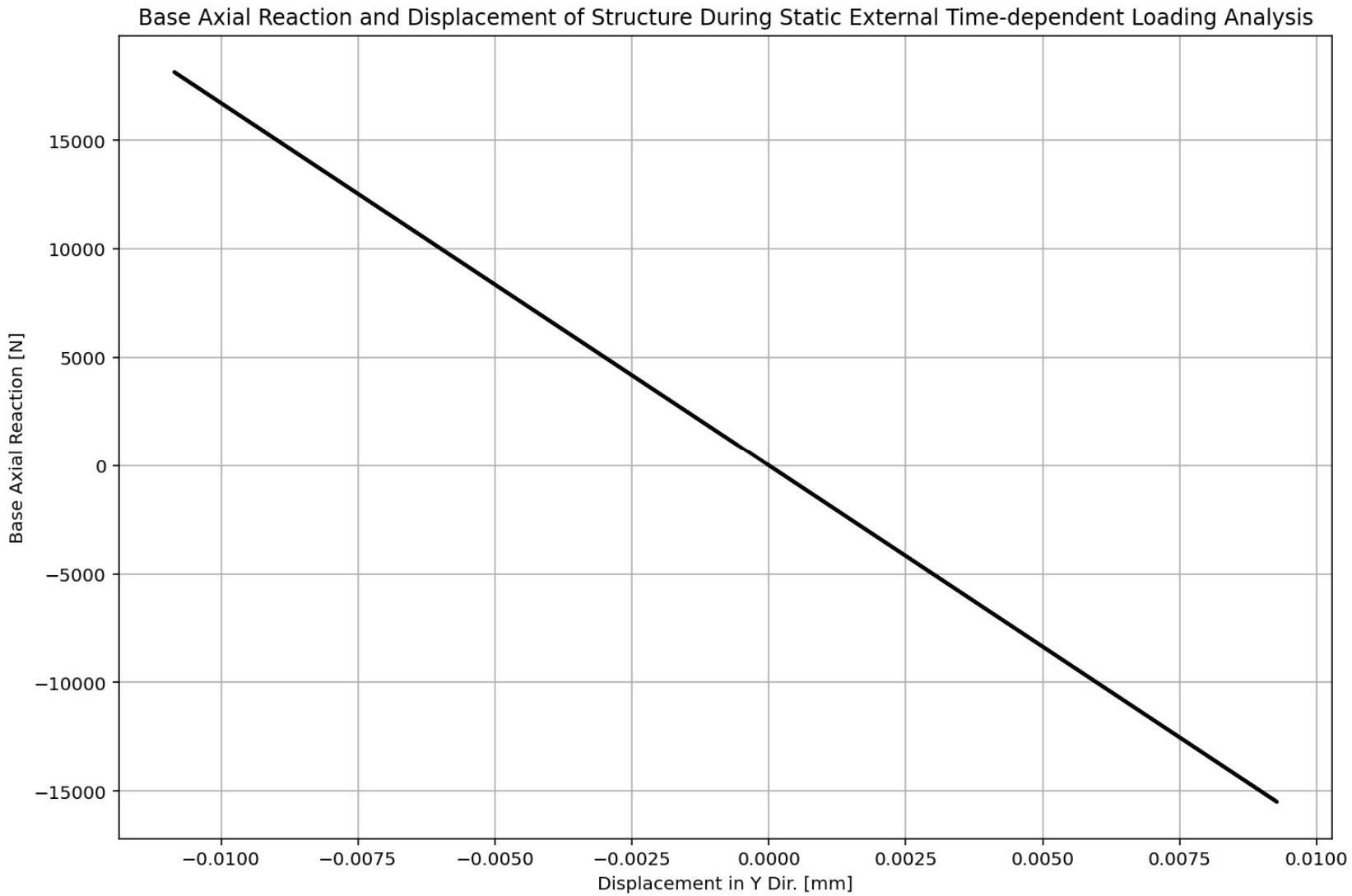


IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

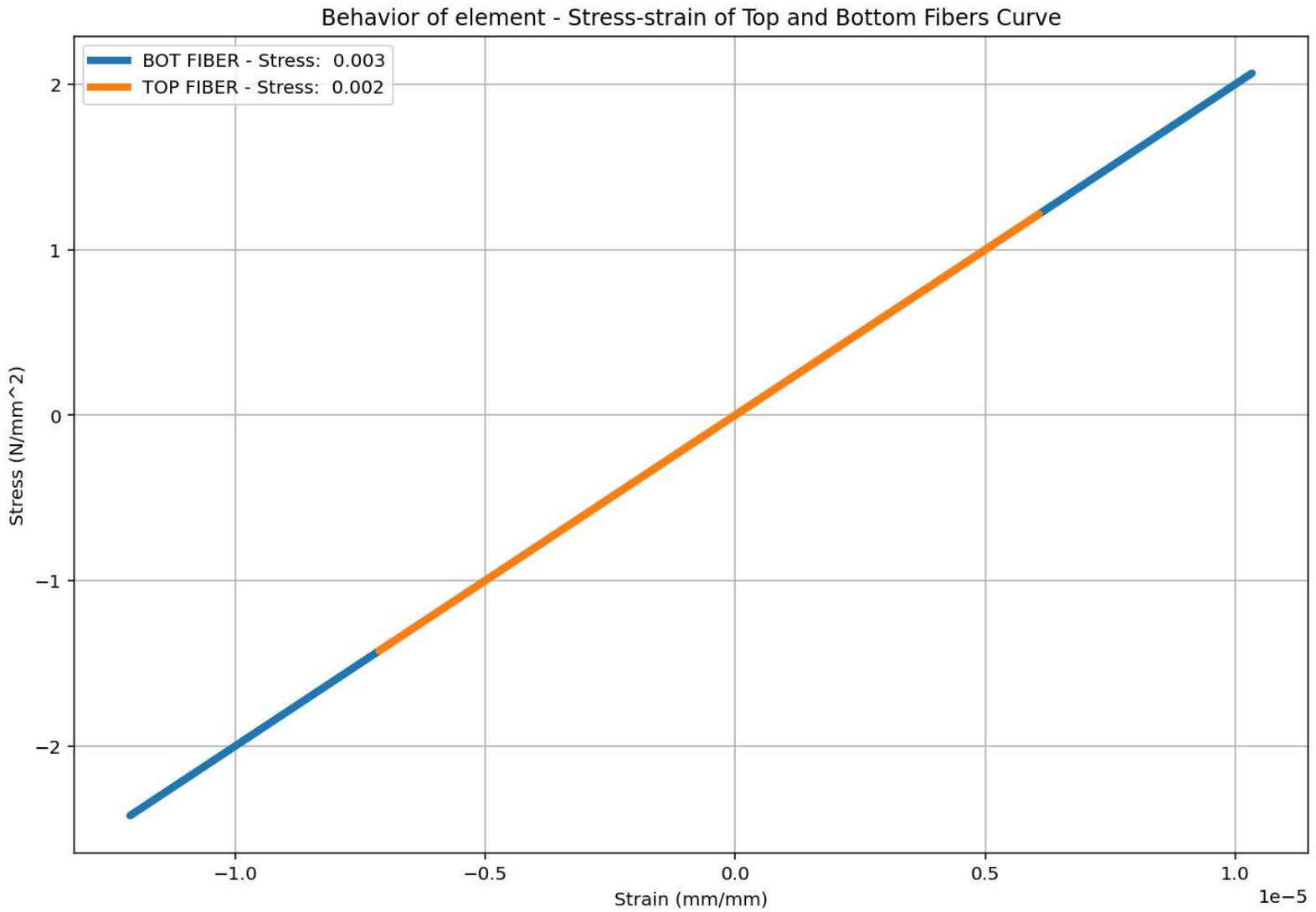
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 780, Col 47 UTF-8 CRLF RW Mem 51%

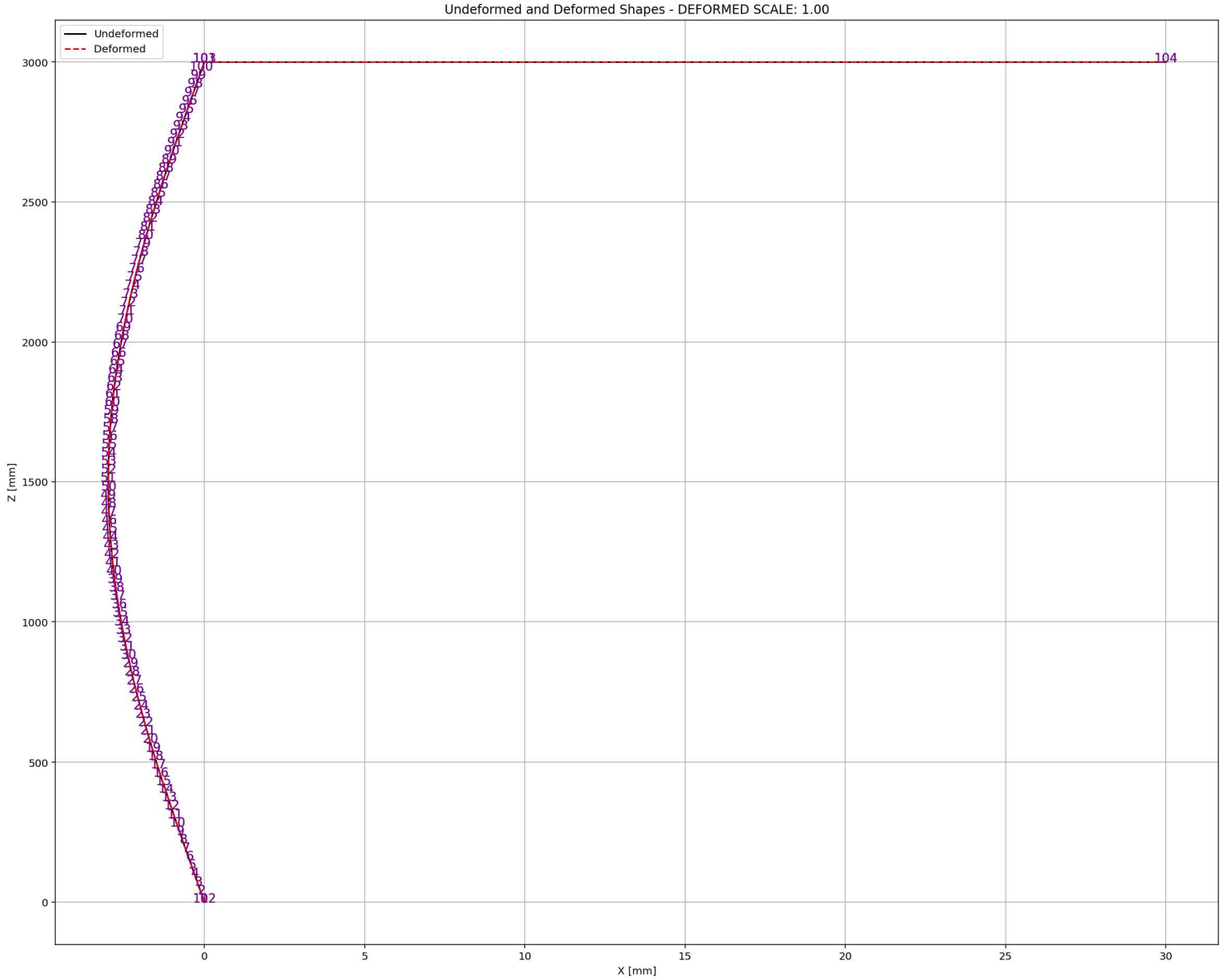

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Shear Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Shear Reaction [N]
150
100
50
0
−50
−100
−150
−0.03
−0.02
−0.01
0.00
0.01
0.02
Displacement in X Dir. [mm]


Base Axial Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Axial Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
0.0100
Displacement in Y Dir. [mm]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.5
0.4
0.3
0.2
0.1
−0.02
−0.01
0.00
0.01
0.02
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.076 (s) - Mean: 0.076 (s) - Max: 0.076 (s)
PERIOD - MAX VALUES: Min: 0.510 (s) - Mean: 0.511 (s) - Max: 0.511 (s)


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.003
TOP FIBER - Stress: 0.002
Stress (N/mm^2)
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
Strain (mm/mm)
1e−5


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
5
10
15
20
25
30
103
104
102

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py

```
# EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'

DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS, MODE_NUM, SEMI_RIGID)

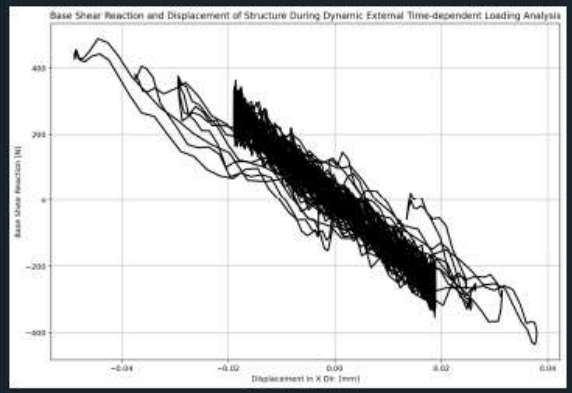
(time_ETDLD, reactionX_ETDLD, reactionY_ETDLD, reactionZ_ETDLD, disp_mid_ETDLD,
ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD_X, damping_ratio_ETDLD_Y,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD, CD_ETDLD, CD_ETDLD) = DATA


XDATA = dispX_ETDLD
YDATA = reactionX_ETDLD
XLABEL = 'Displacement in X Dir. [mm]'
YLABEL = 'Base Shear Reaction [N]'
TITLE = 'Base Shear Reaction and Displacement of Structure During Dynamic External Time-dependent Load
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)


XDATA = dispY_ETDLD
YDATA = reactionY_ETDLD
XLABEL = 'Displacement in Y Dir. [mm]'
YLABEL = 'Base Axial Reaction [N]'
TITLE = 'Base Axial Reaction and Displacement of Structure During Dynamic External Time-dependent Load
```

C:\Users\Dell

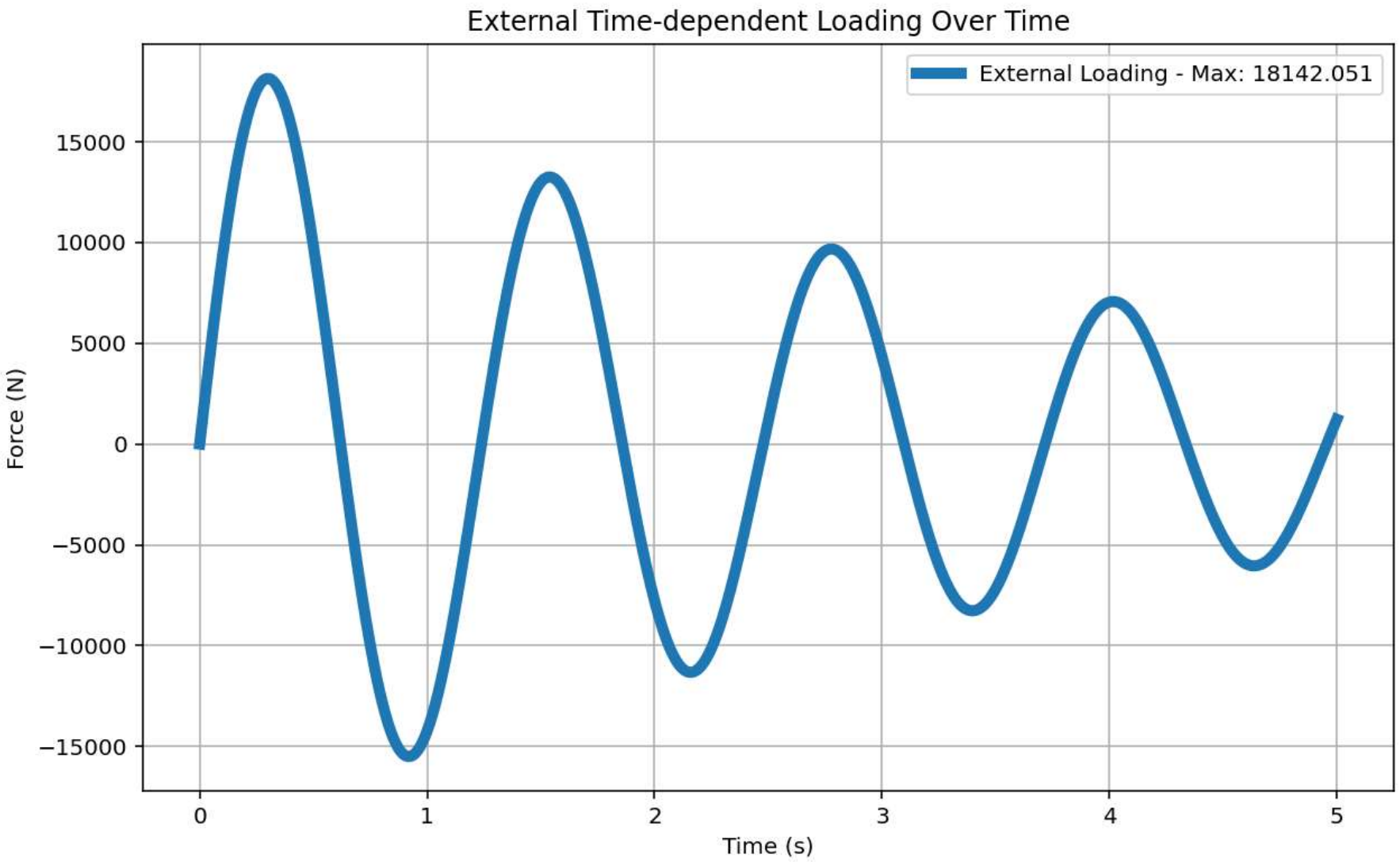
24 %

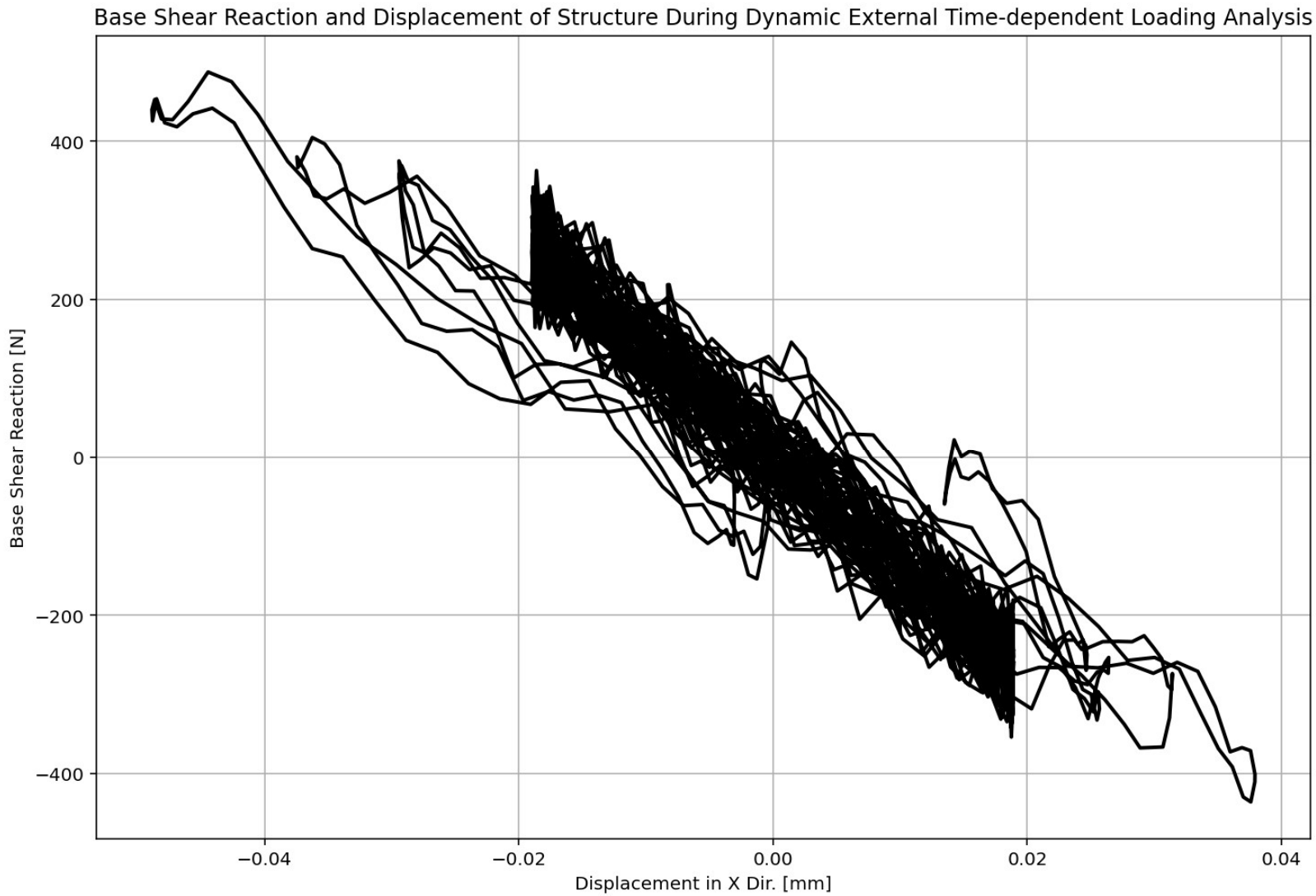


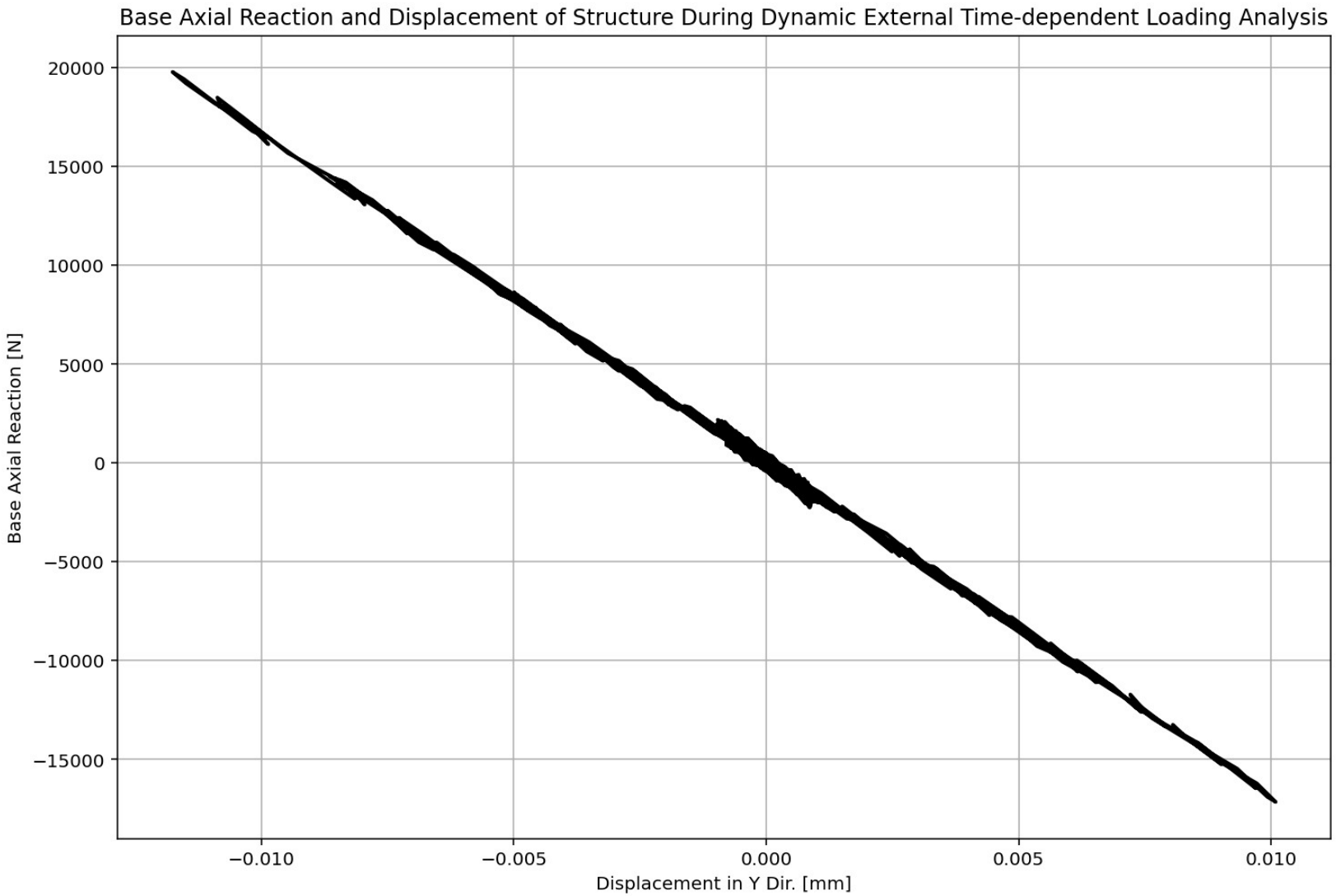


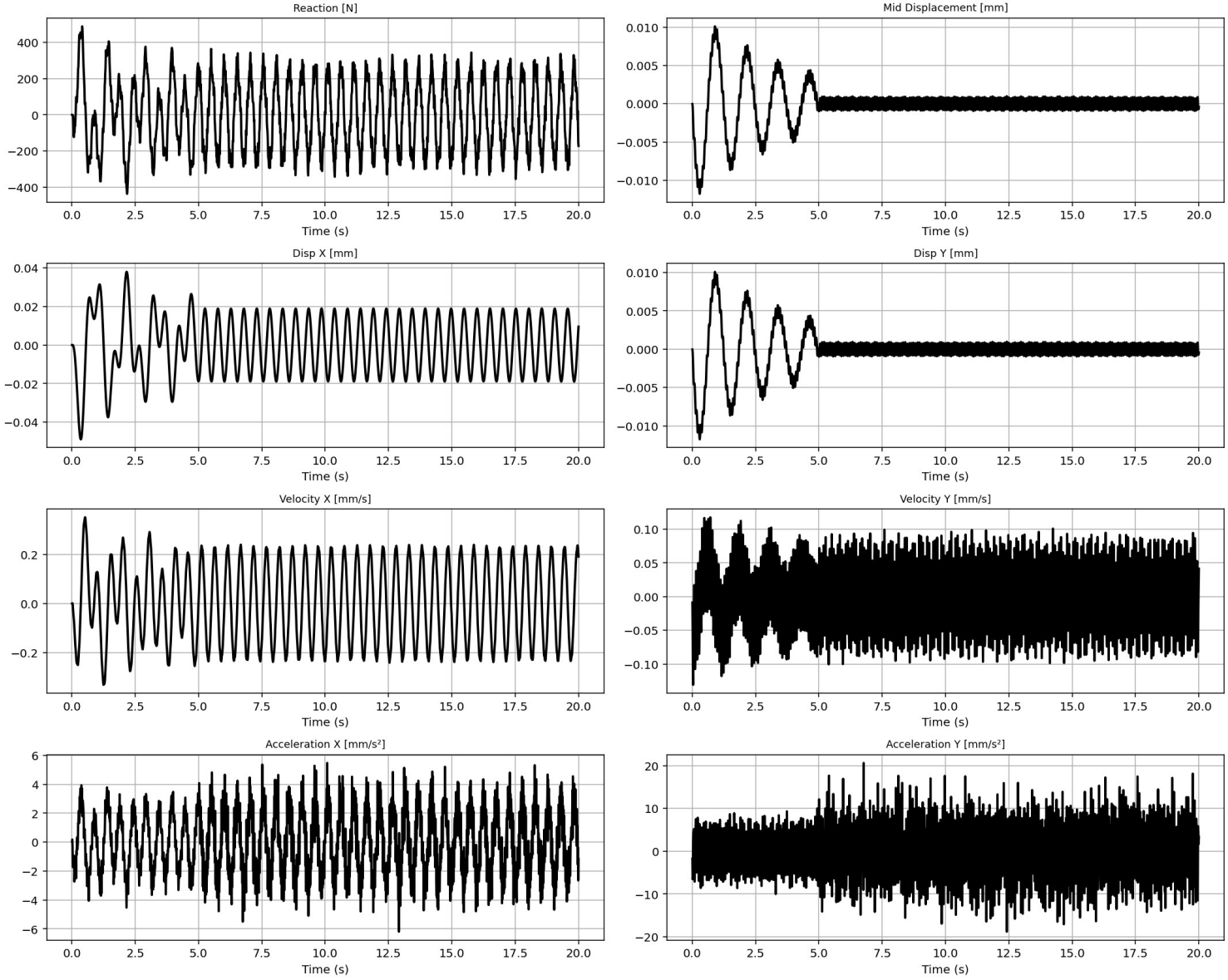
IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 50%

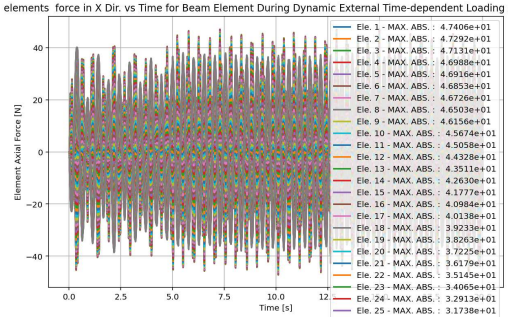

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Shear Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Shear Reaction [N]
400
200
0
−200
−400
−0.04
−0.02
0.00
0.02
0.04
Displacement in X Dir. [mm]

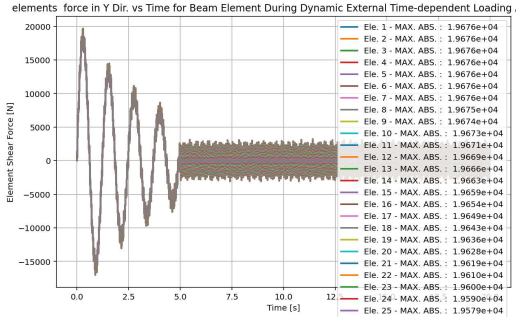

Base Axial Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Axial Reaction [N]
20000
15000
10000
5000
0
−5000
−10000
−15000
−0.010
−0.005
0.000
0.005
0.010
Displacement in Y Dir. [mm]

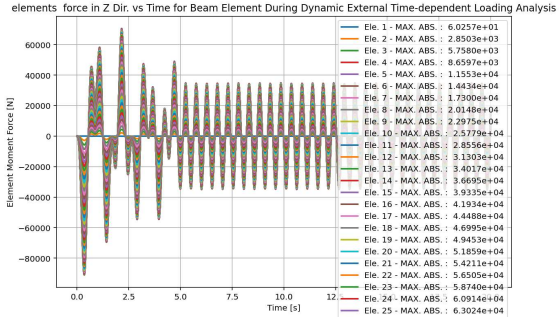

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

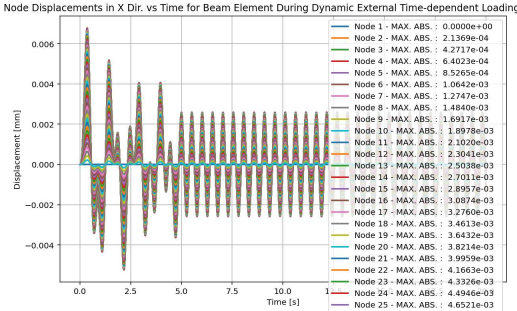


Ele. 1 - MAX. ABS. : 4.7406e+01
Ele. 2 - MAX. ABS. : 4.7292e+01
Ele. 3 - MAX. ABS. : 4.7131e+01
Ele. 4 - MAX. ABS. : 4.6988e+01
Ele. 5 - MAX. ABS. : 4.6916e+01
Ele. 6 - MAX. ABS. : 4.6853e+01
Ele. 7 - MAX. ABS. : 4.6726e+01
Ele. 8 - MAX. ABS. : 4.6503e+01
Ele. 9 - MAX. ABS. : 4.6156e+01
Ele. 10 - MAX. ABS. : 4.5674e+01
Ele. 11 - MAX. ABS. : 4.5058e+01
Ele. 12 - MAX. ABS. : 4.4328e+01
Ele. 13 - MAX. ABS. : 4.3511e+01
Ele. 14 - MAX. ABS. : 4.2630e+01
Ele. 15 - MAX. ABS. : 4.1777e+01
Ele. 16 - MAX. ABS. : 4.0984e+01
Ele. 17 - MAX. ABS. : 4.0138e+01
Ele. 18 - MAX. ABS. : 3.9233e+01
Ele. 19 - MAX. ABS. : 3.8263e+01
Ele. 20 - MAX. ABS. : 3.7225e+01
Ele. 21 - MAX. ABS. : 3.6179e+01
Ele. 22 - MAX. ABS. : 3.5145e+01
Ele. 23 - MAX. ABS. : 3.4065e+01
Ele. 24 - MAX. ABS. : 3.2913e+01
Ele. 25 - MAX. ABS. : 3.1738e+01
Ele. 26 - MAX. ABS. : 3.0550e+01
Ele. 27 - MAX. ABS. : 2.9324e+01
Ele. 28 - MAX. ABS. : 2.8259e+01
Ele. 29 - MAX. ABS. : 2.7200e+01
Ele. 30 - MAX. ABS. : 2.6100e+01
Ele. 31 - MAX. ABS. : 2.4939e+01
Ele. 32 - MAX. ABS. : 2.3719e+01
Ele. 33 - MAX. ABS. : 2.2447e+01
Ele. 34 - MAX. ABS. : 2.1186e+01
Ele. 35 - MAX. ABS. : 1.9884e+01
Ele. 36 - MAX. ABS. : 1.8611e+01
Ele. 37 - MAX. ABS. : 1.7397e+01
Ele. 38 - MAX. ABS. : 1.6188e+01
Ele. 39 - MAX. ABS. : 1.5002e+01
Ele. 40 - MAX. ABS. : 1.3786e+01
Ele. 41 - MAX. ABS. : 1.2534e+01
Ele. 42 - MAX. ABS. : 1.1437e+01
Ele. 43 - MAX. ABS. : 1.0272e+01
Ele. 44 - MAX. ABS. : 9.0796e+00
Ele. 45 - MAX. ABS. : 8.0810e+00
Ele. 46 - MAX. ABS. : 7.0523e+00
Ele. 47 - MAX. ABS. : 6.0895e+00
Ele. 48 - MAX. ABS. : 5.1423e+00
Ele. 49 - MAX. ABS. : 4.1700e+00
Ele. 50 - MAX. ABS. : 3.1307e+00
Ele. 51 - MAX. ABS. : 2.9596e+00
Ele. 52 - MAX. ABS. : 4.3046e+00
Ele. 53 - MAX. ABS. : 5.6393e+00
Ele. 54 - MAX. ABS. : 6.9255e+00
Ele. 55 - MAX. ABS. : 8.1331e+00
Ele. 56 - MAX. ABS. : 9.2504e+00
Ele. 57 - MAX. ABS. : 1.0283e+01
Ele. 58 - MAX. ABS. : 1.1447e+01
Ele. 59 - MAX. ABS. : 1.2628e+01
Ele. 60 - MAX. ABS. : 1.3787e+01
Ele. 61 - MAX. ABS. : 1.4907e+01
Ele. 62 - MAX. ABS. : 1.5957e+01
Ele. 63 - MAX. ABS. : 1.6922e+01
Ele. 64 - MAX. ABS. : 1.7966e+01
Ele. 65 - MAX. ABS. : 1.9028e+01
Ele. 66 - MAX. ABS. : 2.0114e+01
Ele. 67 - MAX. ABS. : 2.1228e+01
Ele. 68 - MAX. ABS. : 2.2290e+01
Ele. 69 - MAX. ABS. : 2.3293e+01
Ele. 70 - MAX. ABS. : 2.4236e+01
Ele. 71 - MAX. ABS. : 2.5087e+01
Ele. 72 - MAX. ABS. : 2.5816e+01
Ele. 73 - MAX. ABS. : 2.6675e+01
Ele. 74 - MAX. ABS. : 2.7651e+01
Ele. 75 - MAX. ABS. : 2.8593e+01
Ele. 76 - MAX. ABS. : 2.9498e+01
Ele. 77 - MAX. ABS. : 3.0367e+01
Ele. 78 - MAX. ABS. : 3.1199e+01
Ele. 79 - MAX. ABS. : 3.1993e+01
Ele. 80 - MAX. ABS. : 3.2749e+01
Ele. 81 - MAX. ABS. : 3.3467e+01
Ele. 82 - MAX. ABS. : 3.4146e+01
Ele. 83 - MAX. ABS. : 3.4786e+01
Ele. 84 - MAX. ABS. : 3.5385e+01
Ele. 85 - MAX. ABS. : 3.5942e+01
Ele. 86 - MAX. ABS. : 3.6458e+01
Ele. 87 - MAX. ABS. : 3.6952e+01
Ele. 88 - MAX. ABS. : 3.7426e+01
Ele. 89 - MAX. ABS. : 3.7857e+01
Ele. 90 - MAX. ABS. : 3.8244e+01
Ele. 91 - MAX. ABS. : 3.8585e+01
Ele. 92 - MAX. ABS. : 3.8881e+01
Ele. 93 - MAX. ABS. : 3.9138e+01
Ele. 94 - MAX. ABS. : 3.9448e+01
Ele. 95 - MAX. ABS. : 3.9715e+01
Ele. 96 - MAX. ABS. : 3.9926e+01
Ele. 97 - MAX. ABS. : 4.0082e+01
Ele. 98 - MAX. ABS. : 4.0193e+01


elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Shear Force [N]
Time [s]
20000
15000
10000
5000
0
−5000
−10000
−15000
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.9676e+04
Ele. 2 - MAX. ABS. : 1.9676e+04
Ele. 3 - MAX. ABS. : 1.9676e+04
Ele. 4 - MAX. ABS. : 1.9676e+04
Ele. 5 - MAX. ABS. : 1.9676e+04
Ele. 6 - MAX. ABS. : 1.9676e+04
Ele. 7 - MAX. ABS. : 1.9676e+04
Ele. 8 - MAX. ABS. : 1.9675e+04
Ele. 9 - MAX. ABS. : 1.9674e+04
Ele. 10 - MAX. ABS. : 1.9673e+04
Ele. 11 - MAX. ABS. : 1.9671e+04
Ele. 12 - MAX. ABS. : 1.9669e+04
Ele. 13 - MAX. ABS. : 1.9666e+04
Ele. 14 - MAX. ABS. : 1.9663e+04
Ele. 15 - MAX. ABS. : 1.9659e+04
Ele. 16 - MAX. ABS. : 1.9654e+04
Ele. 17 - MAX. ABS. : 1.9649e+04
Ele. 18 - MAX. ABS. : 1.9643e+04
Ele. 19 - MAX. ABS. : 1.9636e+04
Ele. 20 - MAX. ABS. : 1.9628e+04
Ele. 21 - MAX. ABS. : 1.9619e+04
Ele. 22 - MAX. ABS. : 1.9610e+04
Ele. 23 - MAX. ABS. : 1.9600e+04
Ele. 24 - MAX. ABS. : 1.9590e+04
Ele. 25 - MAX. ABS. : 1.9579e+04
Ele. 26 - MAX. ABS. : 1.9567e+04
Ele. 27 - MAX. ABS. : 1.9555e+04
Ele. 28 - MAX. ABS. : 1.9543e+04
Ele. 29 - MAX. ABS. : 1.9530e+04
Ele. 30 - MAX. ABS. : 1.9517e+04
Ele. 31 - MAX. ABS. : 1.9504e+04
Ele. 32 - MAX. ABS. : 1.9492e+04
Ele. 33 - MAX. ABS. : 1.9479e+04
Ele. 34 - MAX. ABS. : 1.9466e+04
Ele. 35 - MAX. ABS. : 1.9454e+04
Ele. 36 - MAX. ABS. : 1.9442e+04
Ele. 37 - MAX. ABS. : 1.9430e+04
Ele. 38 - MAX. ABS. : 1.9418e+04
Ele. 39 - MAX. ABS. : 1.9406e+04
Ele. 40 - MAX. ABS. : 1.9395e+04
Ele. 41 - MAX. ABS. : 1.9383e+04
Ele. 42 - MAX. ABS. : 1.9372e+04
Ele. 43 - MAX. ABS. : 1.9361e+04
Ele. 44 - MAX. ABS. : 1.9350e+04
Ele. 45 - MAX. ABS. : 1.9338e+04
Ele. 46 - MAX. ABS. : 1.9326e+04
Ele. 47 - MAX. ABS. : 1.9314e+04
Ele. 48 - MAX. ABS. : 1.9301e+04
Ele. 49 - MAX. ABS. : 1.9288e+04
Ele. 50 - MAX. ABS. : 1.9274e+04
Ele. 51 - MAX. ABS. : 1.9260e+04
Ele. 52 - MAX. ABS. : 1.9245e+04
Ele. 53 - MAX. ABS. : 1.9230e+04
Ele. 54 - MAX. ABS. : 1.9214e+04
Ele. 55 - MAX. ABS. : 1.9198e+04
Ele. 56 - MAX. ABS. : 1.9182e+04
Ele. 57 - MAX. ABS. : 1.9165e+04
Ele. 58 - MAX. ABS. : 1.9148e+04
Ele. 59 - MAX. ABS. : 1.9132e+04
Ele. 60 - MAX. ABS. : 1.9115e+04
Ele. 61 - MAX. ABS. : 1.9099e+04
Ele. 62 - MAX. ABS. : 1.9084e+04
Ele. 63 - MAX. ABS. : 1.9069e+04
Ele. 64 - MAX. ABS. : 1.9055e+04
Ele. 65 - MAX. ABS. : 1.9041e+04
Ele. 66 - MAX. ABS. : 1.9029e+04
Ele. 67 - MAX. ABS. : 1.9016e+04
Ele. 68 - MAX. ABS. : 1.9004e+04
Ele. 69 - MAX. ABS. : 1.8992e+04
Ele. 70 - MAX. ABS. : 1.8980e+04
Ele. 71 - MAX. ABS. : 1.8967e+04
Ele. 72 - MAX. ABS. : 1.8954e+04
Ele. 73 - MAX. ABS. : 1.8940e+04
Ele. 74 - MAX. ABS. : 1.8924e+04
Ele. 75 - MAX. ABS. : 1.8906e+04
Ele. 76 - MAX. ABS. : 1.8887e+04
Ele. 77 - MAX. ABS. : 1.8865e+04
Ele. 78 - MAX. ABS. : 1.8842e+04
Ele. 79 - MAX. ABS. : 1.8817e+04
Ele. 80 - MAX. ABS. : 1.8790e+04
Ele. 81 - MAX. ABS. : 1.8762e+04
Ele. 82 - MAX. ABS. : 1.8733e+04
Ele. 83 - MAX. ABS. : 1.8703e+04
Ele. 84 - MAX. ABS. : 1.8674e+04
Ele. 85 - MAX. ABS. : 1.8646e+04
Ele. 86 - MAX. ABS. : 1.8618e+04
Ele. 87 - MAX. ABS. : 1.8591e+04
Ele. 88 - MAX. ABS. : 1.8564e+04
Ele. 89 - MAX. ABS. : 1.8538e+04
Ele. 90 - MAX. ABS. : 1.8511e+04
Ele. 91 - MAX. ABS. : 1.8482e+04
Ele. 92 - MAX. ABS. : 1.8451e+04
Ele. 93 - MAX. ABS. : 1.8417e+04
Ele. 94 - MAX. ABS. : 1.8379e+04
Ele. 95 - MAX. ABS. : 1.8339e+04
Ele. 96 - MAX. ABS. : 1.8299e+04
Ele. 97 - MAX. ABS. : 1.8260e+04
Ele. 98 - MAX. ABS. : 1.8226e+04

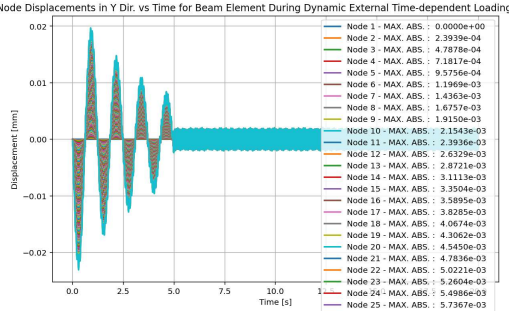

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Moment Force [N]
Time [s]
60000
40000
20000
0
−20000
−40000
−60000
−80000
0.0
2.5
5.0
7.5
10.0
12.
Ele. 1 - MAX. ABS. : 6.0257e+01
Ele. 2 - MAX. ABS. : 2.8503e+03
Ele. 3 - MAX. ABS. : 5.7580e+03
Ele. 4 - MAX. ABS. : 8.6597e+03
Ele. 5 - MAX. ABS. : 1.1553e+04
Ele. 6 - MAX. ABS. : 1.4434e+04
Ele. 7 - MAX. ABS. : 1.7300e+04
Ele. 8 - MAX. ABS. : 2.0148e+04
Ele. 9 - MAX. ABS. : 2.2975e+04
Ele. 10 - MAX. ABS. : 2.5779e+04
Ele. 11 - MAX. ABS. : 2.8556e+04
Ele. 12 - MAX. ABS. : 3.1303e+04
Ele. 13 - MAX. ABS. : 3.4017e+04
Ele. 14 - MAX. ABS. : 3.6695e+04
Ele. 15 - MAX. ABS. : 3.9335e+04
Ele. 16 - MAX. ABS. : 4.1934e+04
Ele. 17 - MAX. ABS. : 4.4488e+04
Ele. 18 - MAX. ABS. : 4.6995e+04
Ele. 19 - MAX. ABS. : 4.9453e+04
Ele. 20 - MAX. ABS. : 5.1859e+04
Ele. 21 - MAX. ABS. : 5.4211e+04
Ele. 22 - MAX. ABS. : 5.6505e+04
Ele. 23 - MAX. ABS. : 5.8740e+04
Ele. 24 - MAX. ABS. : 6.0914e+04
Ele. 25 - MAX. ABS. : 6.3024e+04
Ele. 26 - MAX. ABS. : 6.5069e+04
Ele. 27 - MAX. ABS. : 6.7047e+04
Ele. 28 - MAX. ABS. : 6.8955e+04
Ele. 29 - MAX. ABS. : 7.0792e+04
Ele. 30 - MAX. ABS. : 7.2557e+04
Ele. 31 - MAX. ABS. : 7.4247e+04
Ele. 32 - MAX. ABS. : 7.5862e+04
Ele. 33 - MAX. ABS. : 7.7400e+04
Ele. 34 - MAX. ABS. : 7.8859e+04
Ele. 35 - MAX. ABS. : 8.0239e+04
Ele. 36 - MAX. ABS. : 8.1539e+04
Ele. 37 - MAX. ABS. : 8.2757e+04
Ele. 38 - MAX. ABS. : 8.3893e+04
Ele. 39 - MAX. ABS. : 8.4945e+04
Ele. 40 - MAX. ABS. : 8.5926e+04
Ele. 41 - MAX. ABS. : 8.6832e+04
Ele. 42 - MAX. ABS. : 8.7653e+04
Ele. 43 - MAX. ABS. : 8.8385e+04
Ele. 44 - MAX. ABS. : 8.9030e+04
Ele. 45 - MAX. ABS. : 8.9587e+04
Ele. 46 - MAX. ABS. : 9.0054e+04
Ele. 47 - MAX. ABS. : 9.0431e+04
Ele. 48 - MAX. ABS. : 9.0719e+04
Ele. 49 - MAX. ABS. : 9.0910e+04
Ele. 50 - MAX. ABS. : 9.1026e+04
Ele. 51 - MAX. ABS. : 9.1045e+04
Ele. 52 - MAX. ABS. : 9.0974e+04
Ele. 53 - MAX. ABS. : 9.0814e+04
Ele. 54 - MAX. ABS. : 9.0565e+04
Ele. 55 - MAX. ABS. : 9.0228e+04
Ele. 56 - MAX. ABS. : 8.9802e+04
Ele. 57 - MAX. ABS. : 8.9289e+04
Ele. 58 - MAX. ABS. : 8.8689e+04
Ele. 59 - MAX. ABS. : 8.8002e+04
Ele. 60 - MAX. ABS. : 8.7229e+04
Ele. 61 - MAX. ABS. : 8.6371e+04
Ele. 62 - MAX. ABS. : 8.5429e+04
Ele. 63 - MAX. ABS. : 8.4403e+04
Ele. 64 - MAX. ABS. : 8.3294e+04
Ele. 65 - MAX. ABS. : 8.2104e+04
Ele. 66 - MAX. ABS. : 8.0834e+04
Ele. 67 - MAX. ABS. : 7.9485e+04
Ele. 68 - MAX. ABS. : 7.8058e+04
Ele. 69 - MAX. ABS. : 7.6556e+04
Ele. 70 - MAX. ABS. : 7.4978e+04
Ele. 71 - MAX. ABS. : 7.3327e+04
Ele. 72 - MAX. ABS. : 7.1605e+04
Ele. 73 - MAX. ABS. : 6.9812e+04
Ele. 74 - MAX. ABS. : 6.7951e+04
Ele. 75 - MAX. ABS. : 6.6024e+04
Ele. 76 - MAX. ABS. : 6.4031e+04
Ele. 77 - MAX. ABS. : 6.1976e+04
Ele. 78 - MAX. ABS. : 5.9859e+04
Ele. 79 - MAX. ABS. : 5.7684e+04
Ele. 80 - MAX. ABS. : 5.5452e+04
Ele. 81 - MAX. ABS. : 5.3166e+04
Ele. 82 - MAX. ABS. : 5.0828e+04
Ele. 83 - MAX. ABS. : 4.8440e+04
Ele. 84 - MAX. ABS. : 4.6006e+04
Ele. 85 - MAX. ABS. : 4.3528e+04
Ele. 86 - MAX. ABS. : 4.1007e+04
Ele. 87 - MAX. ABS. : 3.8457e+04
Ele. 88 - MAX. ABS. : 3.5870e+04
Ele. 89 - MAX. ABS. : 3.3248e+04
Ele. 90 - MAX. ABS. : 3.0592e+04
Ele. 91 - MAX. ABS. : 2.7906e+04
Ele. 92 - MAX. ABS. : 2.5193e+04
Ele. 93 - MAX. ABS. : 2.2453e+04
Ele. 94 - MAX. ABS. : 1.9691e+04
Ele. 95 - MAX. ABS. : 1.6907e+04
Ele. 96 - MAX. ABS. : 1.4107e+04
Ele. 97 - MAX. ABS. : 1.1291e+04
Ele. 98 - MAX. ABS. : 8.4634e+03

Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

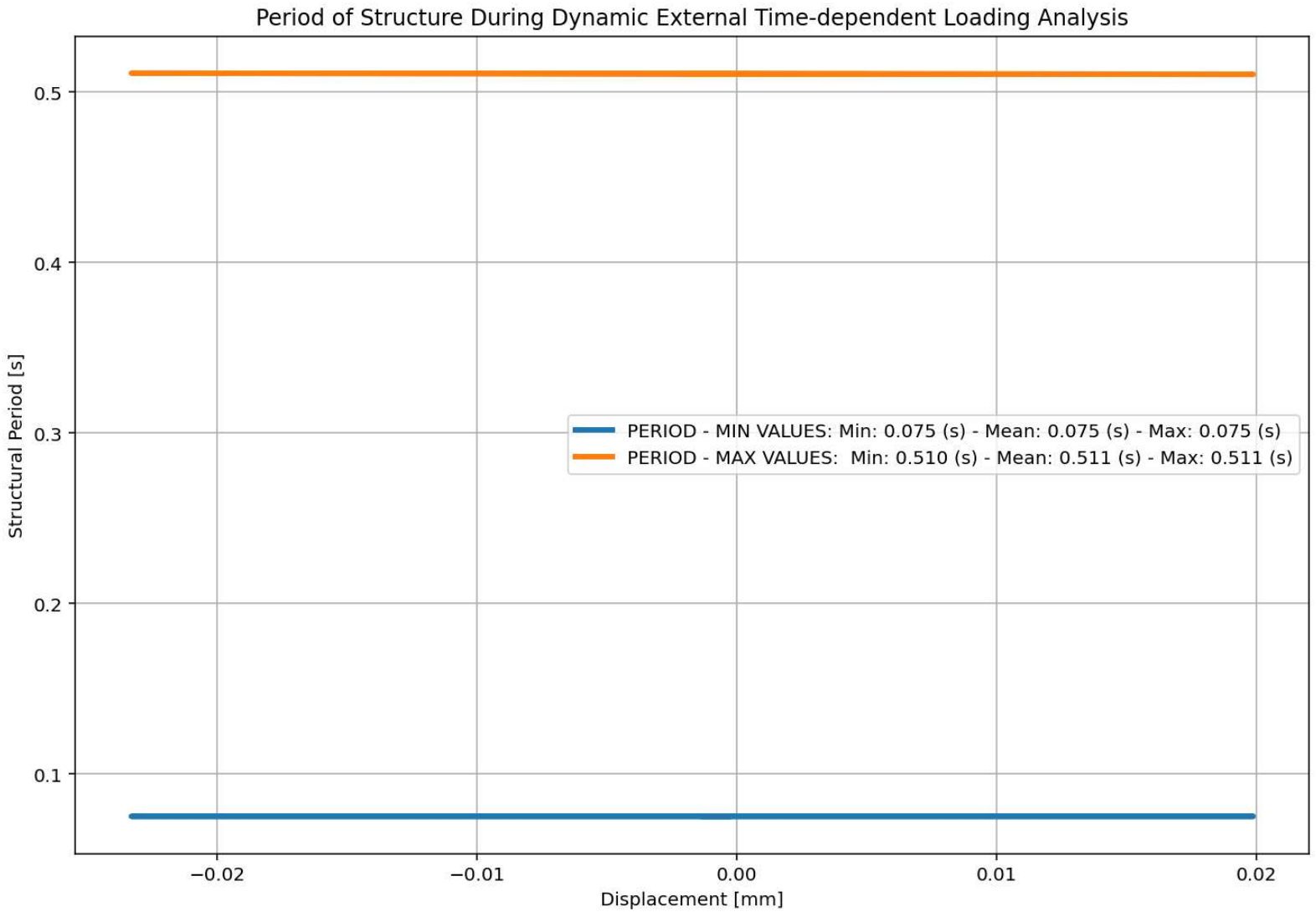


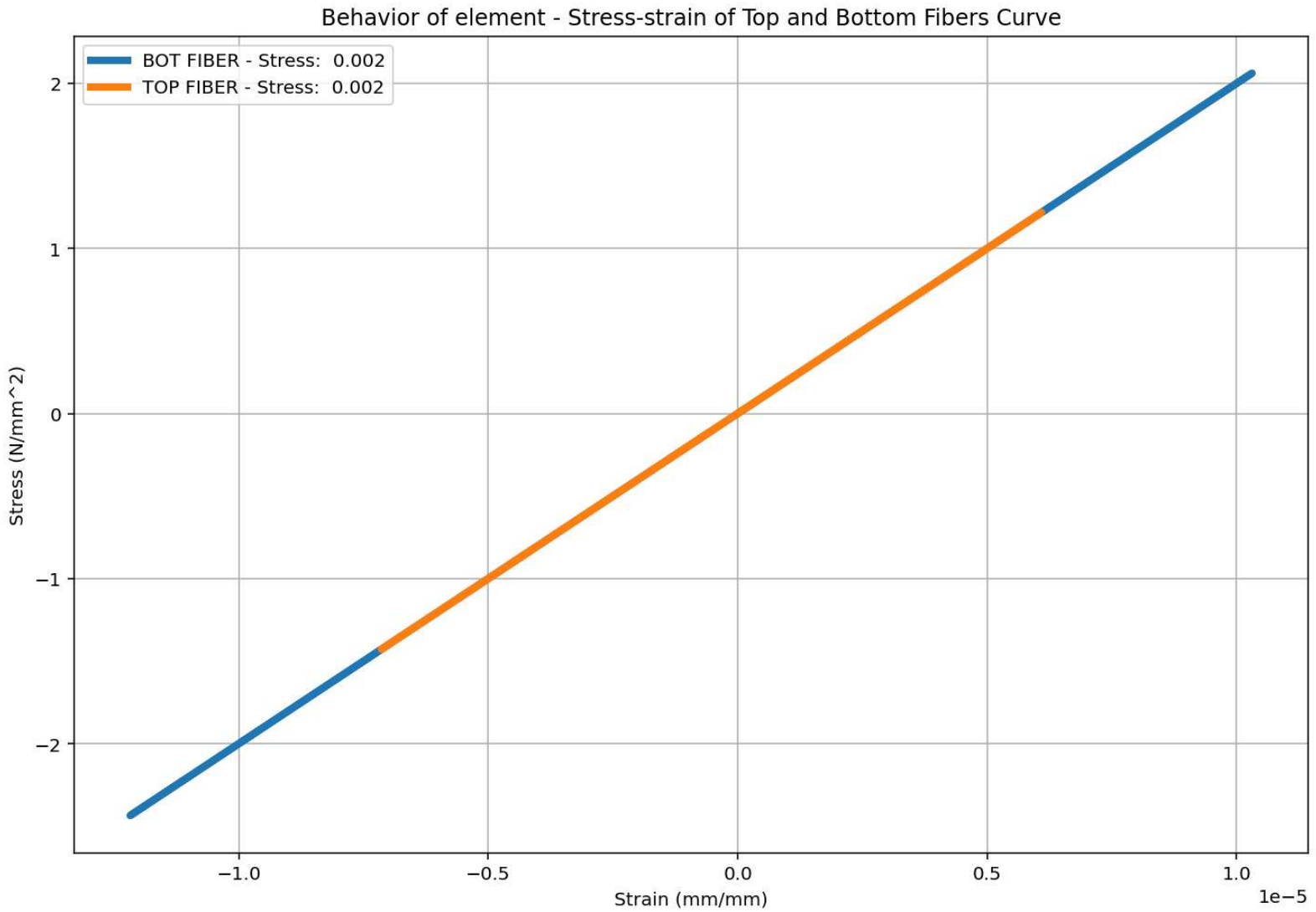
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.1369e-04
Node 3 - MAX. ABS. : 4.2717e-04
Node 4 - MAX. ABS. : 6.4023e-04
Node 5 - MAX. ABS. : 8.5265e-04
Node 6 - MAX. ABS. : 1.0642e-03
Node 7 - MAX. ABS. : 1.2747e-03
Node 8 - MAX. ABS. : 1.4840e-03
Node 9 - MAX. ABS. : 1.6917e-03
Node 10 - MAX. ABS. : 1.8978e-03
Node 11 - MAX. ABS. : 2.1020e-03
Node 12 - MAX. ABS. : 2.3041e-03
Node 13 - MAX. ABS. : 2.5038e-03
Node 14 - MAX. ABS. : 2.7011e-03
Node 15 - MAX. ABS. : 2.8957e-03
Node 16 - MAX. ABS. : 3.0874e-03
Node 17 - MAX. ABS. : 3.2760e-03
Node 18 - MAX. ABS. : 3.4613e-03
Node 19 - MAX. ABS. : 3.6432e-03
Node 20 - MAX. ABS. : 3.8214e-03
Node 21 - MAX. ABS. : 3.9959e-03
Node 22 - MAX. ABS. : 4.1663e-03
Node 23 - MAX. ABS. : 4.3326e-03
Node 24 - MAX. ABS. : 4.4946e-03
Node 25 - MAX. ABS. : 4.6521e-03
Node 26 - MAX. ABS. : 4.8049e-03
Node 27 - MAX. ABS. : 4.9530e-03
Node 28 - MAX. ABS. : 5.0962e-03
Node 29 - MAX. ABS. : 5.2342e-03
Node 30 - MAX. ABS. : 5.3671e-03
Node 31 - MAX. ABS. : 5.4947e-03
Node 32 - MAX. ABS. : 5.6168e-03
Node 33 - MAX. ABS. : 5.7335e-03
Node 34 - MAX. ABS. : 5.8446e-03
Node 35 - MAX. ABS. : 5.9499e-03
Node 36 - MAX. ABS. : 6.0493e-03
Node 37 - MAX. ABS. : 6.1427e-03
Node 38 - MAX. ABS. : 6.2300e-03
Node 39 - MAX. ABS. : 6.3112e-03
Node 40 - MAX. ABS. : 6.3862e-03
Node 41 - MAX. ABS. : 6.4548e-03
Node 42 - MAX. ABS. : 6.5170e-03
Node 43 - MAX. ABS. : 6.5728e-03
Node 44 - MAX. ABS. : 6.6222e-03
Node 45 - MAX. ABS. : 6.6649e-03
Node 46 - MAX. ABS. : 6.7011e-03
Node 47 - MAX. ABS. : 6.7307e-03
Node 48 - MAX. ABS. : 6.7536e-03
Node 49 - MAX. ABS. : 6.7698e-03
Node 50 - MAX. ABS. : 6.7794e-03
Node 51 - MAX. ABS. : 6.7823e-03
Node 52 - MAX. ABS. : 6.7785e-03
Node 53 - MAX. ABS. : 6.7680e-03
Node 54 - MAX. ABS. : 6.7508e-03
Node 55 - MAX. ABS. : 6.7270e-03
Node 56 - MAX. ABS. : 6.6965e-03
Node 57 - MAX. ABS. : 6.6594e-03
Node 58 - MAX. ABS. : 6.6158e-03
Node 59 - MAX. ABS. : 6.5657e-03
Node 60 - MAX. ABS. : 6.5091e-03
Node 61 - MAX. ABS. : 6.4460e-03
Node 62 - MAX. ABS. : 6.3767e-03
Node 63 - MAX. ABS. : 6.3010e-03
Node 64 - MAX. ABS. : 6.2192e-03
Node 65 - MAX. ABS. : 6.1312e-03
Node 66 - MAX. ABS. : 6.0372e-03
Node 67 - MAX. ABS. : 5.9372e-03
Node 68 - MAX. ABS. : 5.8314e-03
Node 69 - MAX. ABS. : 5.7199e-03
Node 70 - MAX. ABS. : 5.6028e-03
Node 71 - MAX. ABS. : 5.4801e-03
Node 72 - MAX. ABS. : 5.3520e-03
Node 73 - MAX. ABS. : 5.2187e-03
Node 74 - MAX. ABS. : 5.0803e-03
Node 75 - MAX. ABS. : 4.9368e-03
Node 76 - MAX. ABS. : 4.7885e-03
Node 77 - MAX. ABS. : 4.6355e-03
Node 78 - MAX. ABS. : 4.4780e-03
Node 79 - MAX. ABS. : 4.3160e-03
Node 80 - MAX. ABS. : 4.1498e-03
Node 81 - MAX. ABS. : 3.9795e-03
Node 82 - MAX. ABS. : 3.8054e-03
Node 83 - MAX. ABS. : 3.6275e-03
Node 84 - MAX. ABS. : 3.4460e-03
Node 85 - MAX. ABS. : 3.2611e-03
Node 86 - MAX. ABS. : 3.0731e-03
Node 87 - MAX. ABS. : 2.8820e-03
Node 88 - MAX. ABS. : 2.6881e-03
Node 89 - MAX. ABS. : 2.4916e-03
Node 90 - MAX. ABS. : 2.2926e-03
Node 91 - MAX. ABS. : 2.0914e-03
Node 92 - MAX. ABS. : 1.8881e-03
Node 93 - MAX. ABS. : 1.6830e-03
Node 94 - MAX. ABS. : 1.4762e-03
Node 95 - MAX. ABS. : 1.2680e-03
Node 96 - MAX. ABS. : 1.0586e-03
Node 97 - MAX. ABS. : 8.4811e-04
Node 98 - MAX. ABS. : 6.3680e-04
Node 99 - MAX. ABS. : 4.2488e-04
Node 100 - MAX. ABS. : 2.1254e-04

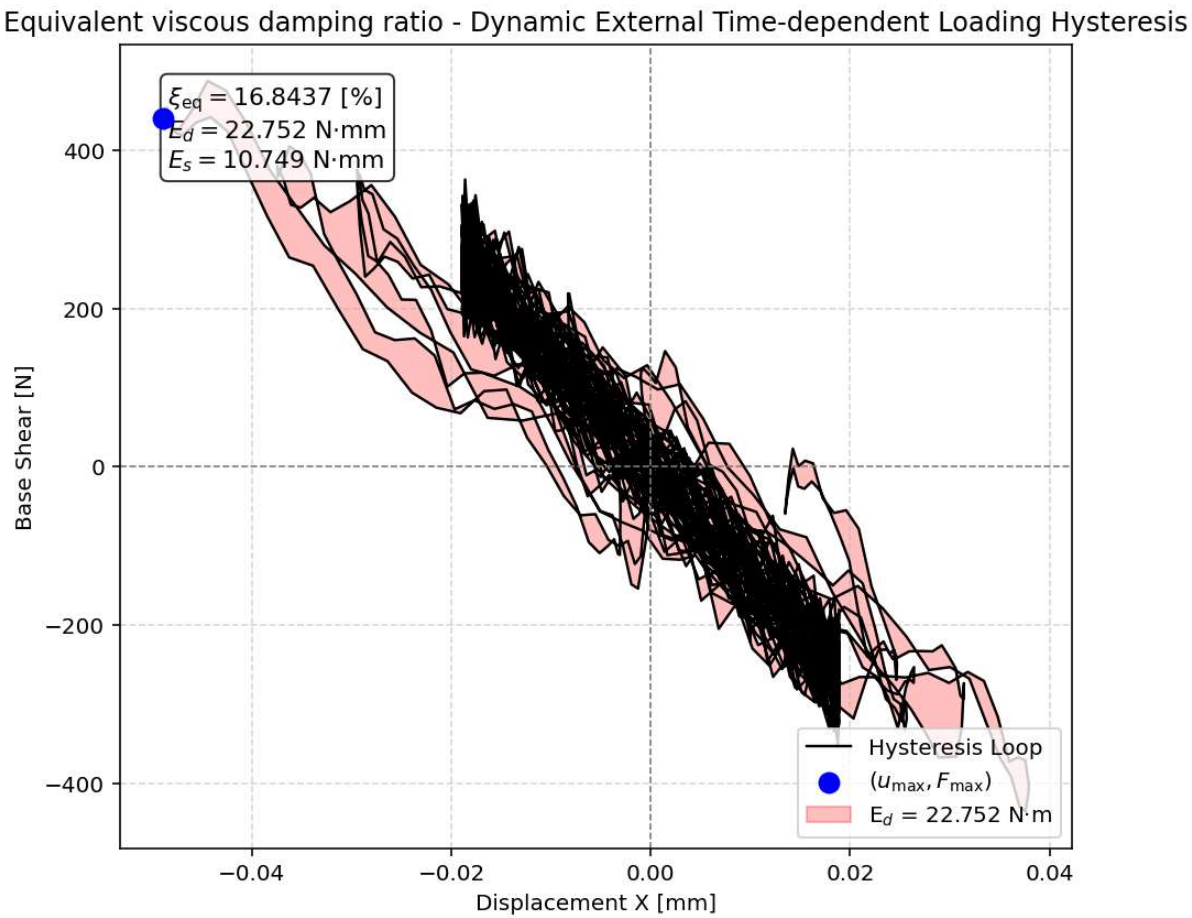
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

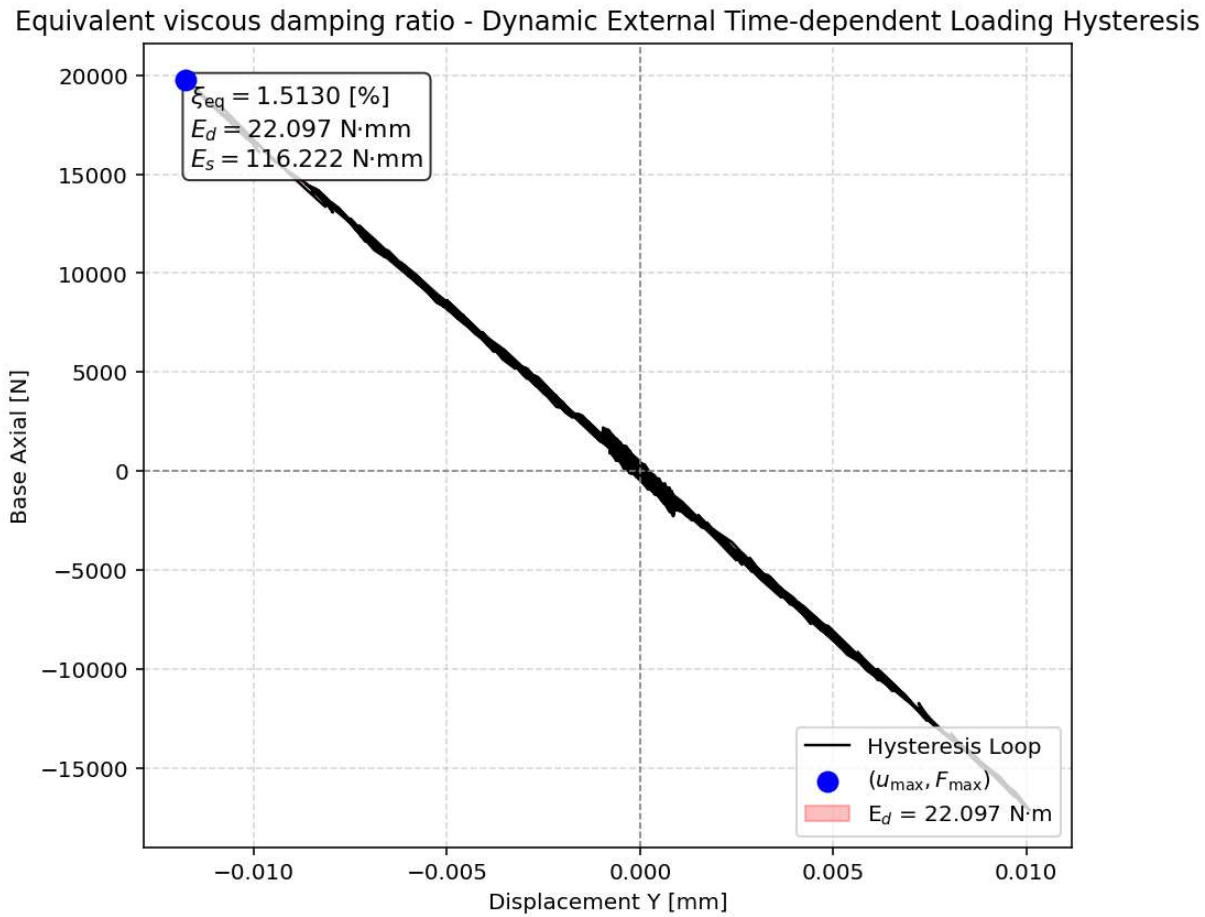


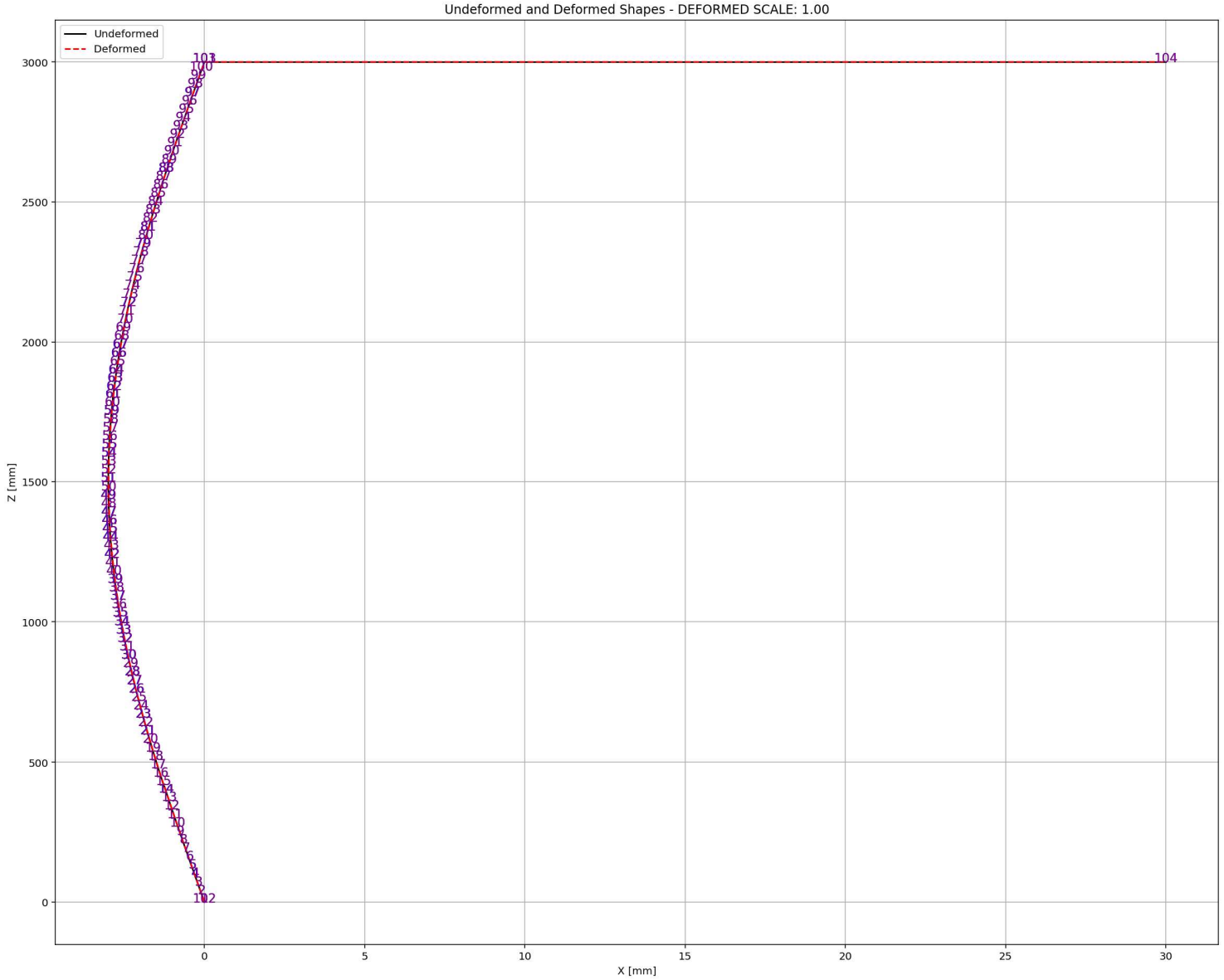
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.3939e-04
Node 3 - MAX. ABS. : 4.7878e-04
Node 4 - MAX. ABS. : 7.1817e-04
Node 5 - MAX. ABS. : 9.5756e-04
Node 6 - MAX. ABS. : 1.1969e-03
Node 7 - MAX. ABS. : 1.4363e-03
Node 8 - MAX. ABS. : 1.6757e-03
Node 9 - MAX. ABS. : 1.9150e-03
Node 10 - MAX. ABS. : 2.1543e-03
Node 11 - MAX. ABS. : 2.3936e-03
Node 12 - MAX. ABS. : 2.6329e-03
Node 13 - MAX. ABS. : 2.8721e-03
Node 14 - MAX. ABS. : 3.1113e-03
Node 15 - MAX. ABS. : 3.3504e-03
Node 16 - MAX. ABS. : 3.5895e-03
Node 17 - MAX. ABS. : 3.8285e-03
Node 18 - MAX. ABS. : 4.0674e-03
Node 19 - MAX. ABS. : 4.3062e-03
Node 20 - MAX. ABS. : 4.5450e-03
Node 21 - MAX. ABS. : 4.7836e-03
Node 22 - MAX. ABS. : 5.0221e-03
Node 23 - MAX. ABS. : 5.2604e-03
Node 24 - MAX. ABS. : 5.4986e-03
Node 25 - MAX. ABS. : 5.7367e-03
Node 26 - MAX. ABS. : 5.9746e-03
Node 27 - MAX. ABS. : 6.2124e-03
Node 28 - MAX. ABS. : 6.4500e-03
Node 29 - MAX. ABS. : 6.6874e-03
Node 30 - MAX. ABS. : 6.9246e-03
Node 31 - MAX. ABS. : 7.1617e-03
Node 32 - MAX. ABS. : 7.3986e-03
Node 33 - MAX. ABS. : 7.6354e-03
Node 34 - MAX. ABS. : 7.8719e-03
Node 35 - MAX. ABS. : 8.1083e-03
Node 36 - MAX. ABS. : 8.3445e-03
Node 37 - MAX. ABS. : 8.5806e-03
Node 38 - MAX. ABS. : 8.8165e-03
Node 39 - MAX. ABS. : 9.0522e-03
Node 40 - MAX. ABS. : 9.2878e-03
Node 41 - MAX. ABS. : 9.5233e-03
Node 42 - MAX. ABS. : 9.7585e-03
Node 43 - MAX. ABS. : 9.9937e-03
Node 44 - MAX. ABS. : 1.0229e-02
Node 45 - MAX. ABS. : 1.0464e-02
Node 46 - MAX. ABS. : 1.0698e-02
Node 47 - MAX. ABS. : 1.0933e-02
Node 48 - MAX. ABS. : 1.1167e-02
Node 49 - MAX. ABS. : 1.1401e-02
Node 50 - MAX. ABS. : 1.1636e-02
Node 51 - MAX. ABS. : 1.1869e-02
Node 52 - MAX. ABS. : 1.2103e-02
Node 53 - MAX. ABS. : 1.2337e-02
Node 54 - MAX. ABS. : 1.2570e-02
Node 55 - MAX. ABS. : 1.2803e-02
Node 56 - MAX. ABS. : 1.3036e-02
Node 57 - MAX. ABS. : 1.3269e-02
Node 58 - MAX. ABS. : 1.3502e-02
Node 59 - MAX. ABS. : 1.3734e-02
Node 60 - MAX. ABS. : 1.3966e-02
Node 61 - MAX. ABS. : 1.4198e-02
Node 62 - MAX. ABS. : 1.4430e-02
Node 63 - MAX. ABS. : 1.4662e-02
Node 64 - MAX. ABS. : 1.4893e-02
Node 65 - MAX. ABS. : 1.5125e-02
Node 66 - MAX. ABS. : 1.5356e-02
Node 67 - MAX. ABS. : 1.5587e-02
Node 68 - MAX. ABS. : 1.5818e-02
Node 69 - MAX. ABS. : 1.6049e-02
Node 70 - MAX. ABS. : 1.6279e-02
Node 71 - MAX. ABS. : 1.6510e-02
Node 72 - MAX. ABS. : 1.6740e-02
Node 73 - MAX. ABS. : 1.6971e-02
Node 74 - MAX. ABS. : 1.7201e-02
Node 75 - MAX. ABS. : 1.7431e-02
Node 76 - MAX. ABS. : 1.7660e-02
Node 77 - MAX. ABS. : 1.7890e-02
Node 78 - MAX. ABS. : 1.8119e-02
Node 79 - MAX. ABS. : 1.8348e-02
Node 80 - MAX. ABS. : 1.8577e-02
Node 81 - MAX. ABS. : 1.8805e-02
Node 82 - MAX. ABS. : 1.9033e-02
Node 83 - MAX. ABS. : 1.9261e-02
Node 84 - MAX. ABS. : 1.9489e-02
Node 85 - MAX. ABS. : 1.9716e-02
Node 86 - MAX. ABS. : 1.9942e-02
Node 87 - MAX. ABS. : 2.0169e-02
Node 88 - MAX. ABS. : 2.0395e-02
Node 89 - MAX. ABS. : 2.0621e-02
Node 90 - MAX. ABS. : 2.0846e-02
Node 91 - MAX. ABS. : 2.1072e-02
Node 92 - MAX. ABS. : 2.1296e-02
Node 93 - MAX. ABS. : 2.1521e-02
Node 94 - MAX. ABS. : 2.1745e-02
Node 95 - MAX. ABS. : 2.1969e-02
Node 96 - MAX. ABS. : 2.2192e-02
Node 97 - MAX. ABS. : 2.2414e-02
Node 98 - MAX. ABS. : 2.2637e-02
Node 99 - MAX. ABS. : 2.2858e-02
Node 100 - MAX. ABS. : 2.3080e-02


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
0.5
0.4
0.3
0.2
0.1
−0.02
−0.01
0.00
0.01
0.02
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.075 (s) - Mean: 0.075 (s) - Max: 0.075 (s)
PERIOD - MAX VALUES: Min: 0.510 (s) - Mean: 0.511 (s) - Max: 0.511 (s)


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.002
TOP FIBER - Stress: 0.002
Stress (N/mm^2)
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
Strain (mm/mm)
1e−5


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 16.8437$ [%]
$E_d = 22.752$ N·mm
$E_s = 10.749$ N·mm
Base Shear [N]
Displacement X [mm]
400
200
0
−200
−400
−0.04
−0.02
0.00
0.02
0.04
Hysteresis Loop
(u_{max}, F_{max})
E_d = 22.752 N·m


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 1.5130$ [%]
$E_d = 22.097$ N·mm
$E_s = 116.222$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 22.097 N·m
Base Axial [N]
Displacement Y [mm]
20000
15000
10000
5000
0
−5000
−10000
−15000
−0.010
−0.005
0.000
0.005
0.010


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
5
10
15
20
25
30
101
102
104

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

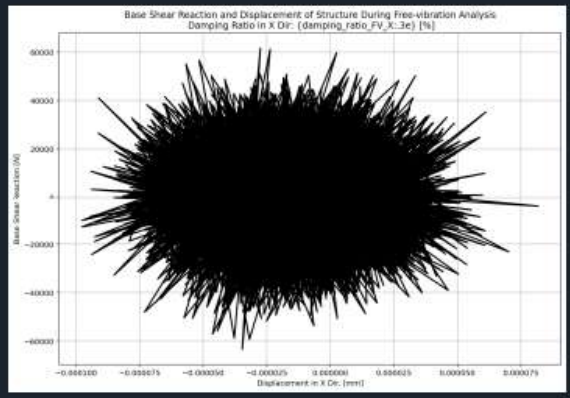
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

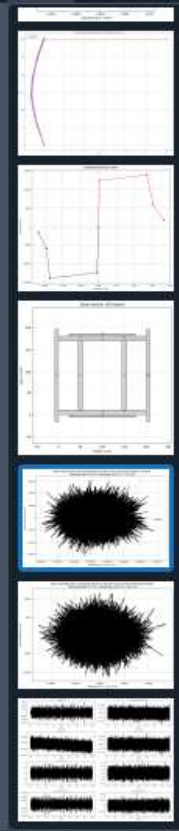
P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py

```
828
829     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
830         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
831         u0 = -0.0035                    # [mm] Initial displacement
832         v0 = 0.015                      # [mm/s] Initial velocity
833         a0 = 0.065                      # [mm/s^2] Initial acceleration
834         IU = True                       # Free Vibration with Initial Displacement
835         IV = True                       # Free Vibration with Initial Velocity
836         IA = True                       # Free Vibration with Initial Acceleration
837         duration = 20.0                 # [s] Analysis duration
838         dt = 0.001                      # [s] Time step
839         DR = 0.03                       # Damping Ratio
840
841         if IU == True:
842             # Define initial displacment
843             ops.setNodeDisp(load_node, 2, u0, '-commit') # INFO LINK: https://openseespydoc.readthedoc
844         if IV == True:
845             # Define initial velocity
846             ops.setNodeVel(load_node, 2, v0, '-commit')  # INFO LINK: https://openseespydoc.readthedoc
847         if IA == True:
848             # Define initial  acceleration
849             ops.setNodeAccel(load_node, 2, a0, '-commit') # INFO LINK: https://openseespydoc.readthedo
850
851         ops.constraints('Plain')
852         ops.numberer('Plain')
853         ops.system('BandGeneral')
854         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://openseespydoc.rea
855         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS - INFO LINK: https://openseesp
856         alpha=0.5; beta=0.25;
857         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthedocs.io/en/la
858         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
859         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.readthedocs.io/e
860         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/src/algor
861         ops.analysis('Transient') # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/src/analy
```

C:\Users\Dell

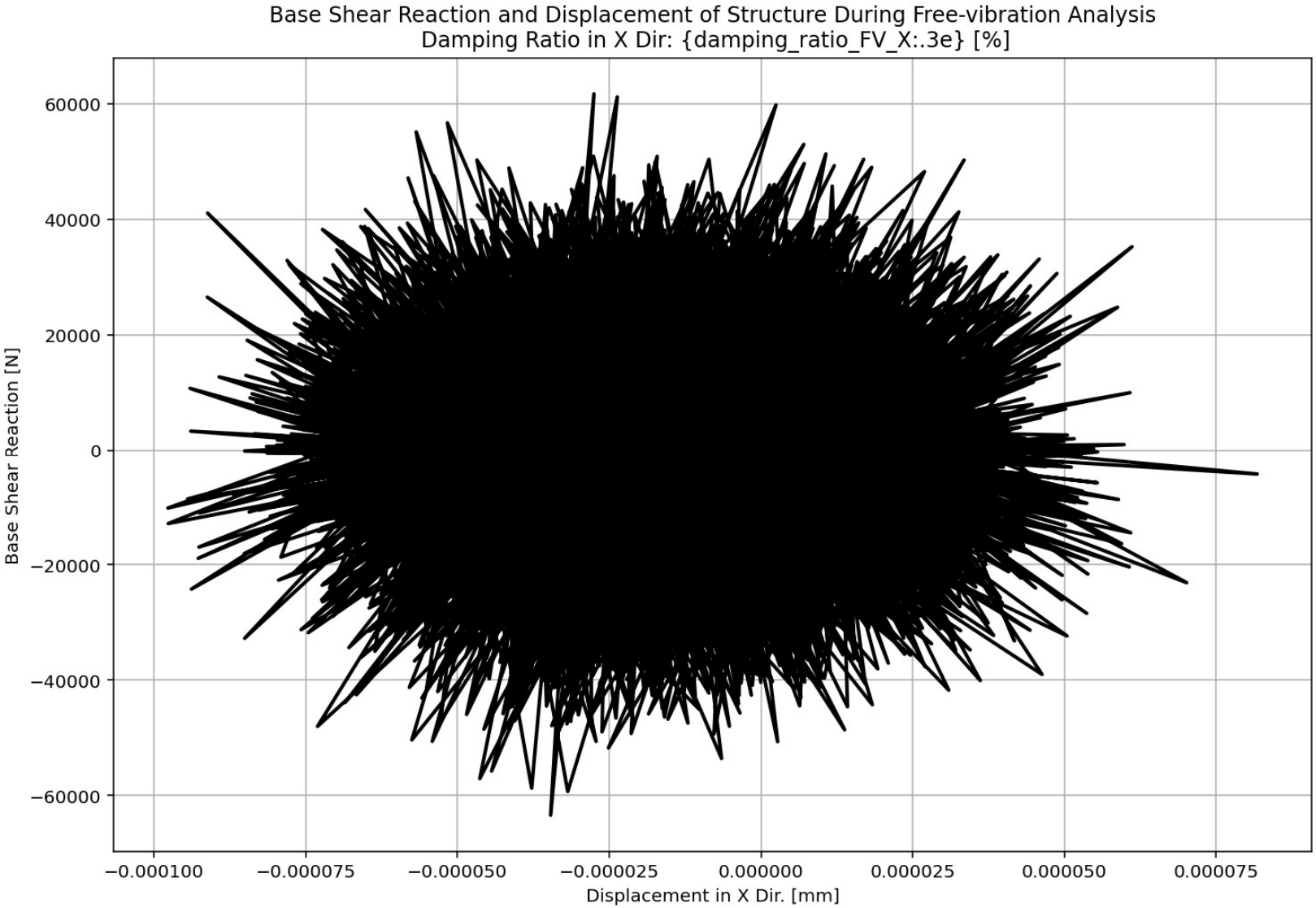
24 %

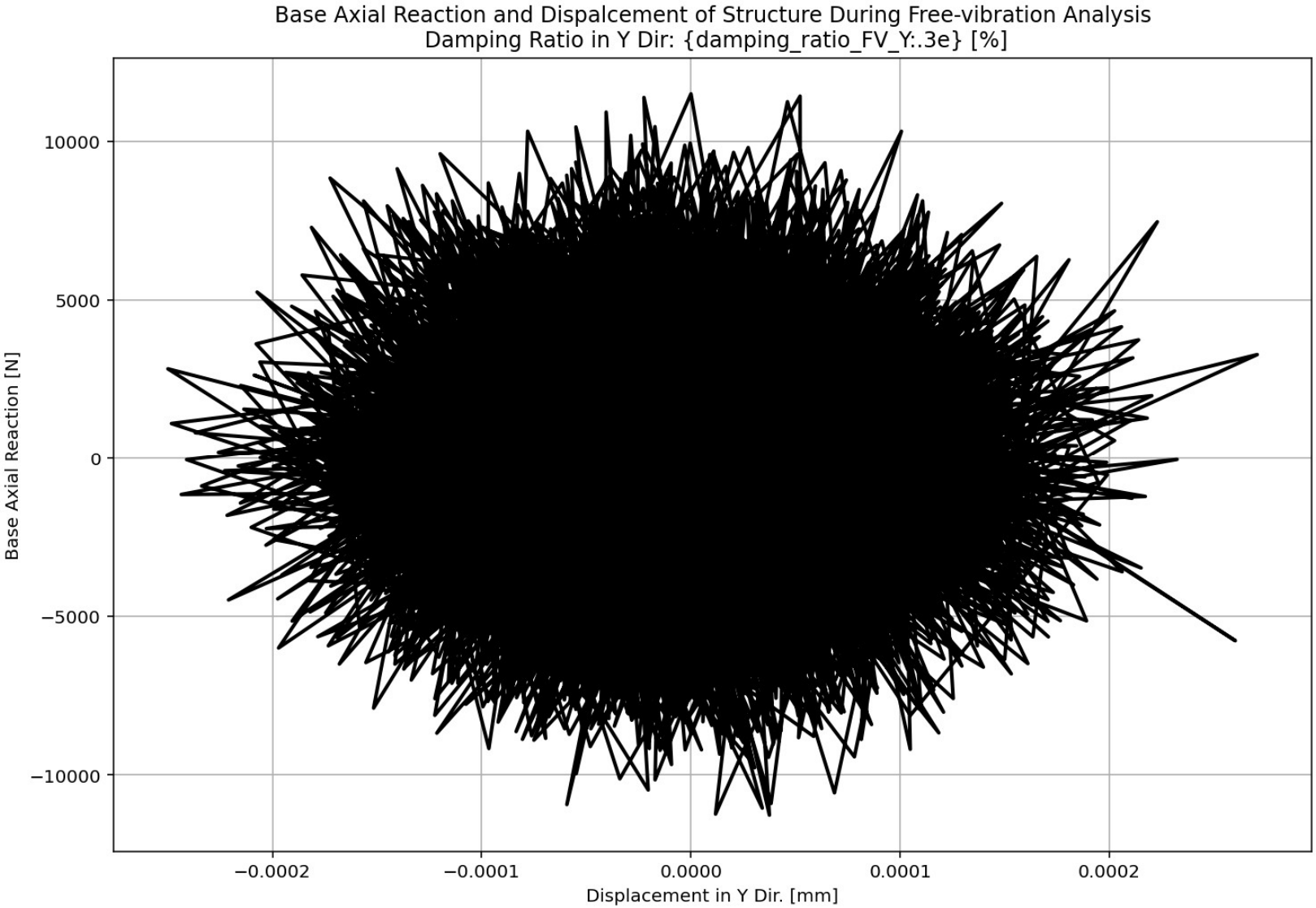


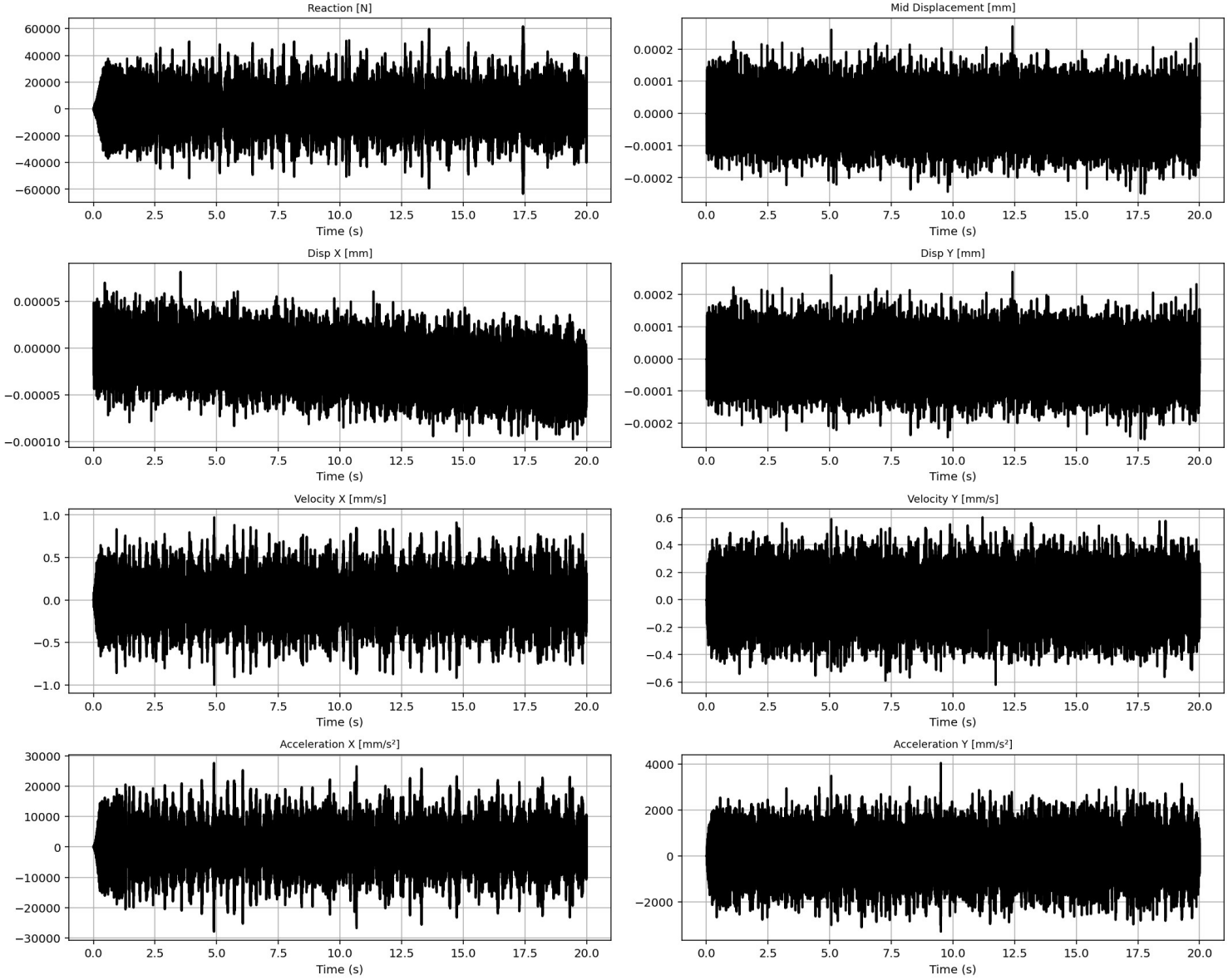


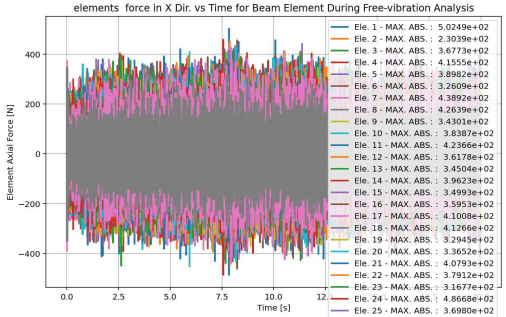
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

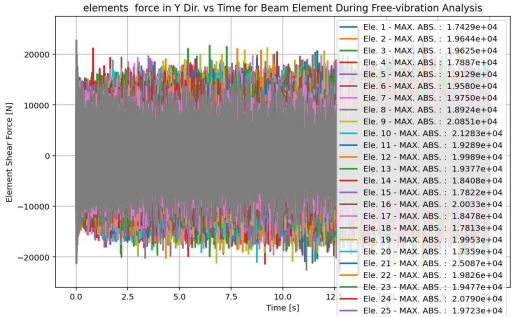
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1548, Col 1 | UTF-8 | CRLF | RW | Mem 46%

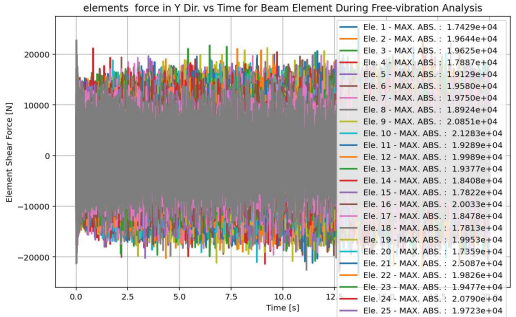

Base Shear Reaction and Displacement of Structure During Free-vibration Analysis
Damping Ratio in X Dir: {damping_ratio_FV_X:.3e} [%]
Base Shear Reaction [N]
60000
40000
20000
0
−20000
−40000
−60000
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
Displacement in X Dir. [mm]

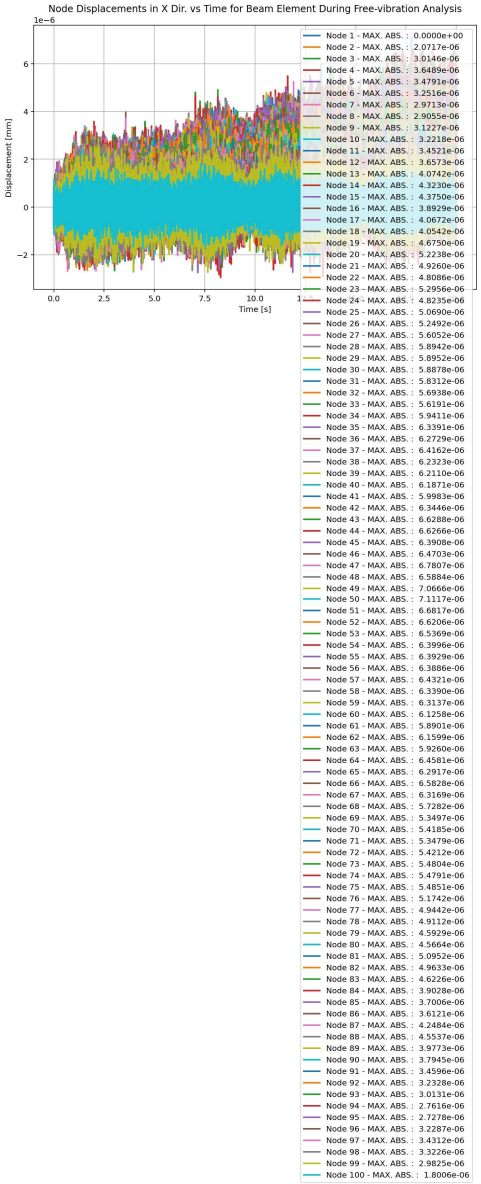

Base Axial Reaction and Dispalcement of Structure During Free-vibration Analysis
Damping Ratio in Y Dir: {damping_ratio_FV_Y:.3e} [%]
Base Axial Reaction [N]
10000
5000
0
−5000
−10000
−0.0002
−0.0001
0.0000
0.0001
0.0002
Displacement in Y Dir. [mm]

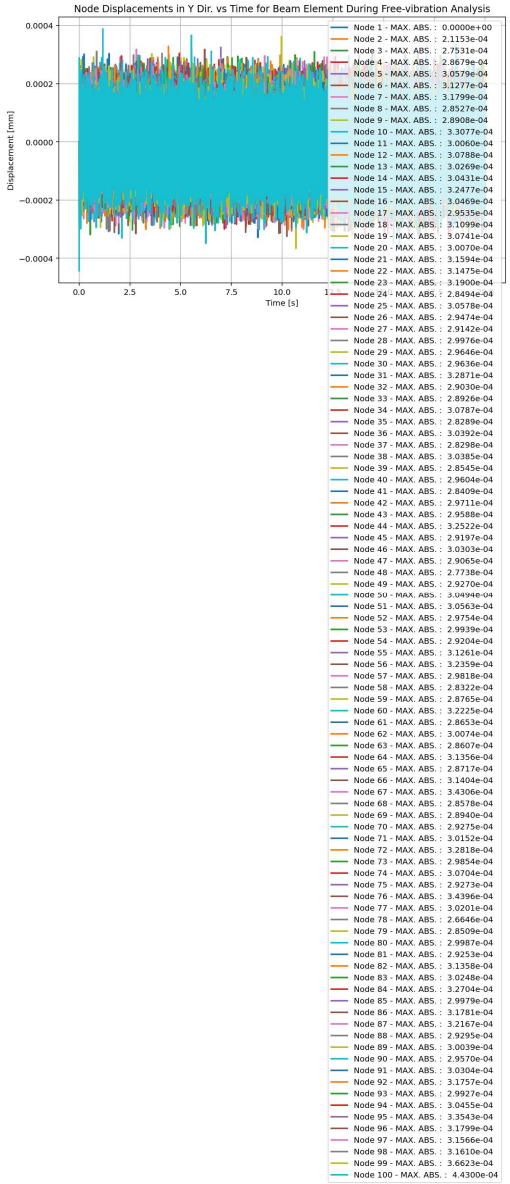

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

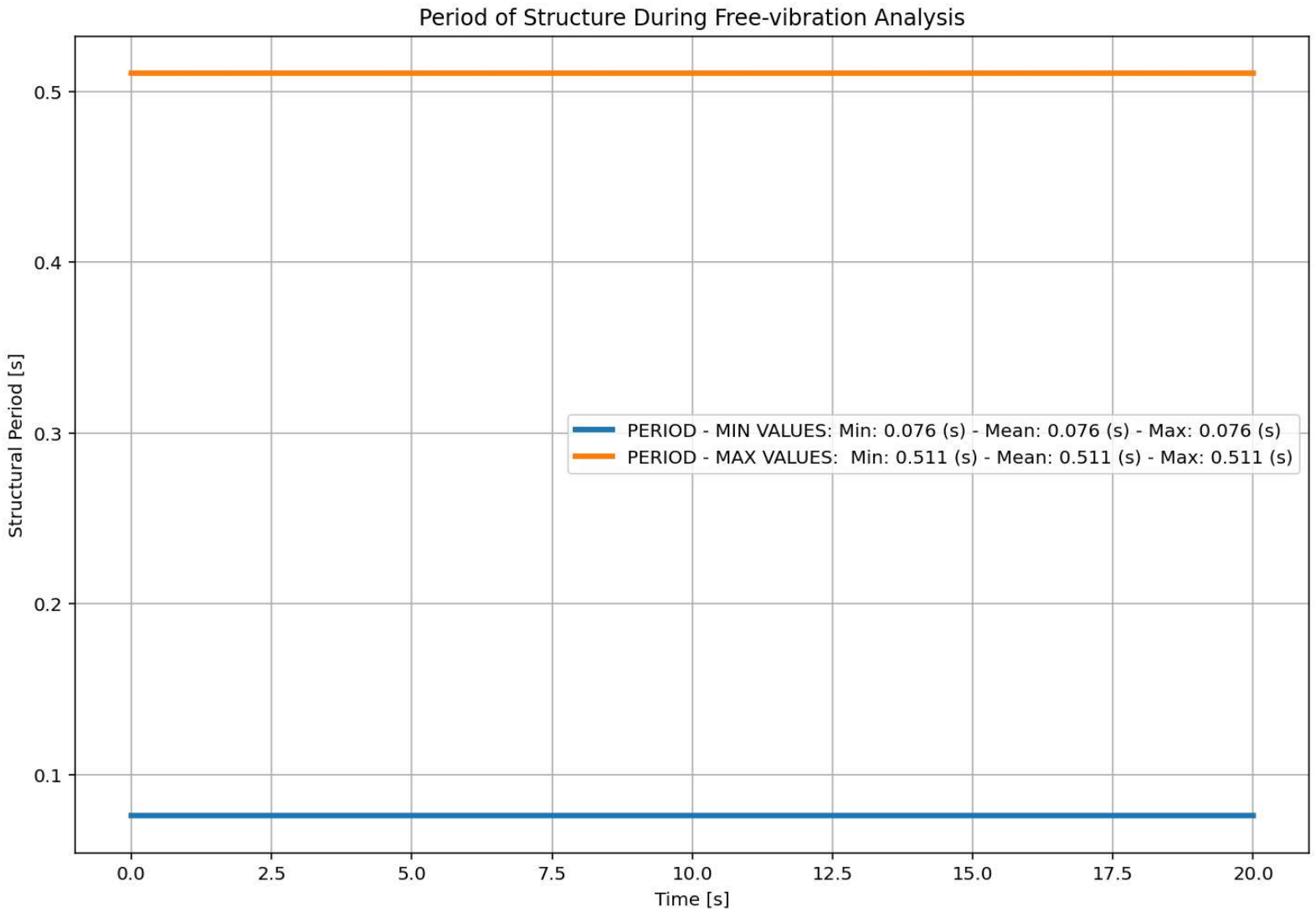

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Element Axial Force [N]
Time [s]
400
200
0
−200
−400
0.0
2.5
5.0
7.5
10.0
12.
Ele. 1 - MAX. ABS. : 5.0249e+02
Ele. 2 - MAX. ABS. : 2.3039e+02
Ele. 3 - MAX. ABS. : 3.6773e+02
Ele. 4 - MAX. ABS. : 4.1555e+02
Ele. 5 - MAX. ABS. : 3.8982e+02
Ele. 6 - MAX. ABS. : 3.2609e+02
Ele. 7 - MAX. ABS. : 4.3892e+02
Ele. 8 - MAX. ABS. : 4.2639e+02
Ele. 9 - MAX. ABS. : 3.4301e+02
Ele. 10 - MAX. ABS. : 3.8387e+02
Ele. 11 - MAX. ABS. : 4.2366e+02
Ele. 12 - MAX. ABS. : 3.6178e+02
Ele. 13 - MAX. ABS. : 3.4504e+02
Ele. 14 - MAX. ABS. : 3.9623e+02
Ele. 15 - MAX. ABS. : 3.4993e+02
Ele. 16 - MAX. ABS. : 3.5953e+02
Ele. 17 - MAX. ABS. : 4.1008e+02
Ele. 18 - MAX. ABS. : 4.1266e+02
Ele. 19 - MAX. ABS. : 3.2945e+02
Ele. 20 - MAX. ABS. : 3.3652e+02
Ele. 21 - MAX. ABS. : 4.0793e+02
Ele. 22 - MAX. ABS. : 3.7912e+02
Ele. 23 - MAX. ABS. : 3.1677e+02
Ele. 24 - MAX. ABS. : 4.8668e+02
Ele. 25 - MAX. ABS. : 3.6980e+02
Ele. 26 - MAX. ABS. : 3.6683e+02
Ele. 27 - MAX. ABS. : 3.4382e+02
Ele. 28 - MAX. ABS. : 3.8370e+02
Ele. 29 - MAX. ABS. : 3.8872e+02
Ele. 30 - MAX. ABS. : 3.5919e+02
Ele. 31 - MAX. ABS. : 3.8990e+02
Ele. 32 - MAX. ABS. : 3.8001e+02
Ele. 33 - MAX. ABS. : 3.5641e+02
Ele. 34 - MAX. ABS. : 4.0308e+02
Ele. 35 - MAX. ABS. : 3.8812e+02
Ele. 36 - MAX. ABS. : 3.3473e+02
Ele. 37 - MAX. ABS. : 3.5985e+02
Ele. 38 - MAX. ABS. : 4.0745e+02
Ele. 39 - MAX. ABS. : 4.0152e+02
Ele. 40 - MAX. ABS. : 3.4833e+02
Ele. 41 - MAX. ABS. : 3.8359e+02
Ele. 42 - MAX. ABS. : 4.4998e+02
Ele. 43 - MAX. ABS. : 3.2880e+02
Ele. 44 - MAX. ABS. : 3.6882e+02
Ele. 45 - MAX. ABS. : 4.2873e+02
Ele. 46 - MAX. ABS. : 3.6517e+02
Ele. 47 - MAX. ABS. : 3.3770e+02
Ele. 48 - MAX. ABS. : 4.3258e+02
Ele. 49 - MAX. ABS. : 3.7466e+02
Ele. 50 - MAX. ABS. : 3.4860e+02
Ele. 51 - MAX. ABS. : 3.4831e+02
Ele. 52 - MAX. ABS. : 3.9448e+02
Ele. 53 - MAX. ABS. : 3.8396e+02
Ele. 54 - MAX. ABS. : 3.1644e+02
Ele. 55 - MAX. ABS. : 4.4097e+02
Ele. 56 - MAX. ABS. : 4.6226e+02
Ele. 57 - MAX. ABS. : 3.6435e+02
Ele. 58 - MAX. ABS. : 3.1420e+02
Ele. 59 - MAX. ABS. : 3.6912e+02
Ele. 60 - MAX. ABS. : 3.8188e+02
Ele. 61 - MAX. ABS. : 4.1300e+02
Ele. 62 - MAX. ABS. : 3.9410e+02
Ele. 63 - MAX. ABS. : 4.3984e+02
Ele. 64 - MAX. ABS. : 3.5540e+02
Ele. 65 - MAX. ABS. : 3.7441e+02
Ele. 66 - MAX. ABS. : 3.8806e+02
Ele. 67 - MAX. ABS. : 3.8944e+02
Ele. 68 - MAX. ABS. : 3.7891e+02
Ele. 69 - MAX. ABS. : 4.1052e+02
Ele. 70 - MAX. ABS. : 4.2689e+02
Ele. 71 - MAX. ABS. : 3.3822e+02
Ele. 72 - MAX. ABS. : 3.5468e+02
Ele. 73 - MAX. ABS. : 3.9008e+02
Ele. 74 - MAX. ABS. : 3.6848e+02
Ele. 75 - MAX. ABS. : 3.4452e+02
Ele. 76 - MAX. ABS. : 4.1948e+02
Ele. 77 - MAX. ABS. : 4.2470e+02
Ele. 78 - MAX. ABS. : 3.3826e+02
Ele. 79 - MAX. ABS. : 4.0535e+02
Ele. 80 - MAX. ABS. : 4.5233e+02
Ele. 81 - MAX. ABS. : 3.9115e+02
Ele. 82 - MAX. ABS. : 3.1055e+02
Ele. 83 - MAX. ABS. : 4.2677e+02
Ele. 84 - MAX. ABS. : 4.1062e+02
Ele. 85 - MAX. ABS. : 3.3857e+02
Ele. 86 - MAX. ABS. : 3.8842e+02
Ele. 87 - MAX. ABS. : 3.9728e+02
Ele. 88 - MAX. ABS. : 3.5964e+02
Ele. 89 - MAX. ABS. : 3.5816e+02
Ele. 90 - MAX. ABS. : 4.3471e+02
Ele. 91 - MAX. ABS. : 4.0845e+02
Ele. 92 - MAX. ABS. : 3.2624e+02
Ele. 93 - MAX. ABS. : 4.4995e+02
Ele. 94 - MAX. ABS. : 4.1999e+02
Ele. 95 - MAX. ABS. : 3.0915e+02
Ele. 96 - MAX. ABS. : 3.7723e+02
Ele. 97 - MAX. ABS. : 4.4081e+02
Ele. 98 - MAX. ABS. : 3.5078e+02

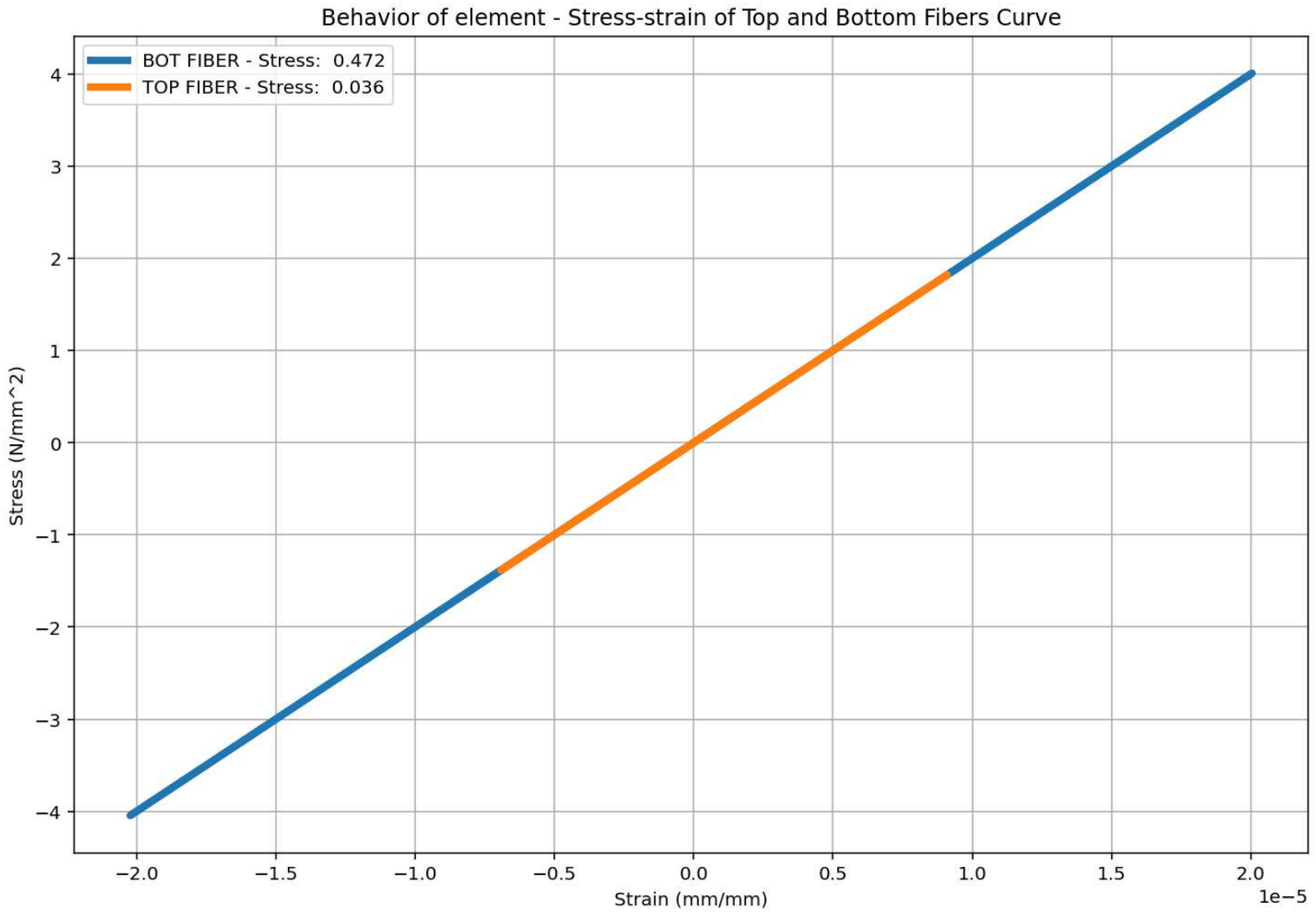

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
20000
10000
0
−10000
−20000
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.7429e+04
Ele. 2 - MAX. ABS. : 1.9644e+04
Ele. 3 - MAX. ABS. : 1.9625e+04
Ele. 4 - MAX. ABS. : 1.7887e+04
Ele. 5 - MAX. ABS. : 1.9129e+04
Ele. 6 - MAX. ABS. : 1.9580e+04
Ele. 7 - MAX. ABS. : 1.9750e+04
Ele. 8 - MAX. ABS. : 1.8924e+04
Ele. 9 - MAX. ABS. : 2.0851e+04
Ele. 10 - MAX. ABS. : 2.1283e+04
Ele. 11 - MAX. ABS. : 1.9289e+04
Ele. 12 - MAX. ABS. : 1.9989e+04
Ele. 13 - MAX. ABS. : 1.9377e+04
Ele. 14 - MAX. ABS. : 1.8408e+04
Ele. 15 - MAX. ABS. : 1.7822e+04
Ele. 16 - MAX. ABS. : 2.0033e+04
Ele. 17 - MAX. ABS. : 1.8478e+04
Ele. 18 - MAX. ABS. : 1.7813e+04
Ele. 19 - MAX. ABS. : 1.9953e+04
Ele. 20 - MAX. ABS. : 1.7359e+04
Ele. 21 - MAX. ABS. : 2.5087e+04
Ele. 22 - MAX. ABS. : 1.9826e+04
Ele. 23 - MAX. ABS. : 1.9477e+04
Ele. 24 - MAX. ABS. : 2.0790e+04
Ele. 25 - MAX. ABS. : 1.9723e+04
Ele. 26 - MAX. ABS. : 1.9850e+04
Ele. 27 - MAX. ABS. : 1.8543e+04
Ele. 28 - MAX. ABS. : 1.9810e+04
Ele. 29 - MAX. ABS. : 1.8626e+04
Ele. 30 - MAX. ABS. : 2.0263e+04
Ele. 31 - MAX. ABS. : 2.1441e+04
Ele. 32 - MAX. ABS. : 1.7280e+04
Ele. 33 - MAX. ABS. : 1.7802e+04
Ele. 34 - MAX. ABS. : 2.1383e+04
Ele. 35 - MAX. ABS. : 1.7936e+04
Ele. 36 - MAX. ABS. : 1.9574e+04
Ele. 37 - MAX. ABS. : 1.9612e+04
Ele. 38 - MAX. ABS. : 1.9585e+04
Ele. 39 - MAX. ABS. : 1.9834e+04
Ele. 40 - MAX. ABS. : 2.1048e+04
Ele. 41 - MAX. ABS. : 1.9971e+04
Ele. 42 - MAX. ABS. : 1.9943e+04
Ele. 43 - MAX. ABS. : 1.9351e+04
Ele. 44 - MAX. ABS. : 1.9343e+04
Ele. 45 - MAX. ABS. : 1.7593e+04
Ele. 46 - MAX. ABS. : 2.1549e+04
Ele. 47 - MAX. ABS. : 1.7653e+04
Ele. 48 - MAX. ABS. : 2.0168e+04
Ele. 49 - MAX. ABS. : 1.7793e+04
Ele. 50 - MAX. ABS. : 1.8357e+04
Ele. 51 - MAX. ABS. : 2.1194e+04
Ele. 52 - MAX. ABS. : 1.8998e+04
Ele. 53 - MAX. ABS. : 2.1098e+04
Ele. 54 - MAX. ABS. : 1.9908e+04
Ele. 55 - MAX. ABS. : 1.9862e+04
Ele. 56 - MAX. ABS. : 1.9687e+04
Ele. 57 - MAX. ABS. : 1.8894e+04
Ele. 58 - MAX. ABS. : 1.8402e+04
Ele. 59 - MAX. ABS. : 2.1244e+04
Ele. 60 - MAX. ABS. : 1.9186e+04
Ele. 61 - MAX. ABS. : 1.9036e+04
Ele. 62 - MAX. ABS. : 1.8533e+04
Ele. 63 - MAX. ABS. : 1.8746e+04
Ele. 64 - MAX. ABS. : 1.8269e+04
Ele. 65 - MAX. ABS. : 2.0022e+04
Ele. 66 - MAX. ABS. : 1.9484e+04
Ele. 67 - MAX. ABS. : 1.9312e+04
Ele. 68 - MAX. ABS. : 2.2998e+04
Ele. 69 - MAX. ABS. : 2.2364e+04
Ele. 70 - MAX. ABS. : 1.9736e+04
Ele. 71 - MAX. ABS. : 2.0916e+04
Ele. 72 - MAX. ABS. : 2.1083e+04
Ele. 73 - MAX. ABS. : 1.9613e+04
Ele. 74 - MAX. ABS. : 2.1187e+04
Ele. 75 - MAX. ABS. : 1.8990e+04
Ele. 76 - MAX. ABS. : 1.8742e+04
Ele. 77 - MAX. ABS. : 1.7761e+04
Ele. 78 - MAX. ABS. : 2.2588e+04
Ele. 79 - MAX. ABS. : 1.8775e+04
Ele. 80 - MAX. ABS. : 1.9907e+04
Ele. 81 - MAX. ABS. : 2.0660e+04
Ele. 82 - MAX. ABS. : 2.1980e+04
Ele. 83 - MAX. ABS. : 2.1745e+04
Ele. 84 - MAX. ABS. : 2.2245e+04
Ele. 85 - MAX. ABS. : 1.8847e+04
Ele. 86 - MAX. ABS. : 2.2357e+04
Ele. 87 - MAX. ABS. : 1.9684e+04
Ele. 88 - MAX. ABS. : 2.2108e+04
Ele. 89 - MAX. ABS. : 2.1177e+04
Ele. 90 - MAX. ABS. : 1.8574e+04
Ele. 91 - MAX. ABS. : 1.9584e+04
Ele. 92 - MAX. ABS. : 1.8374e+04
Ele. 93 - MAX. ABS. : 2.1501e+04
Ele. 94 - MAX. ABS. : 1.7771e+04
Ele. 95 - MAX. ABS. : 2.0240e+04
Ele. 96 - MAX. ABS. : 1.9905e+04
Ele. 97 - MAX. ABS. : 1.9016e+04
Ele. 98 - MAX. ABS. : 2.2779e+04

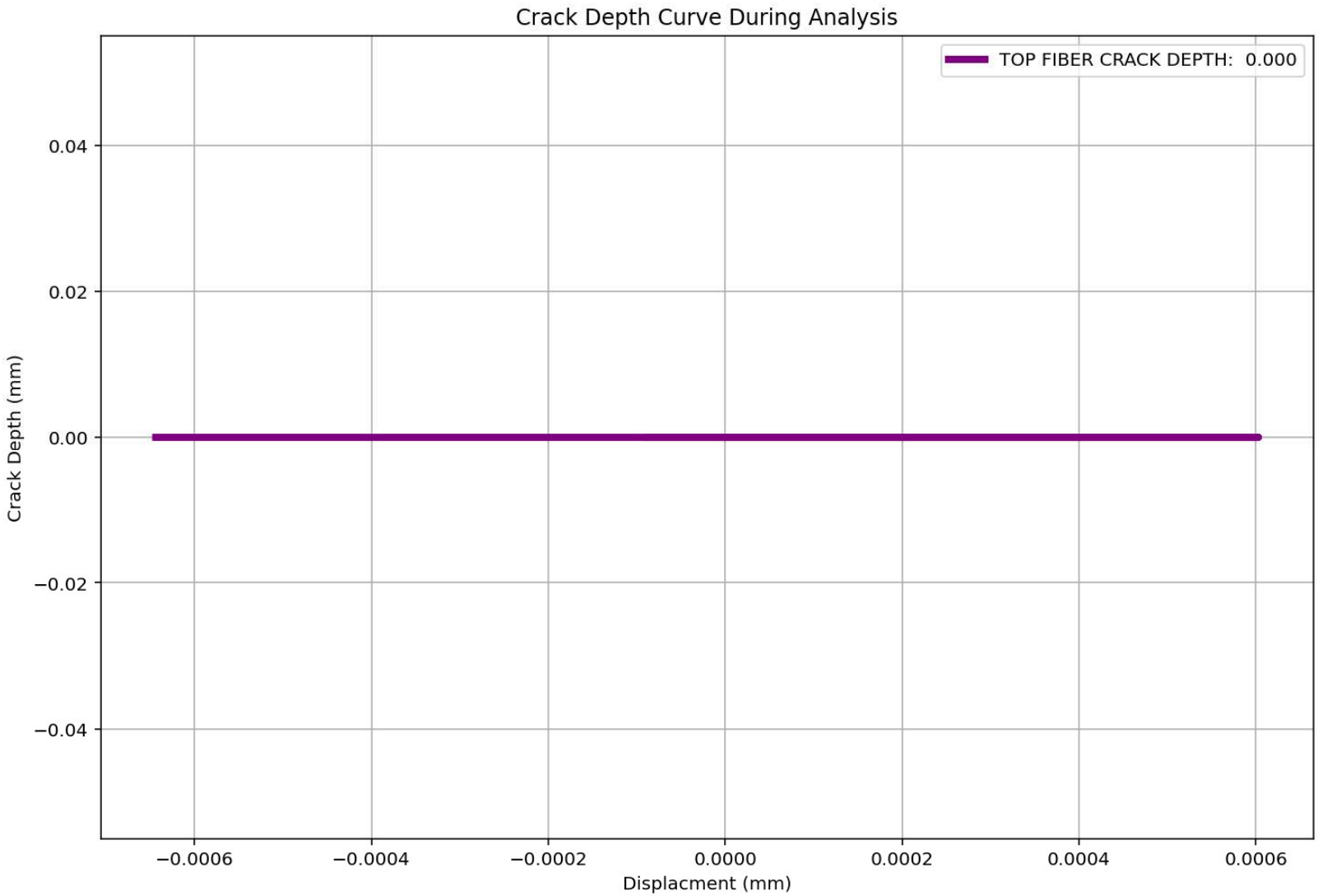

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
20000
10000
0
−10000
−20000
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.7429e+04
Ele. 2 - MAX. ABS. : 1.9644e+04
Ele. 3 - MAX. ABS. : 1.9625e+04
Ele. 4 - MAX. ABS. : 1.7887e+04
Ele. 5 - MAX. ABS. : 1.9129e+04
Ele. 6 - MAX. ABS. : 1.9580e+04
Ele. 7 - MAX. ABS. : 1.9750e+04
Ele. 8 - MAX. ABS. : 1.8924e+04
Ele. 9 - MAX. ABS. : 2.0851e+04
Ele. 10 - MAX. ABS. : 2.1283e+04
Ele. 11 - MAX. ABS. : 1.9289e+04
Ele. 12 - MAX. ABS. : 1.9989e+04
Ele. 13 - MAX. ABS. : 1.9377e+04
Ele. 14 - MAX. ABS. : 1.8408e+04
Ele. 15 - MAX. ABS. : 1.7822e+04
Ele. 16 - MAX. ABS. : 2.0033e+04
Ele. 17 - MAX. ABS. : 1.8478e+04
Ele. 18 - MAX. ABS. : 1.7813e+04
Ele. 19 - MAX. ABS. : 1.9953e+04
Ele. 20 - MAX. ABS. : 1.7359e+04
Ele. 21 - MAX. ABS. : 2.5087e+04
Ele. 22 - MAX. ABS. : 1.9826e+04
Ele. 23 - MAX. ABS. : 1.9477e+04
Ele. 24 - MAX. ABS. : 2.0790e+04
Ele. 25 - MAX. ABS. : 1.9723e+04
Ele. 26 - MAX. ABS. : 1.9850e+04
Ele. 27 - MAX. ABS. : 1.8543e+04
Ele. 28 - MAX. ABS. : 1.9810e+04
Ele. 29 - MAX. ABS. : 1.8626e+04
Ele. 30 - MAX. ABS. : 2.0263e+04
Ele. 31 - MAX. ABS. : 2.1441e+04
Ele. 32 - MAX. ABS. : 1.7280e+04
Ele. 33 - MAX. ABS. : 1.7802e+04
Ele. 34 - MAX. ABS. : 2.1383e+04
Ele. 35 - MAX. ABS. : 1.7936e+04
Ele. 36 - MAX. ABS. : 1.9574e+04
Ele. 37 - MAX. ABS. : 1.9612e+04
Ele. 38 - MAX. ABS. : 1.9585e+04
Ele. 39 - MAX. ABS. : 1.9834e+04
Ele. 40 - MAX. ABS. : 2.1048e+04
Ele. 41 - MAX. ABS. : 1.9971e+04
Ele. 42 - MAX. ABS. : 1.9943e+04
Ele. 43 - MAX. ABS. : 1.9351e+04
Ele. 44 - MAX. ABS. : 1.9343e+04
Ele. 45 - MAX. ABS. : 1.7593e+04
Ele. 46 - MAX. ABS. : 2.1549e+04
Ele. 47 - MAX. ABS. : 1.7653e+04
Ele. 48 - MAX. ABS. : 2.0168e+04
Ele. 49 - MAX. ABS. : 1.7793e+04
Ele. 50 - MAX. ABS. : 1.8357e+04
Ele. 51 - MAX. ABS. : 2.1194e+04
Ele. 52 - MAX. ABS. : 1.8998e+04
Ele. 53 - MAX. ABS. : 2.1098e+04
Ele. 54 - MAX. ABS. : 1.9908e+04
Ele. 55 - MAX. ABS. : 1.9862e+04
Ele. 56 - MAX. ABS. : 1.9687e+04
Ele. 57 - MAX. ABS. : 1.8894e+04
Ele. 58 - MAX. ABS. : 1.8402e+04
Ele. 59 - MAX. ABS. : 2.1244e+04
Ele. 60 - MAX. ABS. : 1.9186e+04
Ele. 61 - MAX. ABS. : 1.9036e+04
Ele. 62 - MAX. ABS. : 1.8533e+04
Ele. 63 - MAX. ABS. : 1.8746e+04
Ele. 64 - MAX. ABS. : 1.8269e+04
Ele. 65 - MAX. ABS. : 2.0022e+04
Ele. 66 - MAX. ABS. : 1.9484e+04
Ele. 67 - MAX. ABS. : 1.9312e+04
Ele. 68 - MAX. ABS. : 2.2998e+04
Ele. 69 - MAX. ABS. : 2.2364e+04
Ele. 70 - MAX. ABS. : 1.9736e+04
Ele. 71 - MAX. ABS. : 2.0916e+04
Ele. 72 - MAX. ABS. : 2.1083e+04
Ele. 73 - MAX. ABS. : 1.9613e+04
Ele. 74 - MAX. ABS. : 2.1187e+04
Ele. 75 - MAX. ABS. : 1.8990e+04
Ele. 76 - MAX. ABS. : 1.8742e+04
Ele. 77 - MAX. ABS. : 1.7761e+04
Ele. 78 - MAX. ABS. : 2.2588e+04
Ele. 79 - MAX. ABS. : 1.8775e+04
Ele. 80 - MAX. ABS. : 1.9907e+04
Ele. 81 - MAX. ABS. : 2.0660e+04
Ele. 82 - MAX. ABS. : 2.1980e+04
Ele. 83 - MAX. ABS. : 2.1745e+04
Ele. 84 - MAX. ABS. : 2.2245e+04
Ele. 85 - MAX. ABS. : 1.8847e+04
Ele. 86 - MAX. ABS. : 2.2357e+04
Ele. 87 - MAX. ABS. : 1.9684e+04
Ele. 88 - MAX. ABS. : 2.2108e+04
Ele. 89 - MAX. ABS. : 2.1177e+04
Ele. 90 - MAX. ABS. : 1.8574e+04
Ele. 91 - MAX. ABS. : 1.9584e+04
Ele. 92 - MAX. ABS. : 1.8374e+04
Ele. 93 - MAX. ABS. : 2.1501e+04
Ele. 94 - MAX. ABS. : 1.7771e+04
Ele. 95 - MAX. ABS. : 2.0240e+04
Ele. 96 - MAX. ABS. : 1.9905e+04
Ele. 97 - MAX. ABS. : 1.9016e+04
Ele. 98 - MAX. ABS. : 2.2779e+04

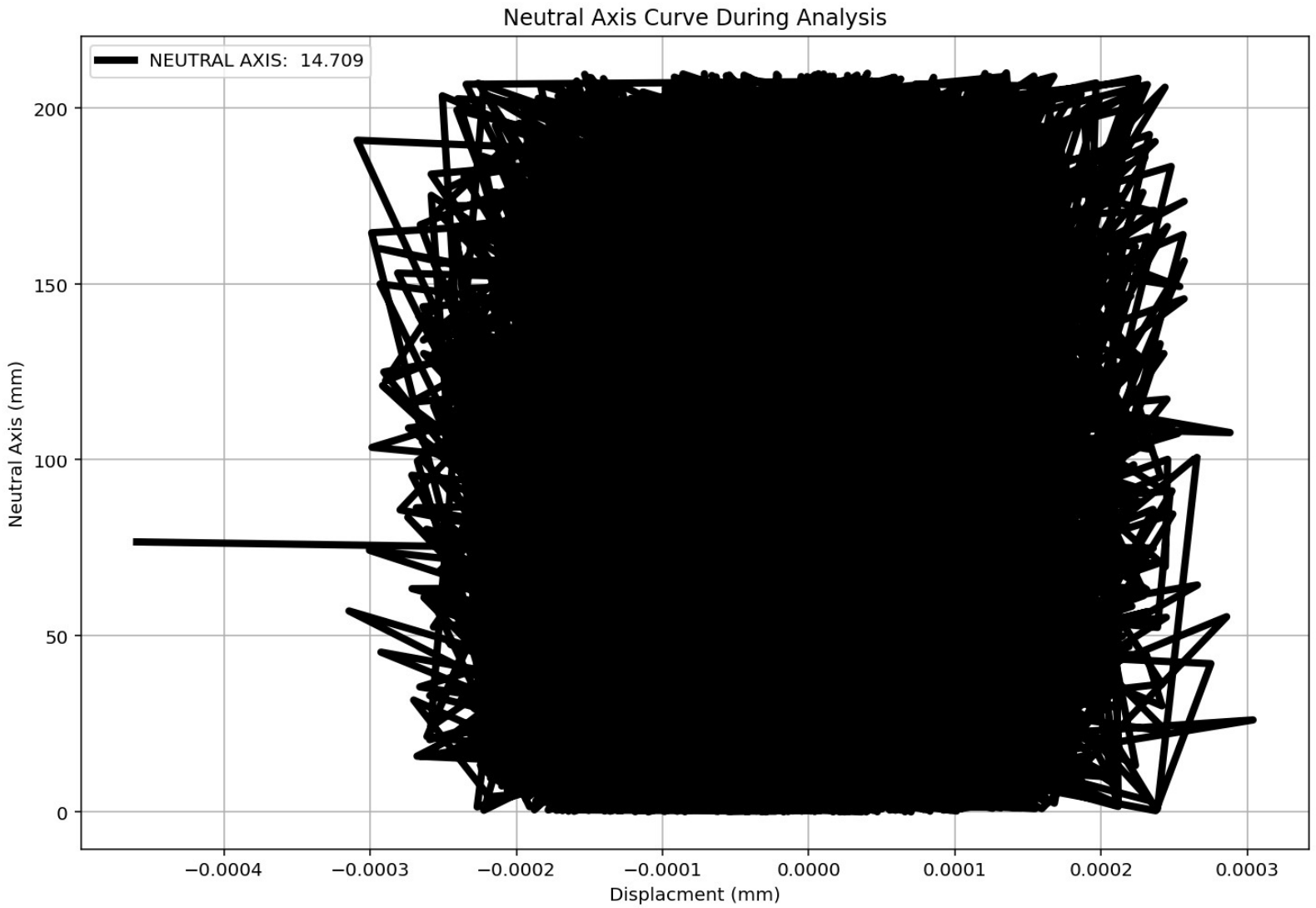

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
1e−6
Displacement [mm]
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.0717e-06
Node 3 - MAX. ABS. : 3.0146e-06
Node 4 - MAX. ABS. : 3.6489e-06
Node 5 - MAX. ABS. : 3.4791e-06
Node 6 - MAX. ABS. : 3.2516e-06
Node 7 - MAX. ABS. : 2.9713e-06
Node 8 - MAX. ABS. : 2.9055e-06
Node 9 - MAX. ABS. : 3.1227e-06
Node 10 - MAX. ABS. : 3.2218e-06
Node 11 - MAX. ABS. : 3.4521e-06
Node 12 - MAX. ABS. : 3.6573e-06
Node 13 - MAX. ABS. : 4.0742e-06
Node 14 - MAX. ABS. : 4.3230e-06
Node 15 - MAX. ABS. : 4.3750e-06
Node 16 - MAX. ABS. : 3.8929e-06
Node 17 - MAX. ABS. : 4.0672e-06
Node 18 - MAX. ABS. : 4.0542e-06
Node 19 - MAX. ABS. : 4.6750e-06
Node 20 - MAX. ABS. : 5.2238e-06
Node 21 - MAX. ABS. : 4.9260e-06
Node 22 - MAX. ABS. : 4.8086e-06
Node 23 - MAX. ABS. : 5.2956e-06
Node 24 - MAX. ABS. : 4.8235e-06
Node 25 - MAX. ABS. : 5.0690e-06
Node 26 - MAX. ABS. : 5.2492e-06
Node 27 - MAX. ABS. : 5.6052e-06
Node 28 - MAX. ABS. : 5.8942e-06
Node 29 - MAX. ABS. : 5.8952e-06
Node 30 - MAX. ABS. : 5.8878e-06
Node 31 - MAX. ABS. : 5.8312e-06
Node 32 - MAX. ABS. : 5.6938e-06
Node 33 - MAX. ABS. : 5.6191e-06
Node 34 - MAX. ABS. : 5.9411e-06
Node 35 - MAX. ABS. : 6.3391e-06
Node 36 - MAX. ABS. : 6.2729e-06
Node 37 - MAX. ABS. : 6.4162e-06
Node 38 - MAX. ABS. : 6.2323e-06
Node 39 - MAX. ABS. : 6.2110e-06
Node 40 - MAX. ABS. : 6.1871e-06
Node 41 - MAX. ABS. : 5.9983e-06
Node 42 - MAX. ABS. : 6.3446e-06
Node 43 - MAX. ABS. : 6.6288e-06
Node 44 - MAX. ABS. : 6.6266e-06
Node 45 - MAX. ABS. : 6.3908e-06
Node 46 - MAX. ABS. : 6.4703e-06
Node 47 - MAX. ABS. : 6.7807e-06
Node 48 - MAX. ABS. : 6.5884e-06
Node 49 - MAX. ABS. : 7.0666e-06
Node 50 - MAX. ABS. : 7.1117e-06
Node 51 - MAX. ABS. : 6.6817e-06
Node 52 - MAX. ABS. : 6.6206e-06
Node 53 - MAX. ABS. : 6.5369e-06
Node 54 - MAX. ABS. : 6.3996e-06
Node 55 - MAX. ABS. : 6.3929e-06
Node 56 - MAX. ABS. : 6.3886e-06
Node 57 - MAX. ABS. : 6.4321e-06
Node 58 - MAX. ABS. : 6.3390e-06
Node 59 - MAX. ABS. : 6.3137e-06
Node 60 - MAX. ABS. : 6.1258e-06
Node 61 - MAX. ABS. : 5.8901e-06
Node 62 - MAX. ABS. : 6.1599e-06
Node 63 - MAX. ABS. : 5.9260e-06
Node 64 - MAX. ABS. : 6.4581e-06
Node 65 - MAX. ABS. : 6.2917e-06
Node 66 - MAX. ABS. : 6.5828e-06
Node 67 - MAX. ABS. : 6.3169e-06
Node 68 - MAX. ABS. : 5.7282e-06
Node 69 - MAX. ABS. : 5.3497e-06
Node 70 - MAX. ABS. : 5.4185e-06
Node 71 - MAX. ABS. : 5.3479e-06
Node 72 - MAX. ABS. : 5.4212e-06
Node 73 - MAX. ABS. : 5.4804e-06
Node 74 - MAX. ABS. : 5.4791e-06
Node 75 - MAX. ABS. : 5.4851e-06
Node 76 - MAX. ABS. : 5.1742e-06
Node 77 - MAX. ABS. : 4.9442e-06
Node 78 - MAX. ABS. : 4.9112e-06
Node 79 - MAX. ABS. : 4.5929e-06
Node 80 - MAX. ABS. : 4.5664e-06
Node 81 - MAX. ABS. : 5.0952e-06
Node 82 - MAX. ABS. : 4.9633e-06
Node 83 - MAX. ABS. : 4.6226e-06
Node 84 - MAX. ABS. : 3.9028e-06
Node 85 - MAX. ABS. : 3.7006e-06
Node 86 - MAX. ABS. : 3.6121e-06
Node 87 - MAX. ABS. : 4.2484e-06
Node 88 - MAX. ABS. : 4.5537e-06
Node 89 - MAX. ABS. : 3.9773e-06
Node 90 - MAX. ABS. : 3.7945e-06
Node 91 - MAX. ABS. : 3.4596e-06
Node 92 - MAX. ABS. : 3.2328e-06
Node 93 - MAX. ABS. : 3.0131e-06
Node 94 - MAX. ABS. : 2.7616e-06
Node 95 - MAX. ABS. : 2.7278e-06
Node 96 - MAX. ABS. : 3.2287e-06
Node 97 - MAX. ABS. : 3.4312e-06
Node 98 - MAX. ABS. : 3.3226e-06
Node 99 - MAX. ABS. : 2.9825e-06
Node 100 - MAX. ABS. : 1.8006e-06

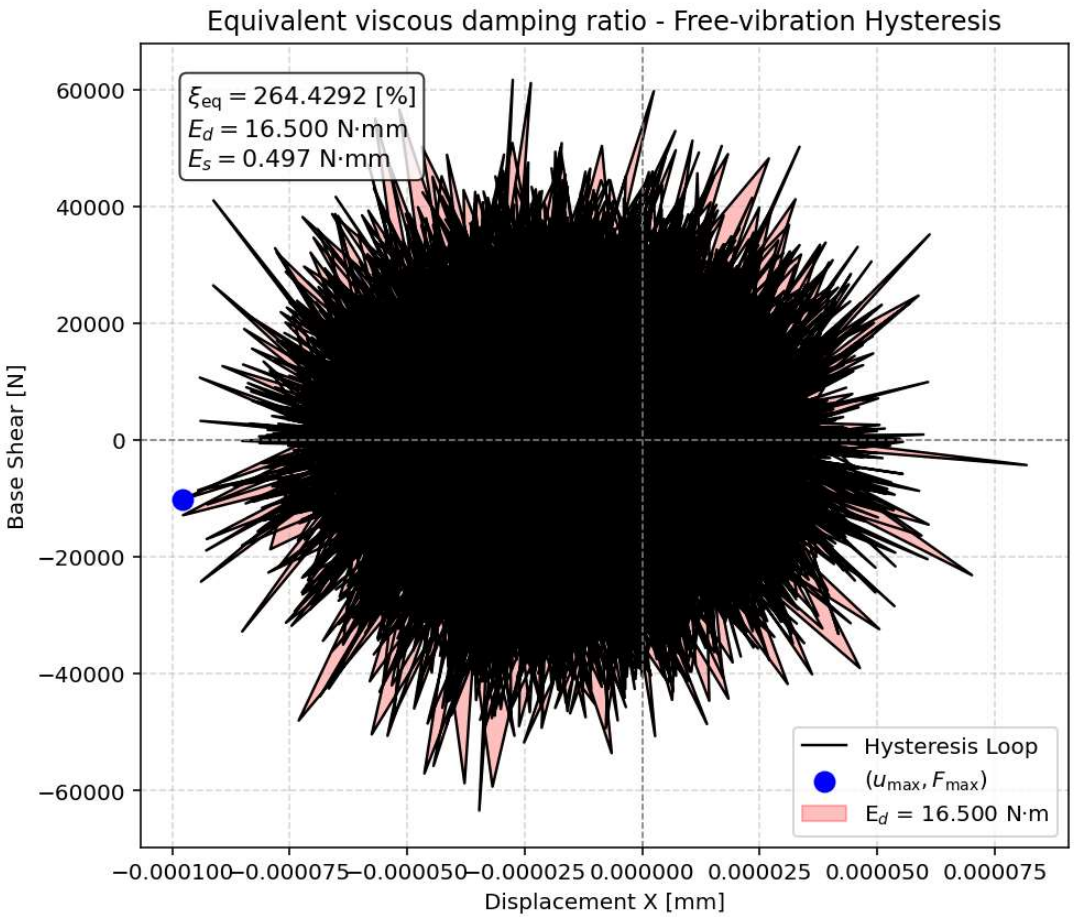

Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.0004
0.0002
0.0000
−0.0002
−0.0004
0.0
2.5
5.0
7.5
10.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.1153e-04
Node 3 - MAX. ABS. : 2.7531e-04
Node 4 - MAX. ABS. : 2.8679e-04
Node 5 - MAX. ABS. : 3.0579e-04
Node 6 - MAX. ABS. : 3.1277e-04
Node 7 - MAX. ABS. : 3.1799e-04
Node 8 - MAX. ABS. : 2.8527e-04
Node 9 - MAX. ABS. : 2.8908e-04
Node 10 - MAX. ABS. : 3.3077e-04
Node 11 - MAX. ABS. : 3.0060e-04
Node 12 - MAX. ABS. : 3.0788e-04
Node 13 - MAX. ABS. : 3.0269e-04
Node 14 - MAX. ABS. : 3.0431e-04
Node 15 - MAX. ABS. : 3.2477e-04
Node 16 - MAX. ABS. : 3.0469e-04
Node 17 - MAX. ABS. : 2.9535e-04
Node 18 - MAX. ABS. : 3.1099e-04
Node 19 - MAX. ABS. : 3.0741e-04
Node 20 - MAX. ABS. : 3.0070e-04
Node 21 - MAX. ABS. : 3.1594e-04
Node 22 - MAX. ABS. : 3.1475e-04
Node 23 - MAX. ABS. : 3.1900e-04
Node 24 - MAX. ABS. : 2.8494e-04
Node 25 - MAX. ABS. : 3.0578e-04
Node 26 - MAX. ABS. : 2.9474e-04
Node 27 - MAX. ABS. : 2.9142e-04
Node 28 - MAX. ABS. : 2.9976e-04
Node 29 - MAX. ABS. : 2.9646e-04
Node 30 - MAX. ABS. : 2.9636e-04
Node 31 - MAX. ABS. : 3.2871e-04
Node 32 - MAX. ABS. : 2.9030e-04
Node 33 - MAX. ABS. : 2.8926e-04
Node 34 - MAX. ABS. : 3.0787e-04
Node 35 - MAX. ABS. : 2.8289e-04
Node 36 - MAX. ABS. : 3.0392e-04
Node 37 - MAX. ABS. : 2.8298e-04
Node 38 - MAX. ABS. : 3.0385e-04
Node 39 - MAX. ABS. : 2.8545e-04
Node 40 - MAX. ABS. : 2.9604e-04
Node 41 - MAX. ABS. : 2.8409e-04
Node 42 - MAX. ABS. : 2.9711e-04
Node 43 - MAX. ABS. : 2.9588e-04
Node 44 - MAX. ABS. : 3.2522e-04
Node 45 - MAX. ABS. : 2.9197e-04
Node 46 - MAX. ABS. : 3.0303e-04
Node 47 - MAX. ABS. : 2.9065e-04
Node 48 - MAX. ABS. : 2.7738e-04
Node 49 - MAX. ABS. : 2.9270e-04
Node 50 - MAX. ABS. : 3.0494e-04
Node 51 - MAX. ABS. : 3.0563e-04
Node 52 - MAX. ABS. : 2.9754e-04
Node 53 - MAX. ABS. : 2.9939e-04
Node 54 - MAX. ABS. : 2.9204e-04
Node 55 - MAX. ABS. : 3.1261e-04
Node 56 - MAX. ABS. : 3.2359e-04
Node 57 - MAX. ABS. : 2.9818e-04
Node 58 - MAX. ABS. : 2.8322e-04
Node 59 - MAX. ABS. : 2.8765e-04
Node 60 - MAX. ABS. : 3.2225e-04
Node 61 - MAX. ABS. : 2.8653e-04
Node 62 - MAX. ABS. : 3.0074e-04
Node 63 - MAX. ABS. : 2.8607e-04
Node 64 - MAX. ABS. : 3.1356e-04
Node 65 - MAX. ABS. : 2.8717e-04
Node 66 - MAX. ABS. : 3.1404e-04
Node 67 - MAX. ABS. : 3.4306e-04
Node 68 - MAX. ABS. : 2.8578e-04
Node 69 - MAX. ABS. : 2.8940e-04
Node 70 - MAX. ABS. : 2.9275e-04
Node 71 - MAX. ABS. : 3.0152e-04
Node 72 - MAX. ABS. : 3.2818e-04
Node 73 - MAX. ABS. : 2.9854e-04
Node 74 - MAX. ABS. : 3.0704e-04
Node 75 - MAX. ABS. : 2.9273e-04
Node 76 - MAX. ABS. : 3.4396e-04
Node 77 - MAX. ABS. : 3.0201e-04
Node 78 - MAX. ABS. : 2.6646e-04
Node 79 - MAX. ABS. : 2.8509e-04
Node 80 - MAX. ABS. : 2.9987e-04
Node 81 - MAX. ABS. : 2.9253e-04
Node 82 - MAX. ABS. : 3.1358e-04
Node 83 - MAX. ABS. : 3.0248e-04
Node 84 - MAX. ABS. : 3.2704e-04
Node 85 - MAX. ABS. : 2.9979e-04
Node 86 - MAX. ABS. : 3.1781e-04
Node 87 - MAX. ABS. : 3.2167e-04
Node 88 - MAX. ABS. : 2.9295e-04
Node 89 - MAX. ABS. : 3.0039e-04
Node 90 - MAX. ABS. : 2.9570e-04
Node 91 - MAX. ABS. : 3.0304e-04
Node 92 - MAX. ABS. : 3.1757e-04
Node 93 - MAX. ABS. : 2.9927e-04
Node 94 - MAX. ABS. : 3.0455e-04
Node 95 - MAX. ABS. : 3.3543e-04
Node 96 - MAX. ABS. : 3.1799e-04
Node 97 - MAX. ABS. : 3.1566e-04
Node 98 - MAX. ABS. : 3.1610e-04
Node 99 - MAX. ABS. : 3.6623e-04
Node 100 - MAX. ABS. : 4.4300e-04

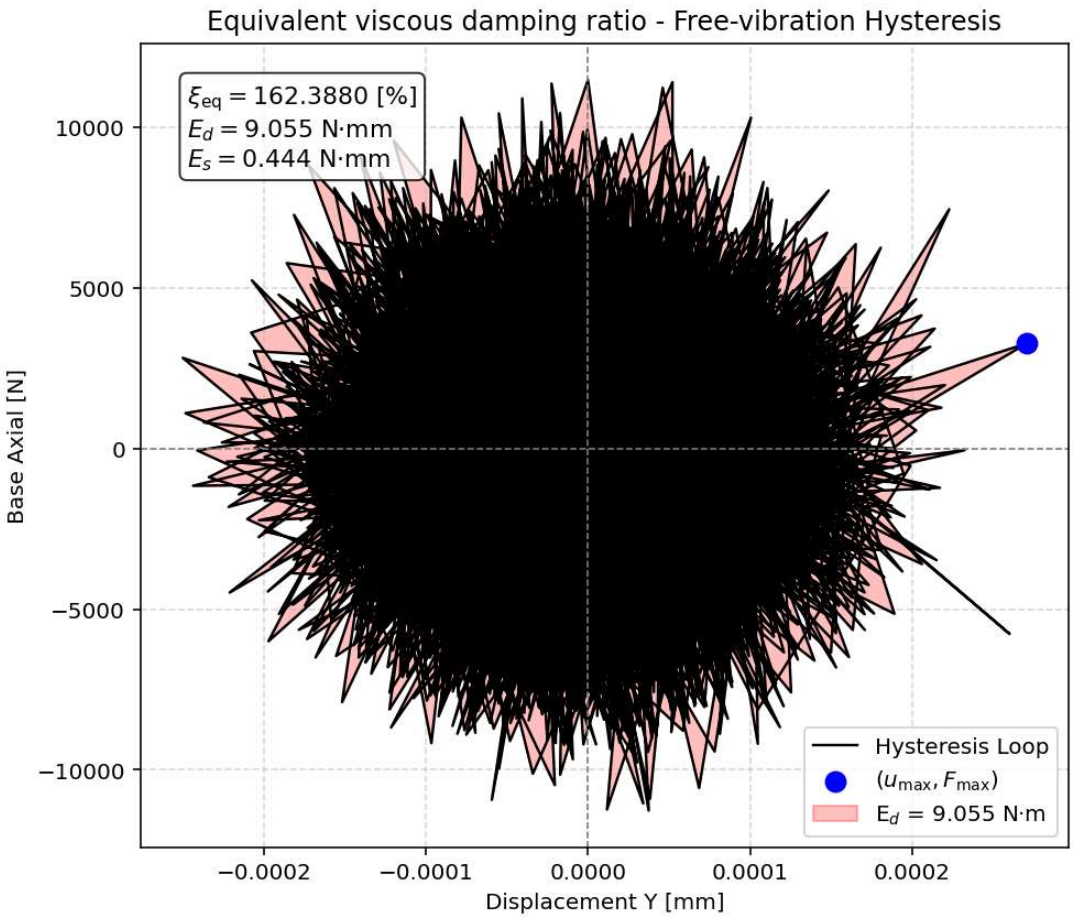

Period of Structure During Free-vibration Analysis
Structural Period [s]
0.5
0.4
0.3
0.2
0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.076 (s) - Mean: 0.076 (s) - Max: 0.076 (s)
PERIOD - MAX VALUES: Min: 0.511 (s) - Mean: 0.511 (s) - Max: 0.511 (s)

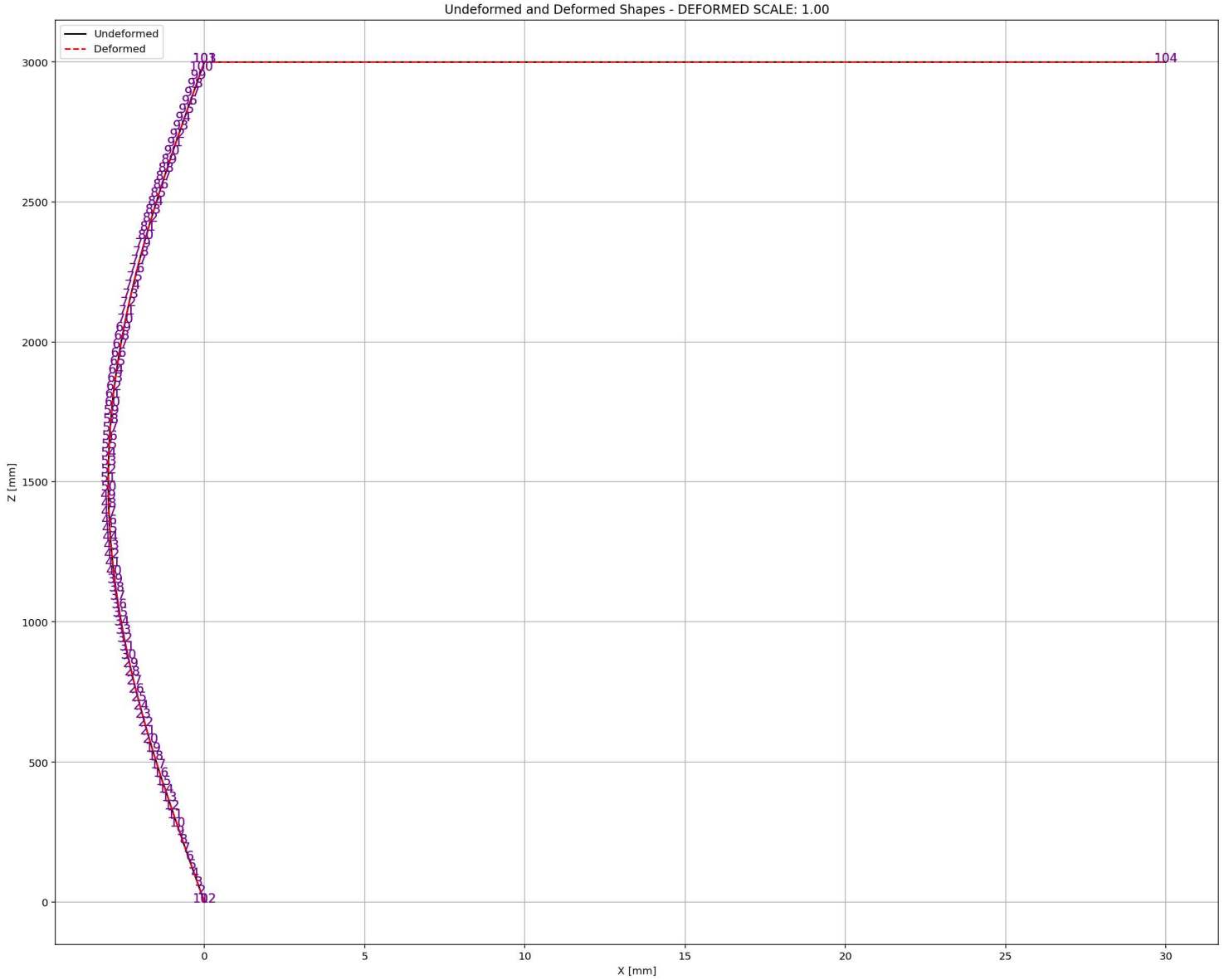

Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.472
TOP FIBER - Stress: 0.036
Stress (N/mm^2)
Strain (mm/mm)
1e−5
4
3
2
1
0
−1
−2
−3
−4
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.0006
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
Displacment (mm)


Neutral Axis Curve During Analysis
NEUTRAL AXIS: 14.709
Neutral Axis (mm)
200
150
100
50
0
−0.0004
−0.0003
−0.0002
−0.0001
0.0000
0.0001
0.0002
0.0003
Displacment (mm)


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 264.4292$ [%]
$E_d = 16.500$ N·mm
$E_s = 0.497$ N·mm
Base Shear [N]
Displacement X [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 16.500 N·m


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 162.3880$ [%]
$E_d = 9.055$ N·mm
$E_s = 0.444$ N·mm
10000
5000
0
−5000
−10000
Base Axial [N]
−0.0002
−0.0001
0.0000
0.0001
0.0002
Displacement Y [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 9.055 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
5
10
15
20
25
30
101
102
104

SEISMIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

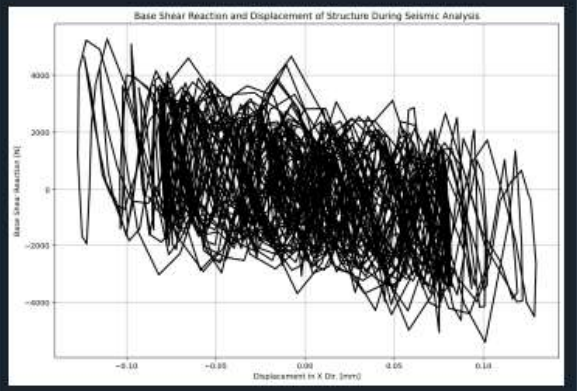
...APACITY_INDEX\55_POST-BUCKLING\EXAMPLE_02_AXIAL_AND_MOMENT

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py

```
948     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
949         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
950         duration = 20.0                    # [s] Analysis duration
951         dt = 0.01                          # [s] Time step
952         GMfact = 9810                      # [mm/s²]standard acceleration of gravity or standard accele
953         SSF_X = 2.50                       # Seismic Acceleration Scale Factor in X Direction
954         SSF_Y = 2.50                       # Seismic Acceleration Scale Factor in Y Direction
955         iv0_X = 0.0005                     # [mm/s] Initial velocity applied to the node  in X Directio
956         iv0_Y = 0.0005                     # [mm/s] Initial velocity applied to the node  in Y Directio
957         st_iv0 = 0.0                       # [s] Initial velocity applied starting time
958         SEI = 'Y'                          # Seismic Direction
959         DR = 0.03                          # Damping ratio
960
961         # Define time series for input motion (Acceleration time history)
962         if SEI == 'X':
963             SEISMIC_TAG_01 = 100
964             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceleration_X.txt'
965             # Define load patterns
966             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fact $cFact>
967             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-vel0', iv0_
968         if SEI == 'Y':
969             SEISMIC_TAG_02 = 200
970             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceleration_Y.txt'
971             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02, '-vel0', iv0_
972         if SEI == 'XY':
973             SEISMIC_TAG_01 = 100
974             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceleration_X.txt'
975             # Define load patterns
976             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fact $cFact>
977             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-vel0', iv0_
978             SEISMIC_TAG_02 = 200
979             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceleration_Y.txt'
980             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02, '-vel0', iv0_
981         print('Seismic Defined Done.')
```

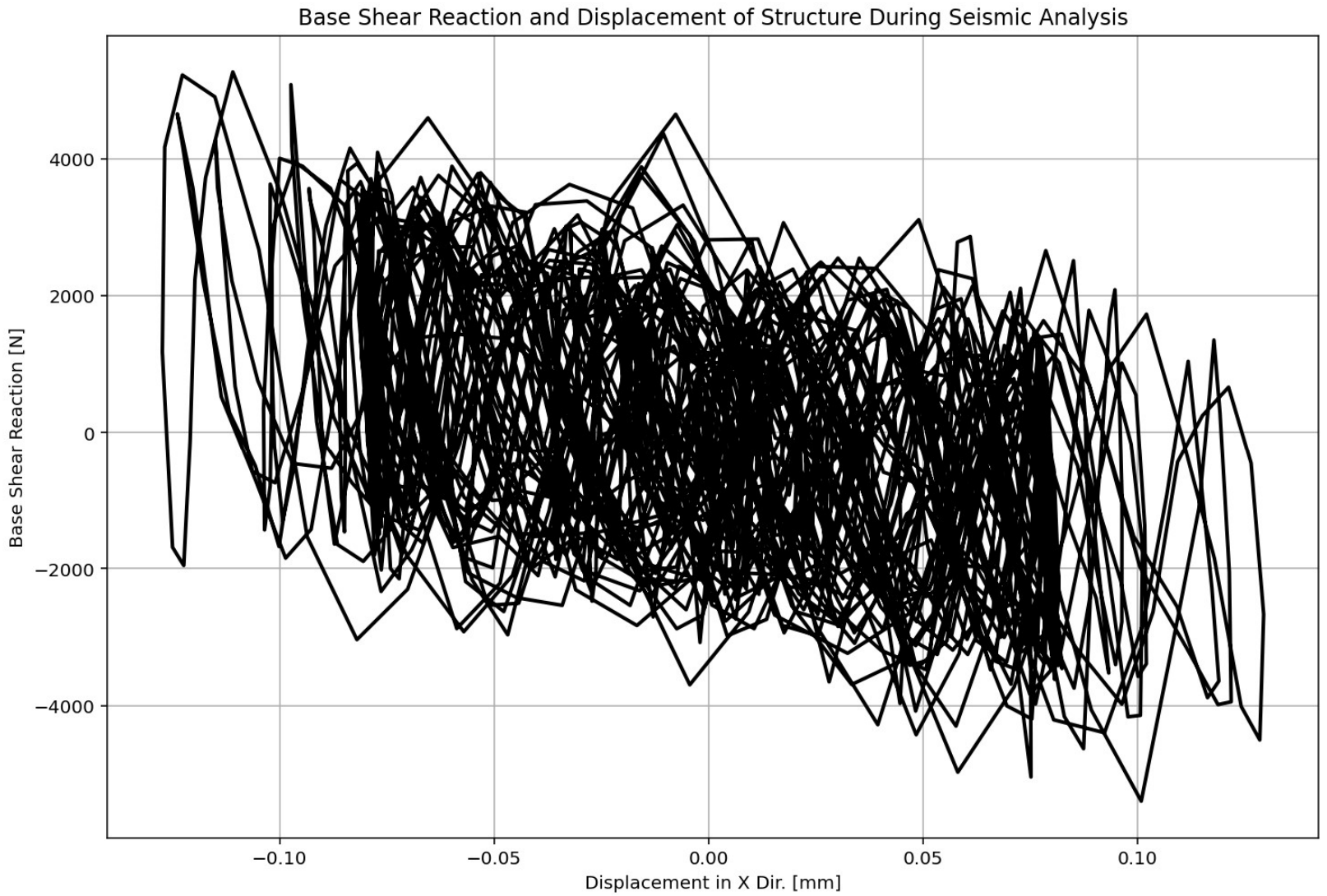
24 %

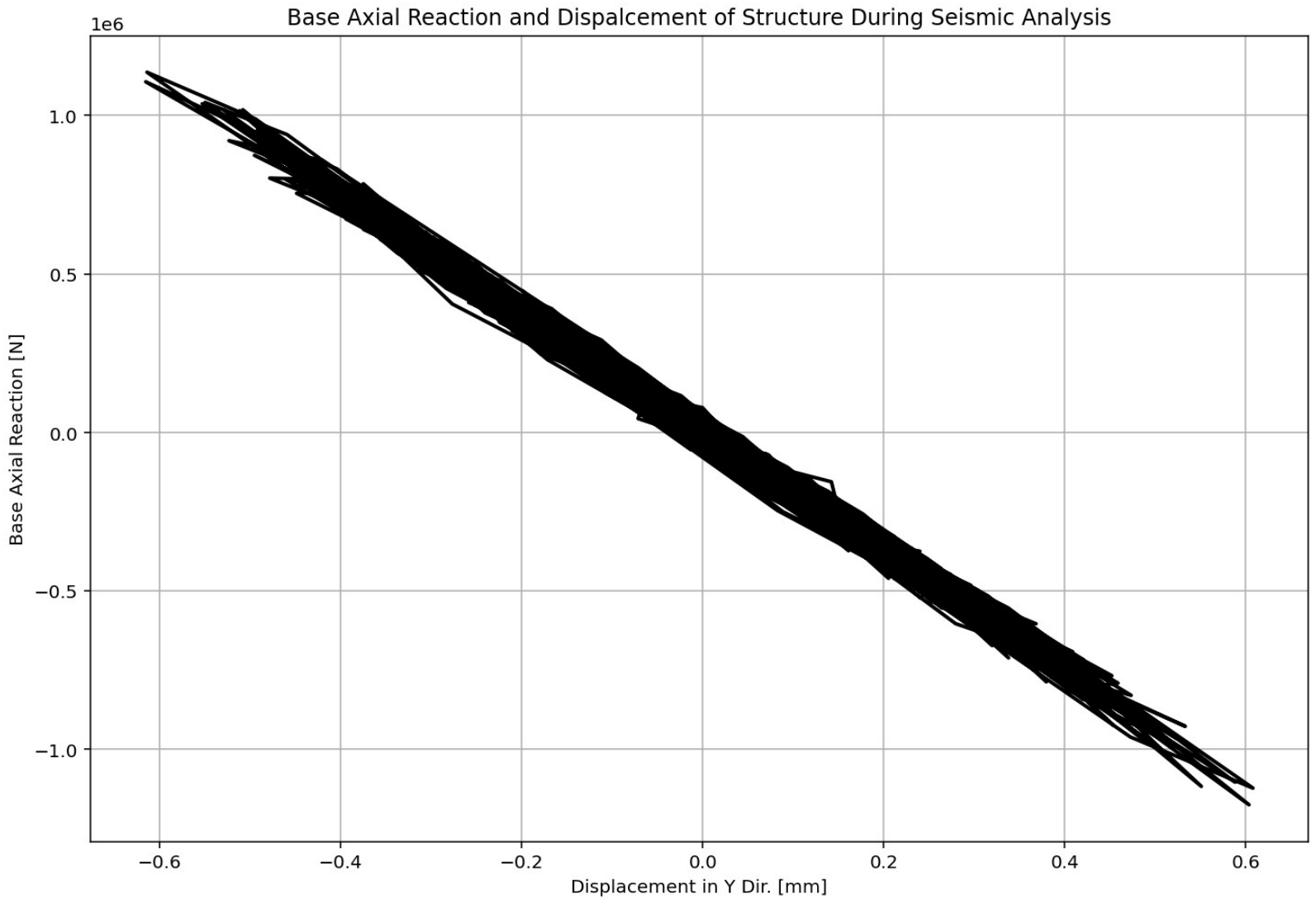


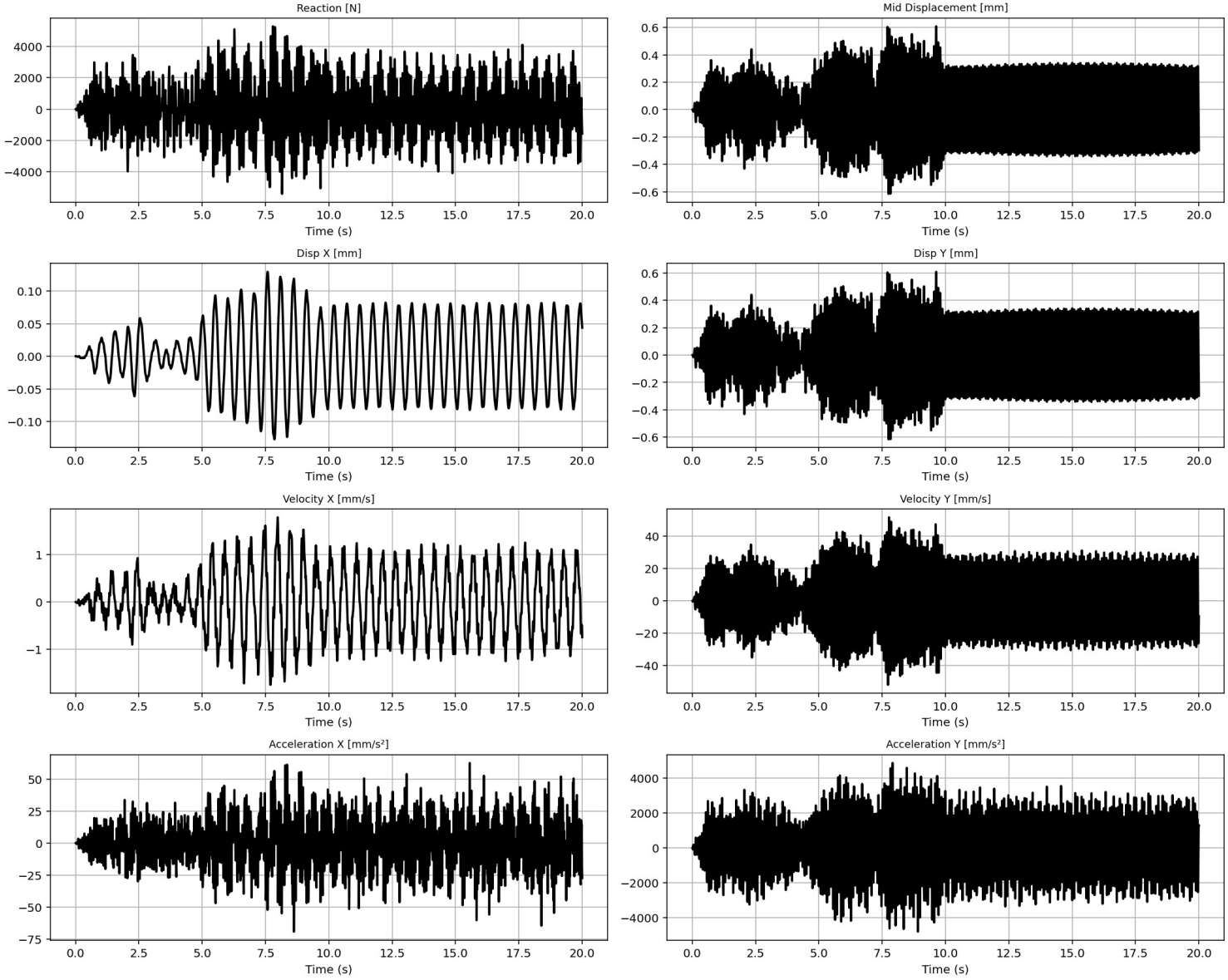


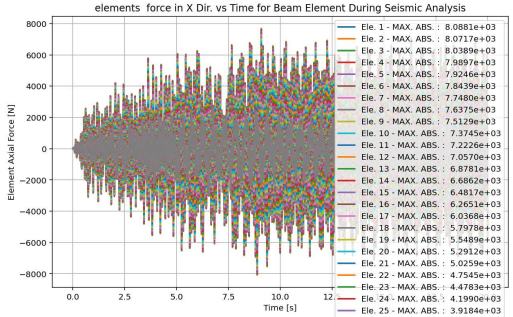
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

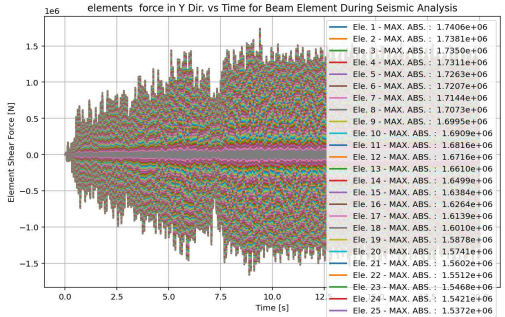
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 947, Col 5 | UTF-8 | CRLF | RW | Mem 51%


Base Shear Reaction and Displacement of Structure During Seismic Analysis
Base Shear Reaction [N]
4000
2000
0
−2000
−4000
−0.10
−0.05
0.00
0.05
0.10
Displacement in X Dir. [mm]

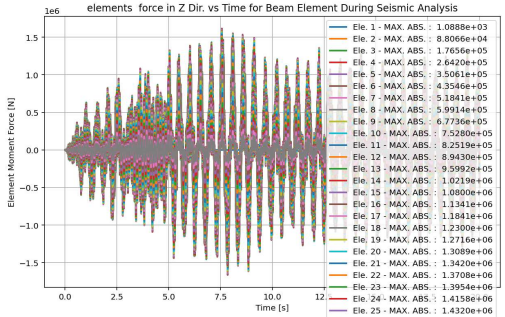

Base Axial Reaction and Dispalcement of Structure During Seismic Analysis
1e6
1.0
0.5
0.0
−0.5
−1.0
Base Axial Reaction [N]
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Displacement in Y Dir. [mm]


Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

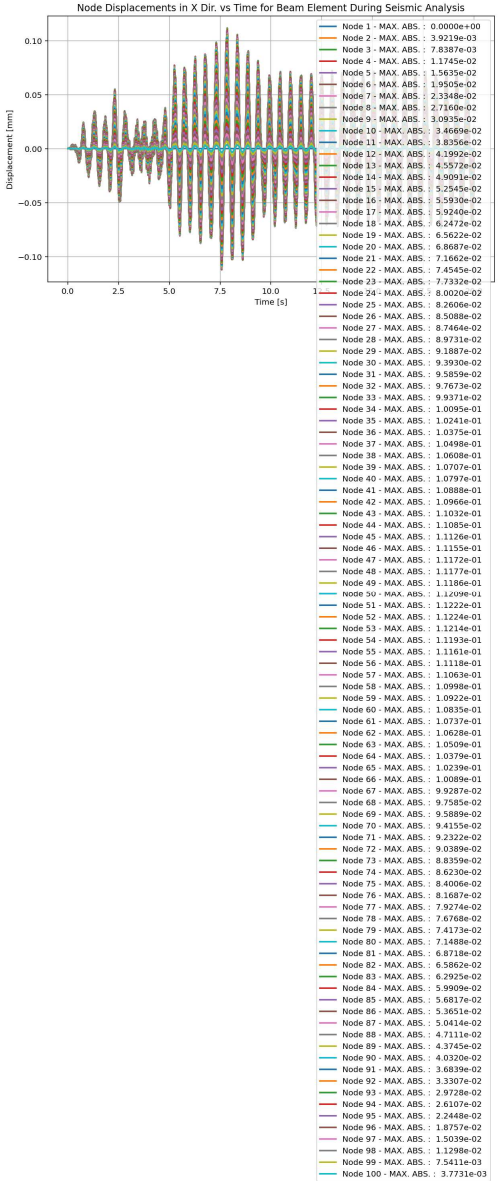

elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Element Axial Force [N]
Time [s]
8000
6000
4000
2000
0
−2000
−4000
−6000
−8000
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 8.0881e+03
Ele. 2 - MAX. ABS. : 8.0717e+03
Ele. 3 - MAX. ABS. : 8.0389e+03
Ele. 4 - MAX. ABS. : 7.9897e+03
Ele. 5 - MAX. ABS. : 7.9246e+03
Ele. 6 - MAX. ABS. : 7.8439e+03
Ele. 7 - MAX. ABS. : 7.7480e+03
Ele. 8 - MAX. ABS. : 7.6375e+03
Ele. 9 - MAX. ABS. : 7.5129e+03
Ele. 10 - MAX. ABS. : 7.3745e+03
Ele. 11 - MAX. ABS. : 7.2226e+03
Ele. 12 - MAX. ABS. : 7.0570e+03
Ele. 13 - MAX. ABS. : 6.8781e+03
Ele. 14 - MAX. ABS. : 6.6862e+03
Ele. 15 - MAX. ABS. : 6.4817e+03
Ele. 16 - MAX. ABS. : 6.2651e+03
Ele. 17 - MAX. ABS. : 6.0368e+03
Ele. 18 - MAX. ABS. : 5.7978e+03
Ele. 19 - MAX. ABS. : 5.5489e+03
Ele. 20 - MAX. ABS. : 5.2912e+03
Ele. 21 - MAX. ABS. : 5.0259e+03
Ele. 22 - MAX. ABS. : 4.7545e+03
Ele. 23 - MAX. ABS. : 4.4783e+03
Ele. 24 - MAX. ABS. : 4.1990e+03
Ele. 25 - MAX. ABS. : 3.9184e+03
Ele. 26 - MAX. ABS. : 3.7392e+03
Ele. 27 - MAX. ABS. : 3.5784e+03
Ele. 28 - MAX. ABS. : 3.4141e+03
Ele. 29 - MAX. ABS. : 3.2461e+03
Ele. 30 - MAX. ABS. : 3.0935e+03
Ele. 31 - MAX. ABS. : 2.9638e+03
Ele. 32 - MAX. ABS. : 2.8338e+03
Ele. 33 - MAX. ABS. : 2.6980e+03
Ele. 34 - MAX. ABS. : 2.5564e+03
Ele. 35 - MAX. ABS. : 2.4088e+03
Ele. 36 - MAX. ABS. : 2.2552e+03
Ele. 37 - MAX. ABS. : 2.0959e+03
Ele. 38 - MAX. ABS. : 1.9391e+03
Ele. 39 - MAX. ABS. : 1.8842e+03
Ele. 40 - MAX. ABS. : 1.8273e+03
Ele. 41 - MAX. ABS. : 1.7684e+03
Ele. 42 - MAX. ABS. : 1.7074e+03
Ele. 43 - MAX. ABS. : 1.6608e+03
Ele. 44 - MAX. ABS. : 1.6338e+03
Ele. 45 - MAX. ABS. : 1.6653e+03
Ele. 46 - MAX. ABS. : 1.6973e+03
Ele. 47 - MAX. ABS. : 1.7307e+03
Ele. 48 - MAX. ABS. : 1.7533e+03
Ele. 49 - MAX. ABS. : 1.7669e+03
Ele. 50 - MAX. ABS. : 1.8220e+03
Ele. 51 - MAX. ABS. : 1.8977e+03
Ele. 52 - MAX. ABS. : 1.9626e+03
Ele. 53 - MAX. ABS. : 2.0161e+03
Ele. 54 - MAX. ABS. : 2.0579e+03
Ele. 55 - MAX. ABS. : 2.0878e+03
Ele. 56 - MAX. ABS. : 2.1056e+03
Ele. 57 - MAX. ABS. : 2.1110e+03
Ele. 58 - MAX. ABS. : 2.1042e+03
Ele. 59 - MAX. ABS. : 2.0852e+03
Ele. 60 - MAX. ABS. : 2.0701e+03
Ele. 61 - MAX. ABS. : 2.0886e+03
Ele. 62 - MAX. ABS. : 2.1343e+03
Ele. 63 - MAX. ABS. : 2.1696e+03
Ele. 64 - MAX. ABS. : 2.2193e+03
Ele. 65 - MAX. ABS. : 2.2745e+03
Ele. 66 - MAX. ABS. : 2.3203e+03
Ele. 67 - MAX. ABS. : 2.3564e+03
Ele. 68 - MAX. ABS. : 2.3826e+03
Ele. 69 - MAX. ABS. : 2.3989e+03
Ele. 70 - MAX. ABS. : 2.4057e+03
Ele. 71 - MAX. ABS. : 2.4033e+03
Ele. 72 - MAX. ABS. : 2.3924e+03
Ele. 73 - MAX. ABS. : 2.3740e+03
Ele. 74 - MAX. ABS. : 2.3488e+03
Ele. 75 - MAX. ABS. : 2.3178e+03
Ele. 76 - MAX. ABS. : 2.2820e+03
Ele. 77 - MAX. ABS. : 2.2419e+03
Ele. 78 - MAX. ABS. : 2.1983e+03
Ele. 79 - MAX. ABS. : 2.1519e+03
Ele. 80 - MAX. ABS. : 2.1031e+03
Ele. 81 - MAX. ABS. : 2.0525e+03
Ele. 82 - MAX. ABS. : 2.0030e+03
Ele. 83 - MAX. ABS. : 2.0520e+03
Ele. 84 - MAX. ABS. : 2.1525e+03
Ele. 85 - MAX. ABS. : 2.2501e+03
Ele. 86 - MAX. ABS. : 2.3444e+03
Ele. 87 - MAX. ABS. : 2.4347e+03
Ele. 88 - MAX. ABS. : 2.5204e+03
Ele. 89 - MAX. ABS. : 2.6011e+03
Ele. 90 - MAX. ABS. : 2.6762e+03
Ele. 91 - MAX. ABS. : 2.7453e+03
Ele. 92 - MAX. ABS. : 2.8078e+03
Ele. 93 - MAX. ABS. : 2.8635e+03
Ele. 94 - MAX. ABS. : 2.9121e+03
Ele. 95 - MAX. ABS. : 2.9535e+03
Ele. 96 - MAX. ABS. : 2.9877e+03
Ele. 97 - MAX. ABS. : 3.0147e+03
Ele. 98 - MAX. ABS. : 3.0347e+03

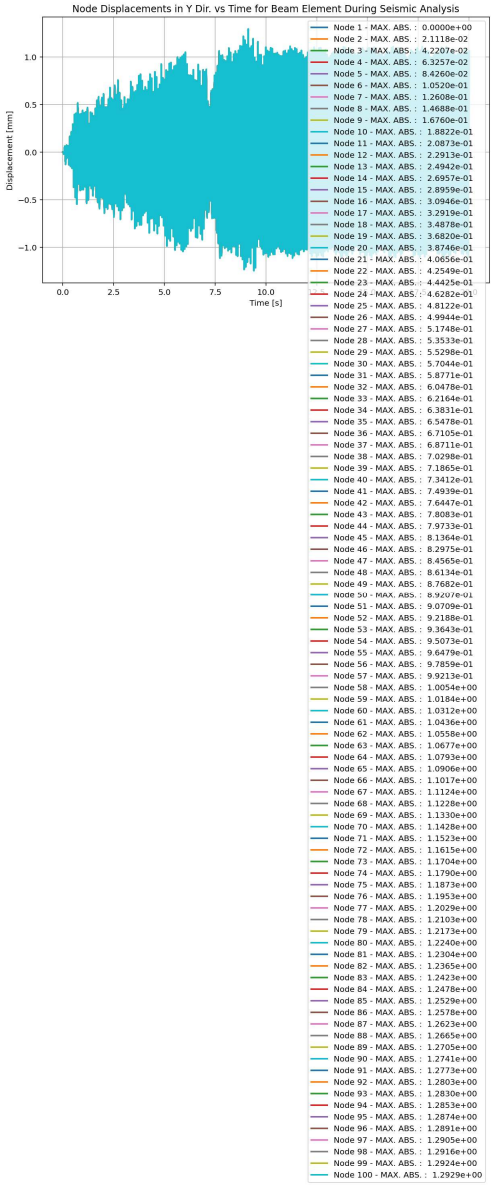


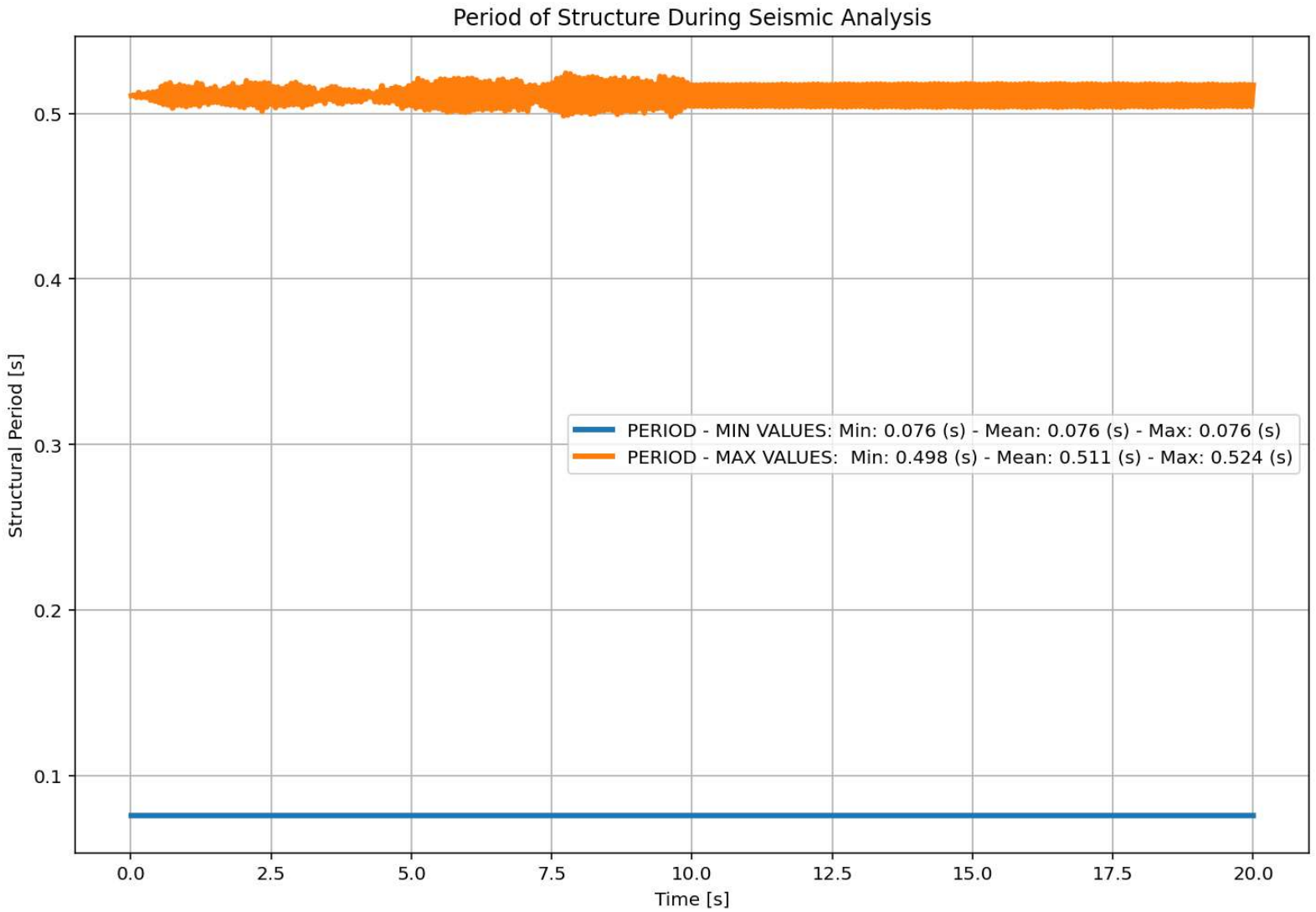
Ele. 1 - MAX. ABS. : 1.7406e+06
Ele. 2 - MAX. ABS. : 1.7381e+06
Ele. 3 - MAX. ABS. : 1.7350e+06
Ele. 4 - MAX. ABS. : 1.7311e+06
Ele. 5 - MAX. ABS. : 1.7263e+06
Ele. 6 - MAX. ABS. : 1.7207e+06
Ele. 7 - MAX. ABS. : 1.7144e+06
Ele. 8 - MAX. ABS. : 1.7073e+06
Ele. 9 - MAX. ABS. : 1.6995e+06
Ele. 10 - MAX. ABS. : 1.6909e+06
Ele. 11 - MAX. ABS. : 1.6816e+06
Ele. 12 - MAX. ABS. : 1.6716e+06
Ele. 13 - MAX. ABS. : 1.6610e+06
Ele. 14 - MAX. ABS. : 1.6499e+06
Ele. 15 - MAX. ABS. : 1.6384e+06
Ele. 16 - MAX. ABS. : 1.6264e+06
Ele. 17 - MAX. ABS. : 1.6139e+06
Ele. 18 - MAX. ABS. : 1.6010e+06
Ele. 19 - MAX. ABS. : 1.5878e+06
Ele. 20 - MAX. ABS. : 1.5741e+06
Ele. 21 - MAX. ABS. : 1.5602e+06
Ele. 22 - MAX. ABS. : 1.5512e+06
Ele. 23 - MAX. ABS. : 1.5468e+06
Ele. 24 - MAX. ABS. : 1.5421e+06
Ele. 25 - MAX. ABS. : 1.5372e+06
Ele. 26 - MAX. ABS. : 1.5318e+06
Ele. 27 - MAX. ABS. : 1.5261e+06
Ele. 28 - MAX. ABS. : 1.5200e+06
Ele. 29 - MAX. ABS. : 1.5135e+06
Ele. 30 - MAX. ABS. : 1.5066e+06
Ele. 31 - MAX. ABS. : 1.4991e+06
Ele. 32 - MAX. ABS. : 1.4911e+06
Ele. 33 - MAX. ABS. : 1.4825e+06
Ele. 34 - MAX. ABS. : 1.4733e+06
Ele. 35 - MAX. ABS. : 1.4635e+06
Ele. 36 - MAX. ABS. : 1.4529e+06
Ele. 37 - MAX. ABS. : 1.4416e+06
Ele. 38 - MAX. ABS. : 1.4297e+06
Ele. 39 - MAX. ABS. : 1.4171e+06
Ele. 40 - MAX. ABS. : 1.4038e+06
Ele. 41 - MAX. ABS. : 1.3899e+06
Ele. 42 - MAX. ABS. : 1.3753e+06
Ele. 43 - MAX. ABS. : 1.3601e+06
Ele. 44 - MAX. ABS. : 1.3443e+06
Ele. 45 - MAX. ABS. : 1.3279e+06
Ele. 46 - MAX. ABS. : 1.3110e+06
Ele. 47 - MAX. ABS. : 1.2935e+06
Ele. 48 - MAX. ABS. : 1.2756e+06
Ele. 49 - MAX. ABS. : 1.2573e+06
Ele. 50 - MAX. ABS. : 1.2384e+06
Ele. 51 - MAX. ABS. : 1.2192e+06
Ele. 52 - MAX. ABS. : 1.1994e+06
Ele. 53 - MAX. ABS. : 1.1792e+06
Ele. 54 - MAX. ABS. : 1.1587e+06
Ele. 55 - MAX. ABS. : 1.1377e+06
Ele. 56 - MAX. ABS. : 1.1164e+06
Ele. 57 - MAX. ABS. : 1.0947e+06
Ele. 58 - MAX. ABS. : 1.0726e+06
Ele. 59 - MAX. ABS. : 1.0502e+06
Ele. 60 - MAX. ABS. : 1.0274e+06
Ele. 61 - MAX. ABS. : 1.0044e+06
Ele. 62 - MAX. ABS. : 9.8107e+05
Ele. 63 - MAX. ABS. : 9.5746e+05
Ele. 64 - MAX. ABS. : 9.3357e+05
Ele. 65 - MAX. ABS. : 9.0941e+05
Ele. 66 - MAX. ABS. : 8.8497e+05
Ele. 67 - MAX. ABS. : 8.6329e+05
Ele. 68 - MAX. ABS. : 8.4341e+05
Ele. 69 - MAX. ABS. : 8.2310e+05
Ele. 70 - MAX. ABS. : 8.0320e+05
Ele. 71 - MAX. ABS. : 7.8410e+05
Ele. 72 - MAX. ABS. : 7.6449e+05
Ele. 73 - MAX. ABS. : 7.4434e+05
Ele. 74 - MAX. ABS. : 7.2364e+05
Ele. 75 - MAX. ABS. : 7.0235e+05
Ele. 76 - MAX. ABS. : 6.8047e+05
Ele. 77 - MAX. ABS. : 6.5796e+05
Ele. 78 - MAX. ABS. : 6.3485e+05
Ele. 79 - MAX. ABS. : 6.1113e+05
Ele. 80 - MAX. ABS. : 5.8684e+05
Ele. 81 - MAX. ABS. : 5.6200e+05
Ele. 82 - MAX. ABS. : 5.3663e+05
Ele. 83 - MAX. ABS. : 5.1074e+05
Ele. 84 - MAX. ABS. : 4.8431e+05
Ele. 85 - MAX. ABS. : 4.5734e+05
Ele. 86 - MAX. ABS. : 4.2987e+05
Ele. 87 - MAX. ABS. : 4.0190e+05
Ele. 88 - MAX. ABS. : 3.7350e+05
Ele. 89 - MAX. ABS. : 3.4474e+05
Ele. 90 - MAX. ABS. : 3.1567e+05
Ele. 91 - MAX. ABS. : 2.8636e+05
Ele. 92 - MAX. ABS. : 2.5803e+05
Ele. 93 - MAX. ABS. : 2.2948e+05
Ele. 94 - MAX. ABS. : 2.0037e+05
Ele. 95 - MAX. ABS. : 1.7072e+05
Ele. 96 - MAX. ABS. : 1.4055e+05
Ele. 97 - MAX. ABS. : 1.0991e+05
Ele. 98 - MAX. ABS. : 7.8846e+04

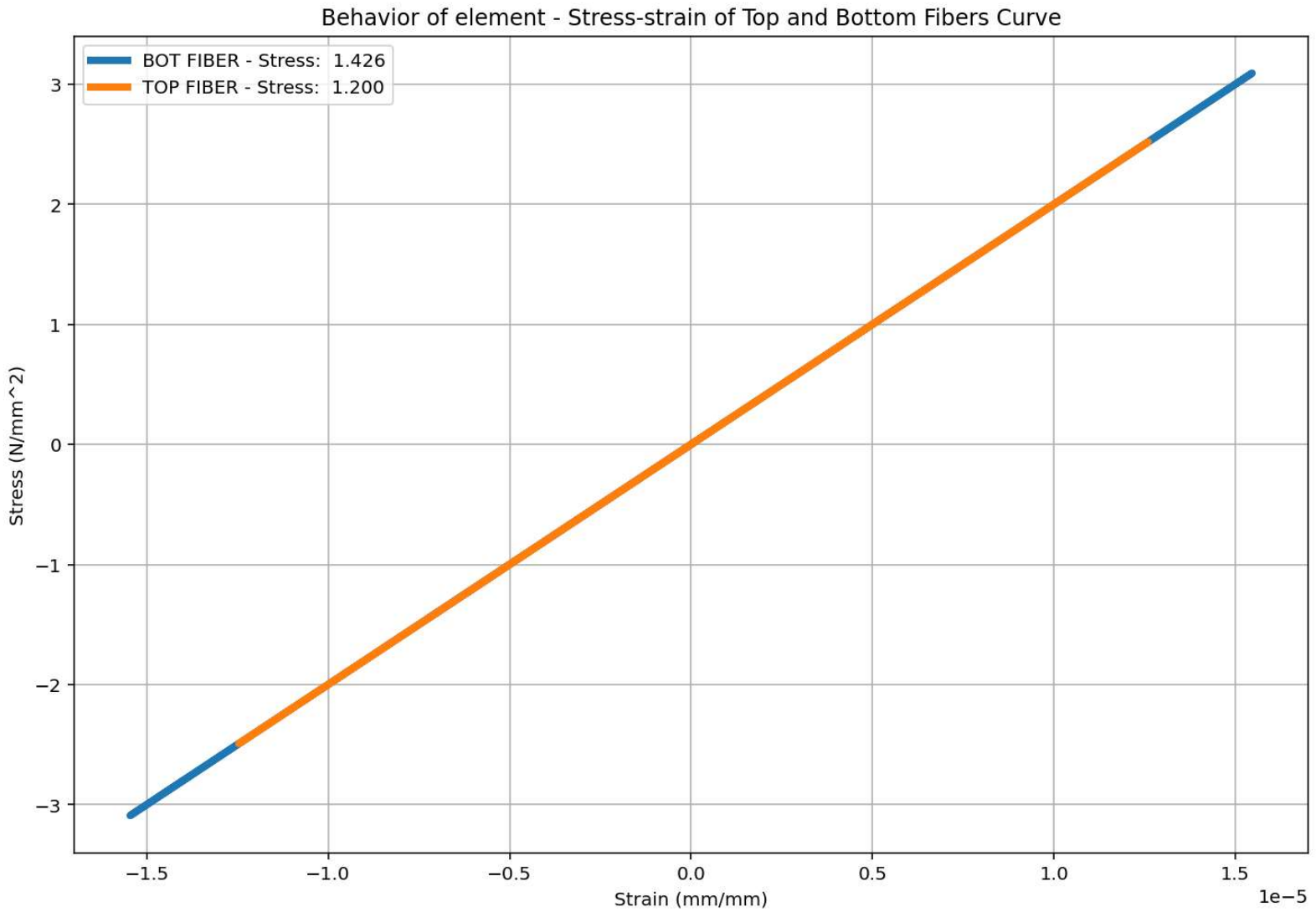


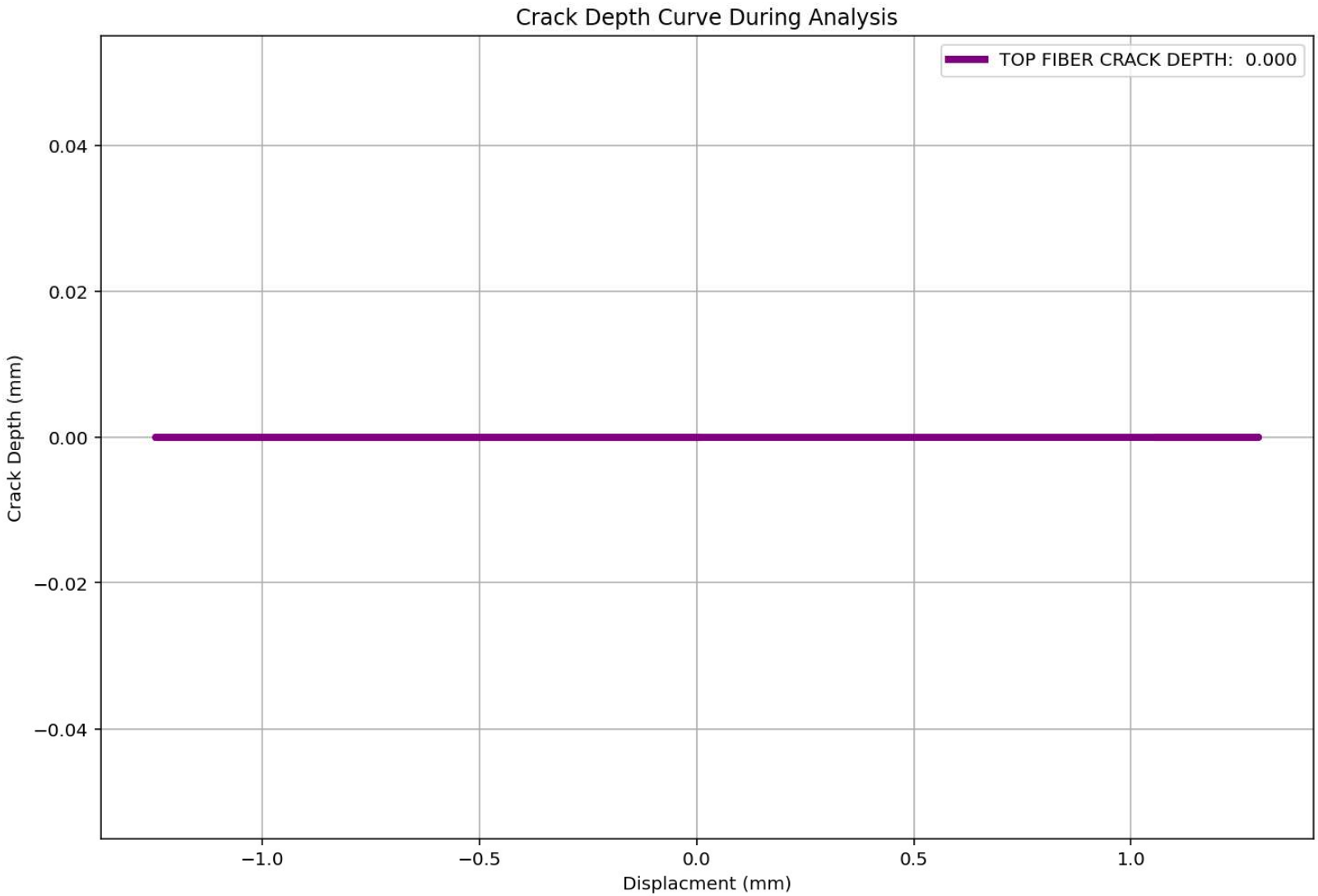
Ele. 1 - MAX. ABS. : 1.0888e+03
Ele. 2 - MAX. ABS. : 8.8066e+04
Ele. 3 - MAX. ABS. : 1.7656e+05
Ele. 4 - MAX. ABS. : 2.6420e+05
Ele. 5 - MAX. ABS. : 3.5061e+05
Ele. 6 - MAX. ABS. : 4.3546e+05
Ele. 7 - MAX. ABS. : 5.1841e+05
Ele. 8 - MAX. ABS. : 5.9914e+05
Ele. 9 - MAX. ABS. : 6.7736e+05
Ele. 10 - MAX. ABS. : 7.5280e+05
Ele. 11 - MAX. ABS. : 8.2519e+05
Ele. 12 - MAX. ABS. : 8.9430e+05
Ele. 13 - MAX. ABS. : 9.5992e+05
Ele. 14 - MAX. ABS. : 1.0219e+06
Ele. 15 - MAX. ABS. : 1.0800e+06
Ele. 16 - MAX. ABS. : 1.1341e+06
Ele. 17 - MAX. ABS. : 1.1841e+06
Ele. 18 - MAX. ABS. : 1.2300e+06
Ele. 19 - MAX. ABS. : 1.2716e+06
Ele. 20 - MAX. ABS. : 1.3089e+06
Ele. 21 - MAX. ABS. : 1.3420e+06
Ele. 22 - MAX. ABS. : 1.3708e+06
Ele. 23 - MAX. ABS. : 1.3954e+06
Ele. 24 - MAX. ABS. : 1.4158e+06
Ele. 25 - MAX. ABS. : 1.4320e+06
Ele. 26 - MAX. ABS. : 1.4441e+06
Ele. 27 - MAX. ABS. : 1.4596e+06
Ele. 28 - MAX. ABS. : 1.4860e+06
Ele. 29 - MAX. ABS. : 1.5095e+06
Ele. 30 - MAX. ABS. : 1.5303e+06
Ele. 31 - MAX. ABS. : 1.5484e+06
Ele. 32 - MAX. ABS. : 1.5638e+06
Ele. 33 - MAX. ABS. : 1.5767e+06
Ele. 34 - MAX. ABS. : 1.5870e+06
Ele. 35 - MAX. ABS. : 1.5949e+06
Ele. 36 - MAX. ABS. : 1.6005e+06
Ele. 37 - MAX. ABS. : 1.6103e+06
Ele. 38 - MAX. ABS. : 1.6269e+06
Ele. 39 - MAX. ABS. : 1.6405e+06
Ele. 40 - MAX. ABS. : 1.6509e+06
Ele. 41 - MAX. ABS. : 1.6582e+06
Ele. 42 - MAX. ABS. : 1.6624e+06
Ele. 43 - MAX. ABS. : 1.6635e+06
Ele. 44 - MAX. ABS. : 1.6616e+06
Ele. 45 - MAX. ABS. : 1.6566e+06
Ele. 46 - MAX. ABS. : 1.6487e+06
Ele. 47 - MAX. ABS. : 1.6380e+06
Ele. 48 - MAX. ABS. : 1.6245e+06
Ele. 49 - MAX. ABS. : 1.6083e+06
Ele. 50 - MAX. ABS. : 1.5997e+06
Ele. 51 - MAX. ABS. : 1.6081e+06
Ele. 52 - MAX. ABS. : 1.6136e+06
Ele. 53 - MAX. ABS. : 1.6164e+06
Ele. 54 - MAX. ABS. : 1.6163e+06
Ele. 55 - MAX. ABS. : 1.6133e+06
Ele. 56 - MAX. ABS. : 1.6144e+06
Ele. 57 - MAX. ABS. : 1.6161e+06
Ele. 58 - MAX. ABS. : 1.6132e+06
Ele. 59 - MAX. ABS. : 1.6057e+06
Ele. 60 - MAX. ABS. : 1.5938e+06
Ele. 61 - MAX. ABS. : 1.5775e+06
Ele. 62 - MAX. ABS. : 1.5571e+06
Ele. 63 - MAX. ABS. : 1.5451e+06
Ele. 64 - MAX. ABS. : 1.5506e+06
Ele. 65 - MAX. ABS. : 1.5533e+06
Ele. 66 - MAX. ABS. : 1.5532e+06
Ele. 67 - MAX. ABS. : 1.5503e+06
Ele. 68 - MAX. ABS. : 1.5444e+06
Ele. 69 - MAX. ABS. : 1.5355e+06
Ele. 70 - MAX. ABS. : 1.5237e+06
Ele. 71 - MAX. ABS. : 1.5089e+06
Ele. 72 - MAX. ABS. : 1.4912e+06
Ele. 73 - MAX. ABS. : 1.4705e+06
Ele. 74 - MAX. ABS. : 1.4468e+06
Ele. 75 - MAX. ABS. : 1.4203e+06
Ele. 76 - MAX. ABS. : 1.3910e+06
Ele. 77 - MAX. ABS. : 1.3588e+06
Ele. 78 - MAX. ABS. : 1.3238e+06
Ele. 79 - MAX. ABS. : 1.2940e+06
Ele. 80 - MAX. ABS. : 1.2675e+06
Ele. 81 - MAX. ABS. : 1.2448e+06
Ele. 82 - MAX. ABS. : 1.2186e+06
Ele. 83 - MAX. ABS. : 1.1883e+06
Ele. 84 - MAX. ABS. : 1.1538e+06
Ele. 85 - MAX. ABS. : 1.1151e+06
Ele. 86 - MAX. ABS. : 1.0723e+06
Ele. 87 - MAX. ABS. : 1.0253e+06
Ele. 88 - MAX. ABS. : 9.7422e+05
Ele. 89 - MAX. ABS. : 9.1913e+05
Ele. 90 - MAX. ABS. : 8.6011e+05
Ele. 91 - MAX. ABS. : 7.9730e+05
Ele. 92 - MAX. ABS. : 7.3084e+05
Ele. 93 - MAX. ABS. : 6.6088e+05
Ele. 94 - MAX. ABS. : 5.8763e+05
Ele. 95 - MAX. ABS. : 5.1126e+05
Ele. 96 - MAX. ABS. : 4.3198e+05
Ele. 97 - MAX. ABS. : 3.4999e+05
Ele. 98 - MAX. ABS. : 2.6551e+05

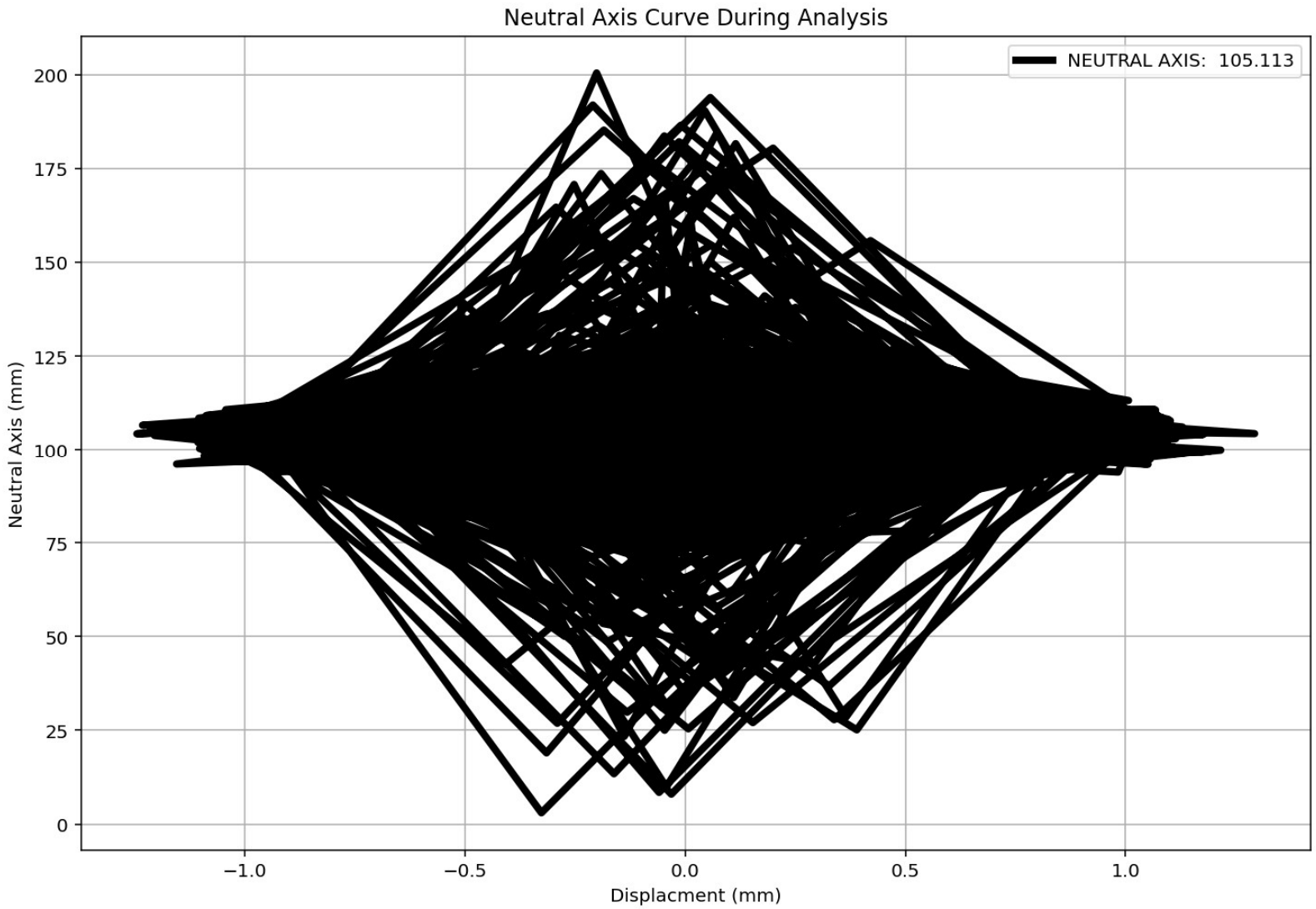

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.9219e-03
Node 3 - MAX. ABS. : 7.8387e-03
Node 4 - MAX. ABS. : 1.1745e-02
Node 5 - MAX. ABS. : 1.5635e-02
Node 6 - MAX. ABS. : 1.9505e-02
Node 7 - MAX. ABS. : 2.3348e-02
Node 8 - MAX. ABS. : 2.7160e-02
Node 9 - MAX. ABS. : 3.0935e-02
Node 10 - MAX. ABS. : 3.4669e-02
Node 11 - MAX. ABS. : 3.8356e-02
Node 12 - MAX. ABS. : 4.1992e-02
Node 13 - MAX. ABS. : 4.5572e-02
Node 14 - MAX. ABS. : 4.9091e-02
Node 15 - MAX. ABS. : 5.2545e-02
Node 16 - MAX. ABS. : 5.5930e-02
Node 17 - MAX. ABS. : 5.9240e-02
Node 18 - MAX. ABS. : 6.2472e-02
Node 19 - MAX. ABS. : 6.5622e-02
Node 20 - MAX. ABS. : 6.8687e-02
Node 21 - MAX. ABS. : 7.1662e-02
Node 22 - MAX. ABS. : 7.4545e-02
Node 23 - MAX. ABS. : 7.7332e-02
Node 24 - MAX. ABS. : 8.0020e-02
Node 25 - MAX. ABS. : 8.2606e-02
Node 26 - MAX. ABS. : 8.5088e-02
Node 27 - MAX. ABS. : 8.7464e-02
Node 28 - MAX. ABS. : 8.9731e-02
Node 29 - MAX. ABS. : 9.1887e-02
Node 30 - MAX. ABS. : 9.3930e-02
Node 31 - MAX. ABS. : 9.5859e-02
Node 32 - MAX. ABS. : 9.7673e-02
Node 33 - MAX. ABS. : 9.9371e-02
Node 34 - MAX. ABS. : 1.0095e-01
Node 35 - MAX. ABS. : 1.0241e-01
Node 36 - MAX. ABS. : 1.0375e-01
Node 37 - MAX. ABS. : 1.0498e-01
Node 38 - MAX. ABS. : 1.0608e-01
Node 39 - MAX. ABS. : 1.0707e-01
Node 40 - MAX. ABS. : 1.0797e-01
Node 41 - MAX. ABS. : 1.0888e-01
Node 42 - MAX. ABS. : 1.0966e-01
Node 43 - MAX. ABS. : 1.1032e-01
Node 44 - MAX. ABS. : 1.1085e-01
Node 45 - MAX. ABS. : 1.1126e-01
Node 46 - MAX. ABS. : 1.1155e-01
Node 47 - MAX. ABS. : 1.1172e-01
Node 48 - MAX. ABS. : 1.1177e-01
Node 49 - MAX. ABS. : 1.1186e-01
Node 50 - MAX. ABS. : 1.1209e-01
Node 51 - MAX. ABS. : 1.1222e-01
Node 52 - MAX. ABS. : 1.1224e-01
Node 53 - MAX. ABS. : 1.1214e-01
Node 54 - MAX. ABS. : 1.1193e-01
Node 55 - MAX. ABS. : 1.1161e-01
Node 56 - MAX. ABS. : 1.1118e-01
Node 57 - MAX. ABS. : 1.1063e-01
Node 58 - MAX. ABS. : 1.0998e-01
Node 59 - MAX. ABS. : 1.0922e-01
Node 60 - MAX. ABS. : 1.0835e-01
Node 61 - MAX. ABS. : 1.0737e-01
Node 62 - MAX. ABS. : 1.0628e-01
Node 63 - MAX. ABS. : 1.0509e-01
Node 64 - MAX. ABS. : 1.0379e-01
Node 65 - MAX. ABS. : 1.0239e-01
Node 66 - MAX. ABS. : 1.0089e-01
Node 67 - MAX. ABS. : 9.9287e-02
Node 68 - MAX. ABS. : 9.7585e-02
Node 69 - MAX. ABS. : 9.5889e-02
Node 70 - MAX. ABS. : 9.4155e-02
Node 71 - MAX. ABS. : 9.2322e-02
Node 72 - MAX. ABS. : 9.0389e-02
Node 73 - MAX. ABS. : 8.8359e-02
Node 74 - MAX. ABS. : 8.6230e-02
Node 75 - MAX. ABS. : 8.4006e-02
Node 76 - MAX. ABS. : 8.1687e-02
Node 77 - MAX. ABS. : 7.9274e-02
Node 78 - MAX. ABS. : 7.6768e-02
Node 79 - MAX. ABS. : 7.4173e-02
Node 80 - MAX. ABS. : 7.1488e-02
Node 81 - MAX. ABS. : 6.8718e-02
Node 82 - MAX. ABS. : 6.5862e-02
Node 83 - MAX. ABS. : 6.2925e-02
Node 84 - MAX. ABS. : 5.9909e-02
Node 85 - MAX. ABS. : 5.6817e-02
Node 86 - MAX. ABS. : 5.3651e-02
Node 87 - MAX. ABS. : 5.0414e-02
Node 88 - MAX. ABS. : 4.7111e-02
Node 89 - MAX. ABS. : 4.3745e-02
Node 90 - MAX. ABS. : 4.0320e-02
Node 91 - MAX. ABS. : 3.6839e-02
Node 92 - MAX. ABS. : 3.3307e-02
Node 93 - MAX. ABS. : 2.9728e-02
Node 94 - MAX. ABS. : 2.6107e-02
Node 95 - MAX. ABS. : 2.2448e-02
Node 96 - MAX. ABS. : 1.8757e-02
Node 97 - MAX. ABS. : 1.5039e-02
Node 98 - MAX. ABS. : 1.1298e-02
Node 99 - MAX. ABS. : 7.5411e-03
Node 100 - MAX. ABS. : 3.7731e-03

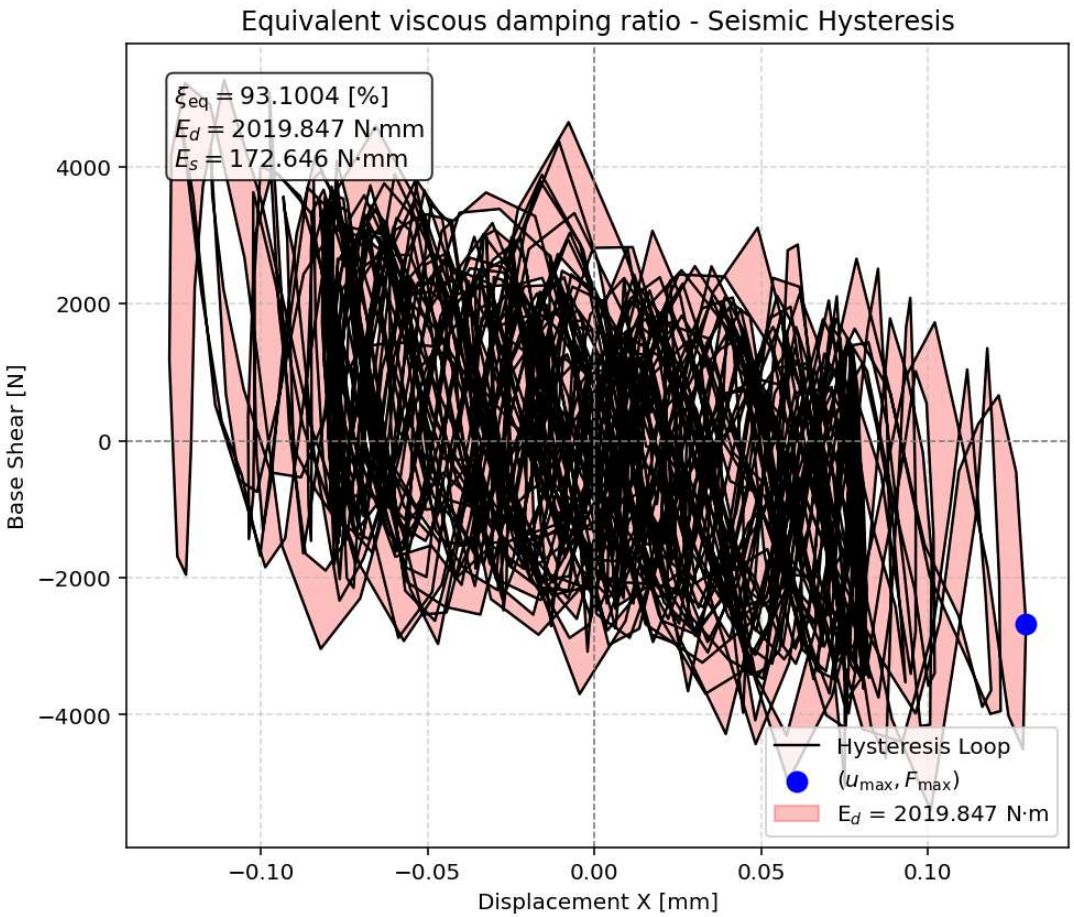

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
1.0
0.5
0.0
−0.5
−1.0
0.0
2.5
5.0
7.5
10.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.1118e-02
Node 3 - MAX. ABS. : 4.2207e-02
Node 4 - MAX. ABS. : 6.3257e-02
Node 5 - MAX. ABS. : 8.4260e-02
Node 6 - MAX. ABS. : 1.0520e-01
Node 7 - MAX. ABS. : 1.2608e-01
Node 8 - MAX. ABS. : 1.4688e-01
Node 9 - MAX. ABS. : 1.6760e-01
Node 10 - MAX. ABS. : 1.8822e-01
Node 11 - MAX. ABS. : 2.0873e-01
Node 12 - MAX. ABS. : 2.2913e-01
Node 13 - MAX. ABS. : 2.4942e-01
Node 14 - MAX. ABS. : 2.6957e-01
Node 15 - MAX. ABS. : 2.8959e-01
Node 16 - MAX. ABS. : 3.0946e-01
Node 17 - MAX. ABS. : 3.2919e-01
Node 18 - MAX. ABS. : 3.4878e-01
Node 19 - MAX. ABS. : 3.6820e-01
Node 20 - MAX. ABS. : 3.8746e-01
Node 21 - MAX. ABS. : 4.0656e-01
Node 22 - MAX. ABS. : 4.2549e-01
Node 23 - MAX. ABS. : 4.4425e-01
Node 24 - MAX. ABS. : 4.6282e-01
Node 25 - MAX. ABS. : 4.8122e-01
Node 26 - MAX. ABS. : 4.9944e-01
Node 27 - MAX. ABS. : 5.1748e-01
Node 28 - MAX. ABS. : 5.3533e-01
Node 29 - MAX. ABS. : 5.5298e-01
Node 30 - MAX. ABS. : 5.7044e-01
Node 31 - MAX. ABS. : 5.8771e-01
Node 32 - MAX. ABS. : 6.0478e-01
Node 33 - MAX. ABS. : 6.2164e-01
Node 34 - MAX. ABS. : 6.3831e-01
Node 35 - MAX. ABS. : 6.5478e-01
Node 36 - MAX. ABS. : 6.7105e-01
Node 37 - MAX. ABS. : 6.8711e-01
Node 38 - MAX. ABS. : 7.0298e-01
Node 39 - MAX. ABS. : 7.1865e-01
Node 40 - MAX. ABS. : 7.3412e-01
Node 41 - MAX. ABS. : 7.4939e-01
Node 42 - MAX. ABS. : 7.6447e-01
Node 43 - MAX. ABS. : 7.8083e-01
Node 44 - MAX. ABS. : 7.9733e-01
Node 45 - MAX. ABS. : 8.1364e-01
Node 46 - MAX. ABS. : 8.2975e-01
Node 47 - MAX. ABS. : 8.4565e-01
Node 48 - MAX. ABS. : 8.6134e-01
Node 49 - MAX. ABS. : 8.7682e-01
Node 50 - MAX. ABS. : 8.9207e-01
Node 51 - MAX. ABS. : 9.0709e-01
Node 52 - MAX. ABS. : 9.2188e-01
Node 53 - MAX. ABS. : 9.3643e-01
Node 54 - MAX. ABS. : 9.5073e-01
Node 55 - MAX. ABS. : 9.6479e-01
Node 56 - MAX. ABS. : 9.7859e-01
Node 57 - MAX. ABS. : 9.9213e-01
Node 58 - MAX. ABS. : 1.0054e+00
Node 59 - MAX. ABS. : 1.0184e+00
Node 60 - MAX. ABS. : 1.0312e+00
Node 61 - MAX. ABS. : 1.0436e+00
Node 62 - MAX. ABS. : 1.0558e+00
Node 63 - MAX. ABS. : 1.0677e+00
Node 64 - MAX. ABS. : 1.0793e+00
Node 65 - MAX. ABS. : 1.0906e+00
Node 66 - MAX. ABS. : 1.1017e+00
Node 67 - MAX. ABS. : 1.1124e+00
Node 68 - MAX. ABS. : 1.1228e+00
Node 69 - MAX. ABS. : 1.1330e+00
Node 70 - MAX. ABS. : 1.1428e+00
Node 71 - MAX. ABS. : 1.1523e+00
Node 72 - MAX. ABS. : 1.1615e+00
Node 73 - MAX. ABS. : 1.1704e+00
Node 74 - MAX. ABS. : 1.1790e+00
Node 75 - MAX. ABS. : 1.1873e+00
Node 76 - MAX. ABS. : 1.1953e+00
Node 77 - MAX. ABS. : 1.2029e+00
Node 78 - MAX. ABS. : 1.2103e+00
Node 79 - MAX. ABS. : 1.2173e+00
Node 80 - MAX. ABS. : 1.2240e+00
Node 81 - MAX. ABS. : 1.2304e+00
Node 82 - MAX. ABS. : 1.2365e+00
Node 83 - MAX. ABS. : 1.2423e+00
Node 84 - MAX. ABS. : 1.2478e+00
Node 85 - MAX. ABS. : 1.2529e+00
Node 86 - MAX. ABS. : 1.2578e+00
Node 87 - MAX. ABS. : 1.2623e+00
Node 88 - MAX. ABS. : 1.2665e+00
Node 89 - MAX. ABS. : 1.2705e+00
Node 90 - MAX. ABS. : 1.2741e+00
Node 91 - MAX. ABS. : 1.2773e+00
Node 92 - MAX. ABS. : 1.2803e+00
Node 93 - MAX. ABS. : 1.2830e+00
Node 94 - MAX. ABS. : 1.2853e+00
Node 95 - MAX. ABS. : 1.2874e+00
Node 96 - MAX. ABS. : 1.2891e+00
Node 97 - MAX. ABS. : 1.2905e+00
Node 98 - MAX. ABS. : 1.2916e+00
Node 99 - MAX. ABS. : 1.2924e+00
Node 100 - MAX. ABS. : 1.2929e+00

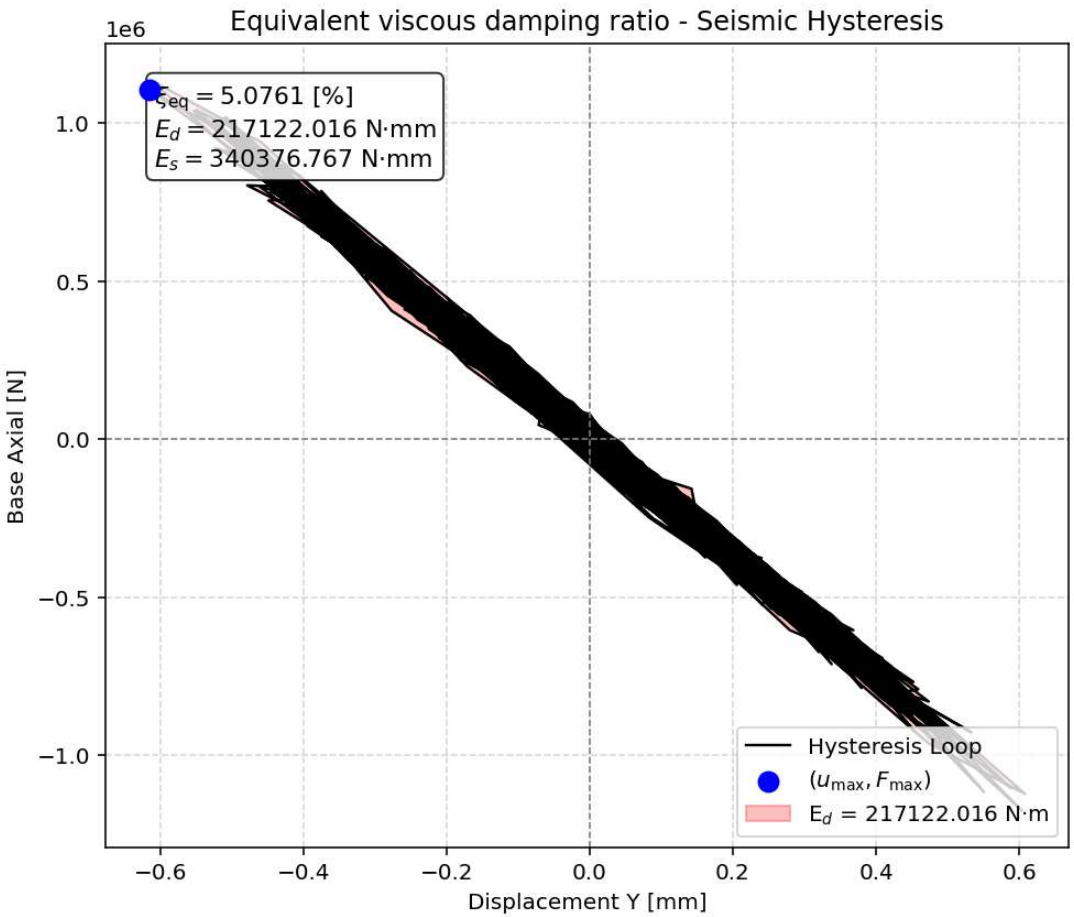

Period of Structure During Seismic Analysis
Structural Period [s]
0.1
0.2
0.3
0.4
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.076 (s) - Mean: 0.076 (s) - Max: 0.076 (s)
PERIOD - MAX VALUES: Min: 0.498 (s) - Mean: 0.511 (s) - Max: 0.524 (s)

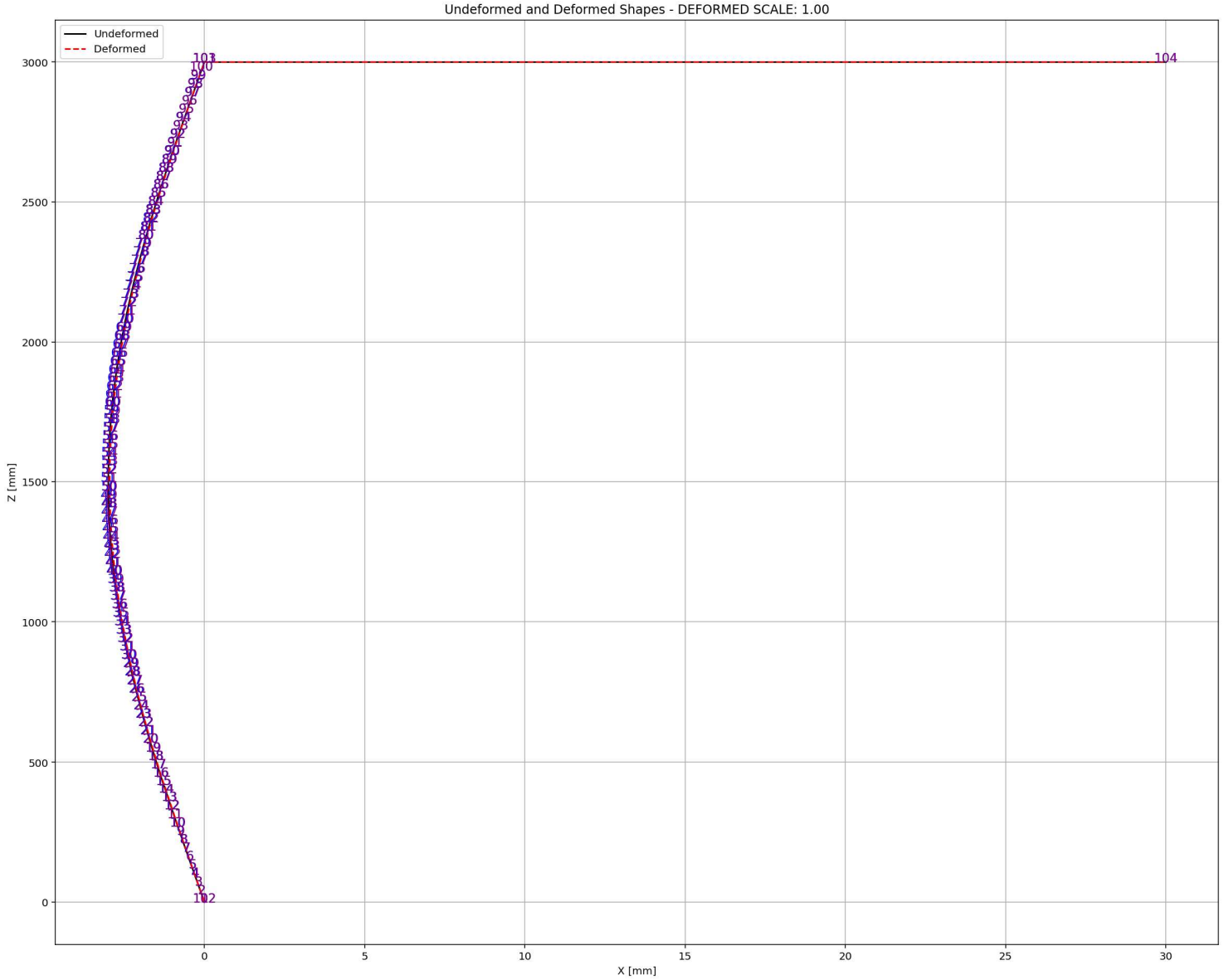

Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 1.426
TOP FIBER - Stress: 1.200
Stress (N/mm^2)
3
2
1
0
−1
−2
−3
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Strain (mm/mm)
1e−5

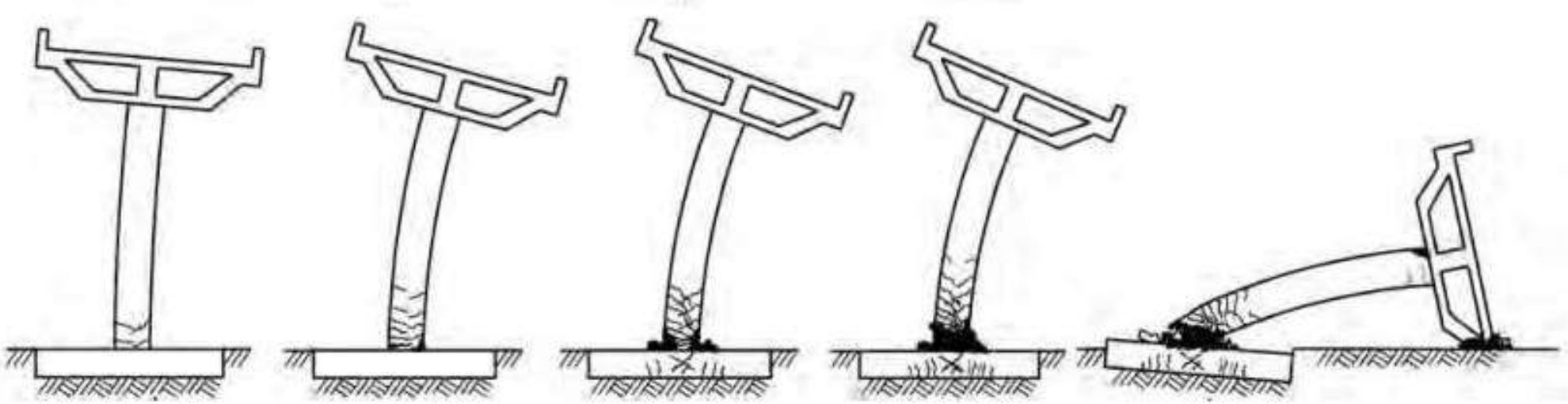

Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−1.0
−0.5
0.0
0.5
1.0
Displacment (mm)

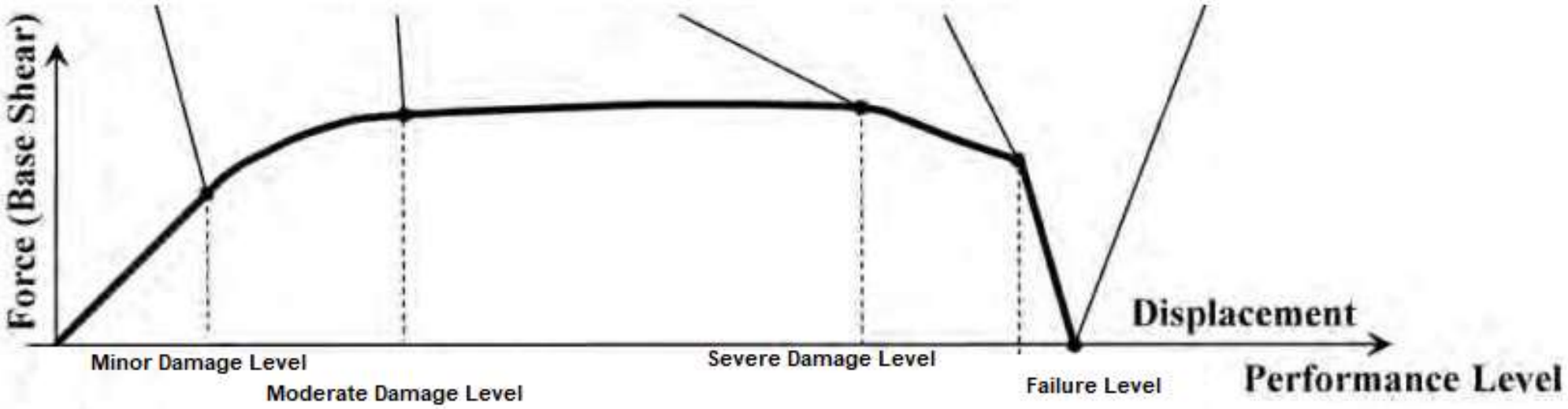

Neutral Axis Curve During Analysis
NEUTRAL AXIS: 105.113
Neutral Axis (mm)
200
175
150
125
100
75
50
25
0
−1.0
−0.5
0.0
0.5
1.0
Displacment (mm)

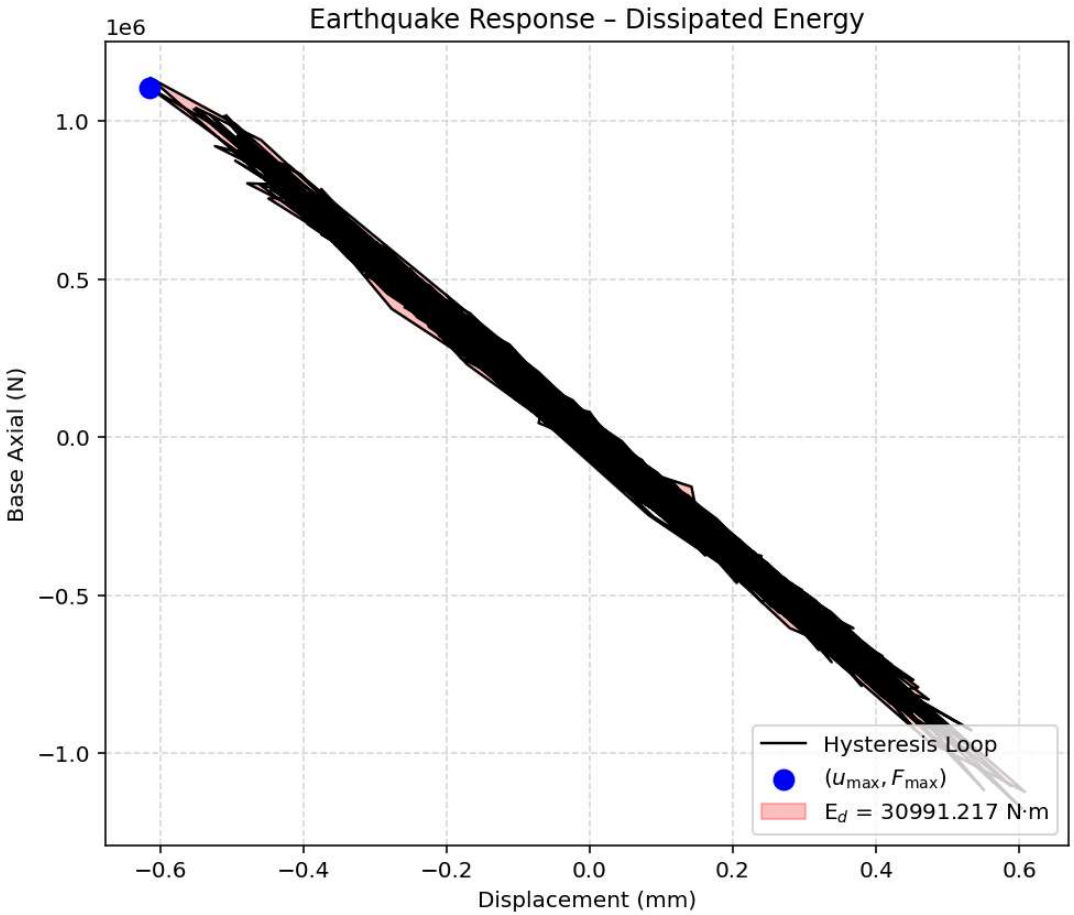

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 93.1004$ [%]
$E_d = 2019.847$ N·mm
$E_s = 172.646$ N·mm
Base Shear [N]
4000
2000
0
−2000
−4000
−0.10
−0.05
0.00
0.05
0.10
Displacement X [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2019.847 N·m

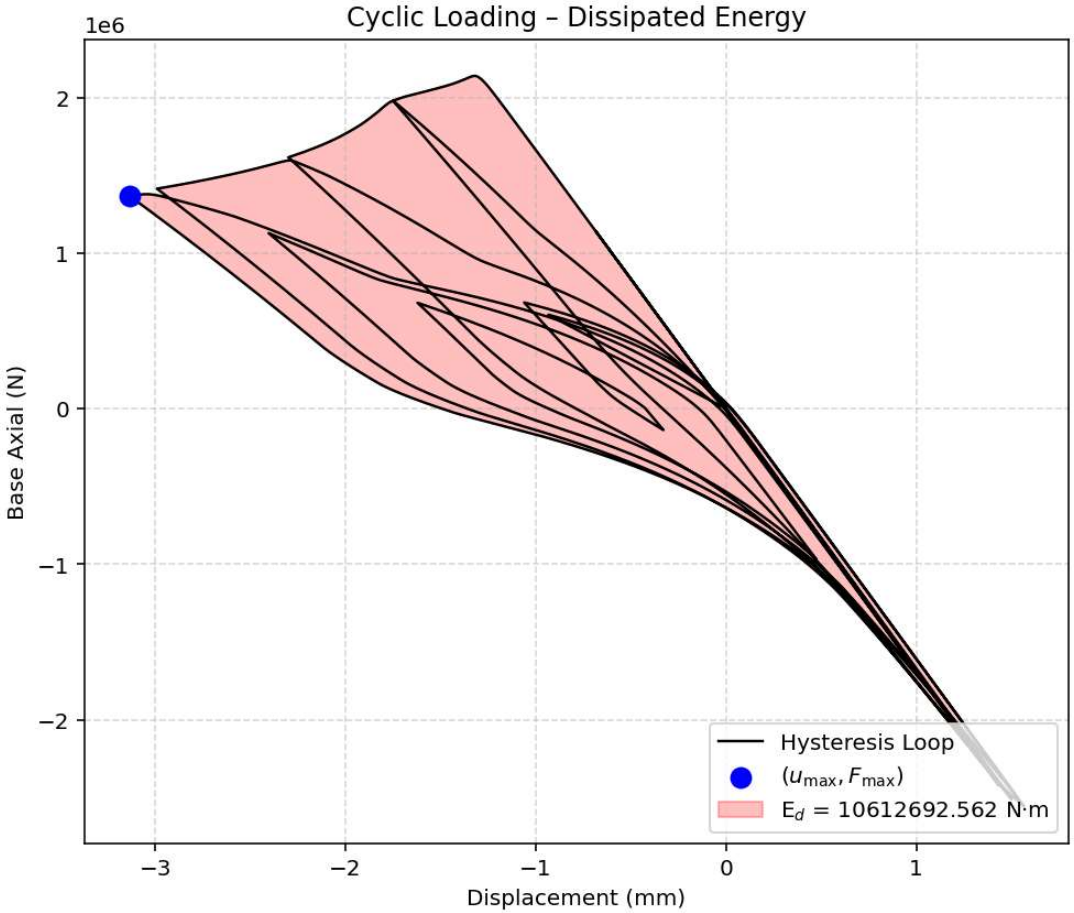

Equivalent viscous damping ratio - Seismic Hysteresis
1e6
$\zeta_{eq} = 5.0761$ [%]
$E_d = 217122.016$ N·mm
$E_s = 340376.767$ N·mm
Base Axial [N]
Displacement Y [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 217122.016 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
5
10
15
20
25
30
103
104
102



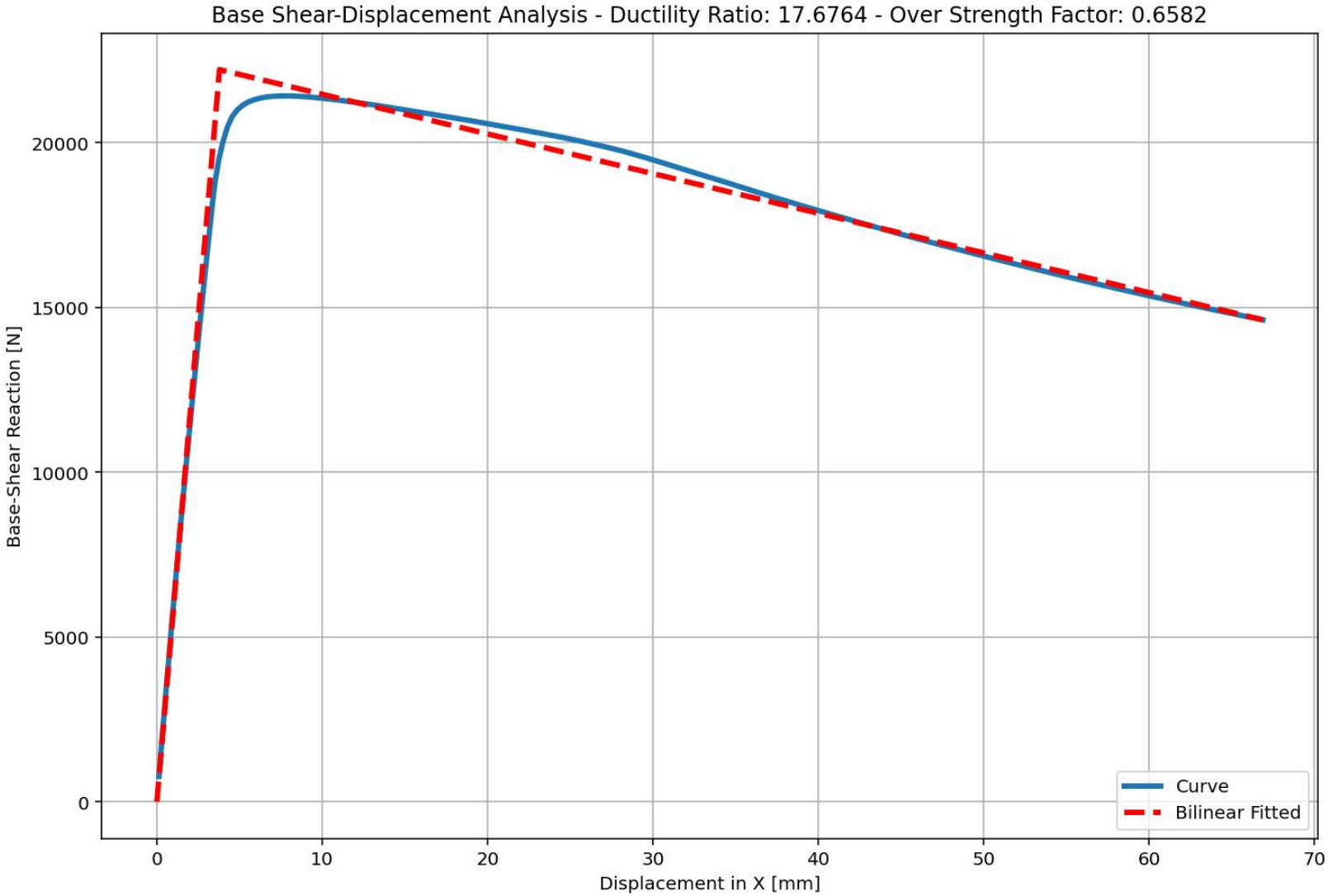

Force (Base Shear)
Displacement
Performance Level
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level


Earthquake Response – Dissipated Energy
1e6
1.0
0.5
0.0
−0.5
−1.0
Base Axial (N)
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Displacement (mm)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 30991.217 N·m

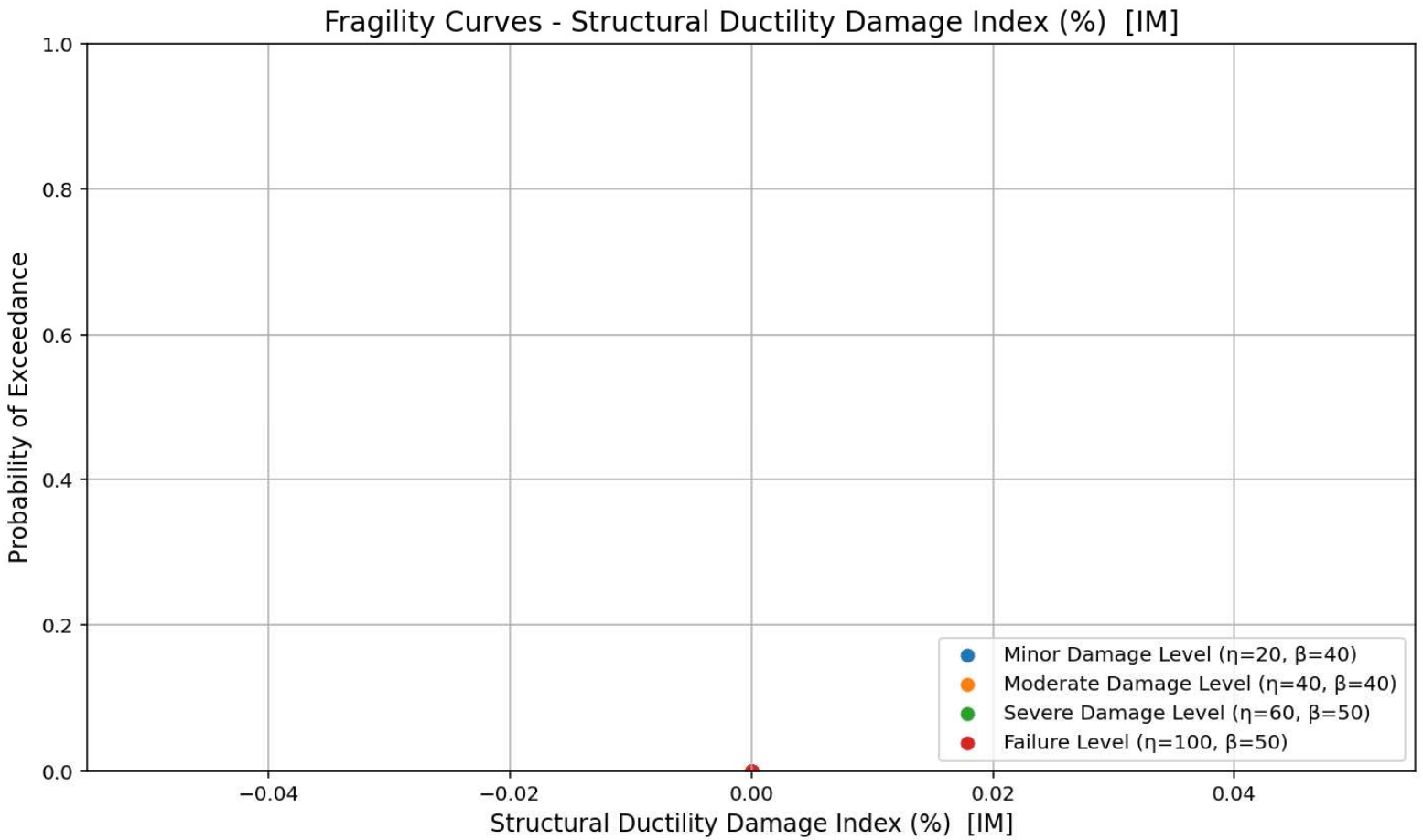

Cyclic Loading – Dissipated Energy
1e6
2
1
0
−1
−2
Base Axial (N)
−3
−2
−1
0
1
Displacement (mm)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 10612692.562 N·m

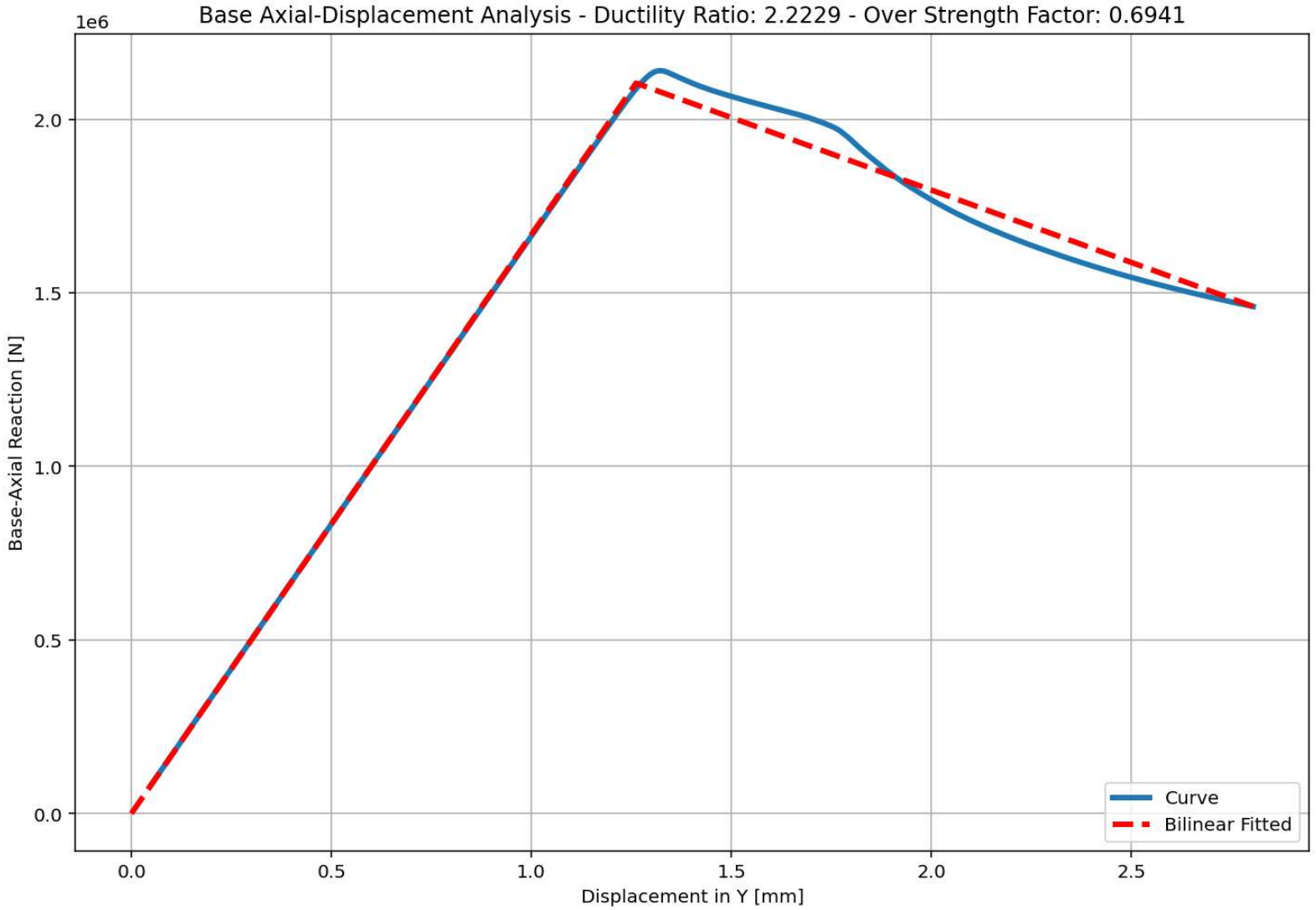
Dissipated Energy from Earthquake= 489440.55 N·mm
Dissipated Energy from Cyclic Displacement= 10612692.56 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.292 [%]
ZONE 1: VERY MINOR DAMAGE

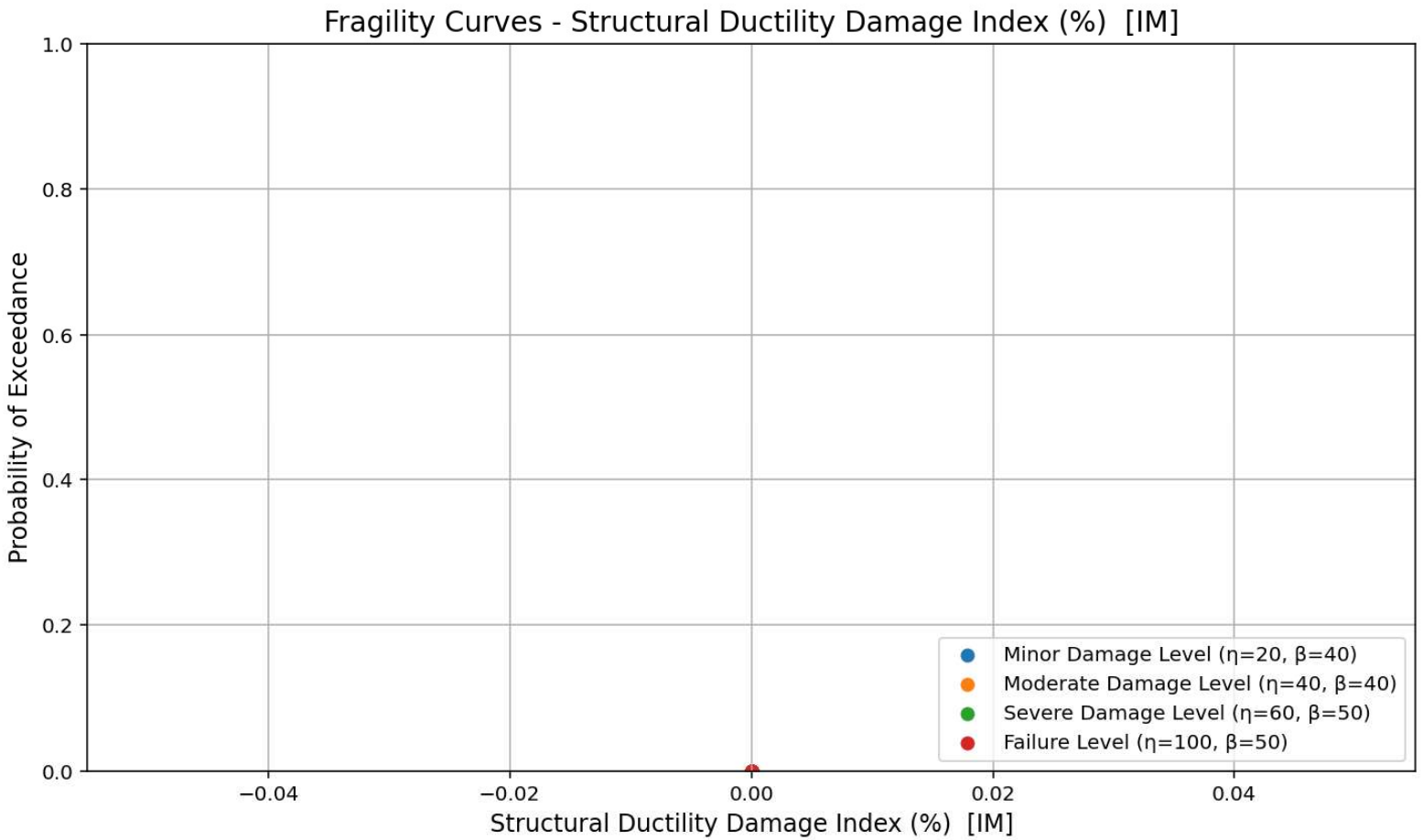

Base Shear-Displacement Analysis - Ductility Ratio: 17.6764 - Over Strength Factor: 0.6582
Base-Shear Reaction [N]
20000
15000
10000
5000
0
0
10
20
30
40
50
60
70
Displacement in X [mm]
Curve
Bilinear Fitted

```
+==========================+
=  Analysis curve fitted =
   Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [2.80101456e+00 2.30592094e+06]
 [2.50000000e+01 1.16951808e+06]]
+==========================+
+------------------------------------------------+
 Structure Elastic Stiffness :    823244.89
 Structure Plastic Stiffness :    46780.72
 Structure Tangent Stiffness :    -51191.66
 Structure Ductility Ratio :      8.93
 Structure Over Strength Factor:   0.51
+------------------------------------------------+
Over Strength Coefficient (Ω0):     0.6582
Displacement Ductility Ratio (μ):   17.6764
Ductility Coefficient (Rμ):         17.6764
Structural Behavior Coefficient (R): 11.6351
Structural Ductility Damage Index in X Direction: -5.7914 (%)
```


Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
−0.04
−0.02
0.00
0.02
0.04
Structural Ductility Damage Index (%) [IM]
Minor Damage Level (η=20, β=40)
Moderate Damage Level (η=40, β=40)
Severe Damage Level (η=60, β=50)
Failure Level (η=100, β=50)


Base Axial-Displacement Analysis - Ductility Ratio: 2.2229 - Over Strength Factor: 0.6941
1e6
2.0
1.5
1.0
0.5
0.0
Base-Axial Reaction [N]
0.0
0.5
1.0
1.5
2.0
2.5
Displacement in Y [mm]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [2.80101456e+00 2.30592094e+06]
 [2.50000000e+01 1.16951808e+06]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :     823244.89
 Structure Plastic Stiffness :     46780.72
 Structure Tangent Stiffness :     -51191.66
 Structure Ductility Ratio :      8.93
 Structure Over Strength Factor:   0.51
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     0.6941
Displacement Ductility Ratio (μ):    2.2229
Ductility Coefficient (Rμ):         2.2229
Structural Behavior Coefficient (R): 1.5430
Structural Ductility Damage Index in Y Direction: -41.8758 (%)
```


Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
−0.04
−0.02
0.00
0.02
0.04
Structural Ductility Damage Index (%) [IM]
Minor Damage Level (η=20, β=40)
Moderate Damage Level (η=40, β=40)
Severe Damage Level (η=60, β=50)
Failure Level (η=100, β=50)